



B. Tech. ELECTRICAL ENGINEERING

Syllabus AY: 2022-23



CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Authored By: Charusat



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY



ACADEMIC REGULATIONS & SYLLABUS 2022-23 (Choice Based Credit System)

Faculty of Technology & Engineering
M & V Patel Department of Electrical Engineering
Bachelor of Technology Programme
(Electrical Engineering)

Chandubhai S Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives

M & V Patel Department of Electrical Engineering

Vision

To become an internationally recognized, research intensive education entity, serving as a workforce developer for the nation by providing the best education in close collaboration with society.

Mission

To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.

THE PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	Preparation: To prepare the students with sound academically background, intensive development of knowledge and skills, and with an uncompromising attitude to achieve excellence in the electrical engineering.
PEO 2	Core Competence: To build a broad engineering core competence and to enable the students to solve the problems systematically by providing them exposure to different dimensions of specializations.
PEO 3	Learning Environment: To provide conducive learning environment instilling sound foundation in mathematics, computing techniques, science and humanities along with professional courses for futuristic thinking in an increasingly technology dependent society.
PEO 4	Professionalism: To provide a platform to the students to plan and shape their career in the chosen field and to imbibe ethical values with thorough professionalism and encouragement towards entrepreneurial thinking and traits.

PROGRAM ARTICULATION MATRIX

(-Not Matching, 1-Low, 2-Medium, 3-High)

	PEO 1	PEO 2	PEO 3	PEO 4
To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.	3	3	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO 1	Engineering knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO 4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	The engineer and society: Apply reasoning obtained by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO 1	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.
PSO 2	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Electrical Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-265011, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2022-2023

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING ACADEMIC REGULATIONS

Bachelor of Technology Programmes

Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

Undergraduate programme	(B. Tech.)
Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for admissions:

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – I attached

6. Attendance

6.1 All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

6.2 Student's attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, the continuous assessment will be conducted by the respective department /institute.
- 7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these.
- 7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.
- 7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.
- 7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examination

- 7.2.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

- 7.2.2 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grade Point System

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	<80 ≥73	<73 ≥66	<66 ≥60	<60 ≥55	<55 ≥50	<50 ≥45	<45
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9 Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
Distinction	$CGPA \geq 7.5 \text{ \& } \leq 10.0$
First Class	$CGPA \geq 6.0 \text{ \& } < 7.5$
Second Class	$CGPA \geq 5.0 \text{ \& } < 6.0$
Pass Class	$CGPA < 5.0$

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9.1 Maximum duration allowed for Completion of a programme

- Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10 *Detention Criteria*

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

11 *Transcript:*

- The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical/Tutorial	Total		Internal	External	Internal	External	
FY SEM-I	HS101.02A	Communicative English	0	2	2	2			30	70	100
	MA143	Engineering Mathematics I	4	1T	5	4	30	70	0	0	100
	ME146	Engineering Graphics	2	2+1T	5	3	30	70	25	25	150
	PY142	Engineering Physics- I	0	2	2	2			50	50	100
	CL144.01A	Environmental Science	0	2	2	2			30	70	100
	CL143	Engineering Mechanics	3	2	5	4	30	70	25	25	150
	ME142	Workshop Practices	0	2	2	1			25	25	50
	FS101A	Foundation Course on Mathematics and Physics	0	2	2	2			50	50	100
			9	14	25	20	90	210	235	315	850
FY SEM-II	HS201.02A- HS210.02	A Course on Liberal Arts	2		2	2			30	70	100
	MA144	Engineering Mathematics II	4	1	5	4	30	70			100
	IT143	Fundamentals of Computer Programming	2	2	4	3	30	70	25	25	150
	ME147	Basics of Civil & Mechanical Engineering	3	2	5	4	30	70	25	25	150
	PY143	Engineering Physics- II	0	2	2	2			50	50	100
	EE145	Basics of Electronics & Electrical Engineering	3	2	5	4	30	70	25	25	150
	FS102A	Foundation Course on Chemistry and Biology	0	2	2	2			50	50	100
			14	11	25	21	120	280	205	245	850

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – I)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS101.02A: COMMUNICATIVE ENGLISH

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

B. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Communicative English	03
2.	Communication Functions	06
3.	Basic Communication Skills I – Listening	03
4.	Basic Communication Skills II – Reading	03
5.	Basic Communication Skills III – Speaking	06
6.	Basic Communication Skills IV – Writing	06
7.	Developing Vocabulary	03

Total hours (Theory): 00 Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

B. Detailed Syllabus:

- | | | |
|--|-----------------|------------|
| 1. Introduction to Communicative English | 03 Hours | 10% |
| English as a Window Language; Varieties of English: British English, American English, Indian English; Language Variations; Importance of English for Academic and Professional Development; Strategies for Language Acquisition; Formal VS Informal English | | |
| 2. Communication Functions | 6 Hours | 20% |
| Greeting and Introducing; Making Requests and Asking for Information; Expressing Likes and Dislikes; Seeking Permission; Giving and Taking Advice; Describing People, Place, Things; Retelling Past Events; Comparing and Contrasting; Persuading; Describing Cause and Effect | | |
| 3. Basic Communication Skills I – Listening | 3 Hours | 10% |

Importance of Listening as a Language Skill; Basic Listening Skills; Types of Listening; Barriers to Listening; Strategies for Effective Listening; Listening Practice

4. Basic Communication Skills II – Reading 3 Hours 10%

Importance of Reading as a Language Skill, Reading Strategies: Skimming, Scanning, Intensive Reading, Extensive Reading, Strategies for Effective Reading Comprehension, Reading Practice

5. Basic Communication Skills III – Speaking 6 Hours 20%

Importance of Speaking as a Language Skill, Basic Speaking Skills; Paralanguage for Effective Speaking; Strategies for Oral Communication; Extempore and Public Speaking

6. Basic Communication Skills IV – Writing 6 Hours 20%

Importance of Writing as a Language Skill; Process of writing: Prewriting, Drafting, Revision, Editing, Publication; Seven C's of Writing; Sentence Construction – Complex, Compound, Paragraph Development; Letter Writing (Academic Context)

7. Developing Vocabulary 3 Hours 10%

High Frequency Vocabulary (Everyday and Academic use); Words Often Confused and Misused; Useful Phrasal Verbs, Idioms and Proverbs, Homonyms and Homographs; Lexical Range, Word Games

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Communicate with people using English language functions including greetings, introductions, making and responding to requests, suggestions, invitations and apologies, conducting simple transactions in shops and offices, asking for and giving directions, etc
CO2	Go through a text and identify specific and global information.
CO3	Become more knowledgeable about speaking strategies and speak effectively using appropriate words, expression, tone and pronunciation.
CO4	Be aware about various reading strategies and read and comprehend academic and non-academic prose (text).
CO5	Write systematically using nuances of writing
CO6	Express their opinion and likes and dislikes, advice and convince others in a more polite and accepted way.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	3	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-

CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	2	3	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Text book:

1. Sanjay Kumar and PushpLata (Second Edition, 2015), Communication Skills, Oxford University Press, New Delhi
2. M V Rodriques (2013), Effective Business Communication, Concept Publishing Company (P) Ltd., New Delhi
3. Krishna Mohan and Meera Banerji (2010), Developing Communication Skills, Macmillan Publications India Ltd., New Delhi

❖ Reference book:

1. Mohan and Meenakshi Raman (2006), Effective English Communication, Mcgraw-Hill Publishing Company Limited, New Delhi
2. Geoffrey Leech & Jan Swartvik (1994), A Communicative Grammar of English, Longman Publications, New York
3. Jones Leo (1979), Functions of English, Cambridge University Press, UK
4. European Journal of Language and Literature Studies Vol.1 Nr, 1 April 2015
5. English for Academic Purpose: A Tool for Enhancing Students' Proficiency in English Language Skills

❖ Web material:

1. <https://www.futurelearn.com/courses/language-assessment>
2. <https://www.coursera.org/learn/importance-of-listening?#syllabus>
3. <https://www.futurelearn.com/courses/english-academic-study>

**FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES**

MA143: ENGINEERING MATHEMATICS – I

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	0	1	5	4
Marks	100	0	-	100	

B. Outline of the course:

Sr. No.	Title of the unit	Minimum number of Hours
1.	Higher order derivatives and applications	16
2.	Complex numbers and Roots of polynomial Equations	14
3.	Matrix Algebra –I	12
4.	Partial differentiations	08
5.	Applications of Partial differentiations	10
	Total Hrs	60

C. Detailed Syllabus:

		Hours
1	Higher order derivatives and applications:	16
1.1	Set theory and Function	
1.2	Limit, Continuity, Differentiability for function of single variable and its uses. Mean Value Theorem, Local Maxima and Minima	
1.3	Successive differentiation: n^{th} derivative of elementary functions: rational, logarithmic, trigonometric, exponential and hyperbolic etc.	
1.4	Leibnitz rule for the n^{th} order derivatives of product of two functions	
		Hours
1.5	Tests of convergence of series viz., comparison test, ratio test, root test, Leibnitz test. Power series expansion of a function: Maclaurin's and Taylor's series expansion (one and two variables),	
1.6	L'Hospital's rule and related applications, Indeterminate forms	

2	Complex numbers and Roots of polynomial Equations:	14
2.1	Complex numbers and their geometric representation	
2.2	Complex numbers in polar and exponential forms	
2.3	De Moivre's theorem and its applications	
2.4	Exponential, Logarithmic, Trigonometric and hyperbolic functions.	
2.5	Statement of fundamental theorem of Algebra, Analytical solution of cubic equation by Cardan's method	
2.6	Analytic solution of Biquadratic equations by Ferrari's method with their applications.	
3.	Matrix Algebra- I:	12
3.1	Definition of Matrix, types of matrices and their properties	
3.2	Determinant and their properties	
3.3	Rank and nullity of a matrix	
3.4	Determination of rank	
3.5	The inverse of a matrix by Gauss Jordan method.	
3.6	Solution of a system of linear equations by Gauss elimination and Gauss Jordan Methods.	
4.	Partial differentiations:	08
4.1	Partial derivative and geometrical interpretation	
4.2	Euler's theorem with corollaries and their applications	
4.3	Chain rule	
4.4	Implicit functions	
4.5	Total differentials	
5.	Applications of Partial differentiations:	10
5.1	Tangent plane and normal line to a surface	
5.2	Maxima and Minima	
5.3	Lagrange's method of multiplier	
5.4	Jacobian	
5.5	Errors and approximations	

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/tutorials which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.

- Quiz (surprise test) /Oral tests/ Viva/Assignment will be conducted which carries 10% component of the overall evaluation.

E. Course Learning Outcomes:

At the end of the course, the students will be able to

CO1	develop skill of successive differentiation, utilize appropriate theory and computational techniques to construct Taylor's series with its interval of convergence for using in a variety of applications such as approximating values, creating series representation and behaviour of a functions, use L'Hospital's rule to compute limits of the indeterminate forms.
CO2	Perform basic mathematical operations with complex numbers in Cartesian and polar forms; know methods of finding the n^{th} roots of a complex number and solutions of simple polynomial equations, work with functions of complex variable.
CO3	Find determinant and inverse of a square matrix, evaluate rank and nullity of a matrix, solve system of linear equations by using concept of matrices, which are useful in various fields of engineering.
CO4	Evaluate partial derivatives including higher order derivatives; solve problems using the chain rules, Euler's theorem with corollaries, implicit function and total differentials.
CO5	Expand any function of two variables in ascending power of variables; solve problems using the techniques of multivariable calculus in various branches of engineering.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	--	-	-	-	2	-
CO2	3	1	-	-	-	-	-	-	--	-	-	-	2	-
CO3	3	2	1	-	2	-	-	-	--	-	-	-	3	1
CO4	3	-	-	-	1	-	-	-	--	-	-	-	2	-
CO5	3	2	1	1	1	-	-	-	--	-	-	-	3	1

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. Recommended Study Material:

a. Text Books:

- Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
- H. K. Dass and Rajnish Verma, Higher Engineering Mathematics, S Chand & Co Pvt Ltd. 2012.

b. Reference Books:

- M. D. Weir et al: Thomas' Calculus, 11th Ed., Pearson Education, 2008.
- Stewart James: Calculus Early Transcendental, 5th Ed., Thomson India, 2007

- iii. Wylie, C. R., and L. C. Barrett. Advanced Engineering Mathematics. 1982., McGrawHill Book Company.
- iv. Greenberg, Michael D. Advanced Engineering Mathematics. Prentice-Hall, 1988.

c. URL Links:

- <https://ocw.mit.edu/ans7870/resources/Strang/Edited/Calculus/Calculus.pdf>
- <http://nptel.ac.in/courses/111107108/>
- <http://nptel.ac.in/courses/122101003/>
- <http://nptel.ac.in/courses/111104085/>

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL
ENGINEERING

ME146: ENGINEERING GRAPHICS

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	1	5	3
Marks	100	50	00	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of Engineering Graphics	05
2.	Projections of Points and Lines	03
3.	Projections of Planes	03
4.	Projections & Section of Solid	04
5.	Orthographic Projection*	05
6.	Isometric Projections*	04
7.	Computer Graphics*	03
8.	Development of Lateral Surfaces	03

* indicates the topics are discussed in laboratory

Total hours (Theory): 30

Total hours (Practical): 30

Total hours: 60

C. Detailed Syllabus:

1. **Fundamentals of Engineering Drawing 05 Hours 17%**
 - 1.1 Importance of engineering drawing, drawing instruments and materials, BIS and ISO
 - 1.2 Different types of lines used in engineering practice, methods of projections as per SP 46-1988.
 - 1.3 Engineering scale.
 - 1.4 Engineering curve.

2. Projections of Points and Lines	03 Hours	10%
2.1 Introduction to methods of projections		
2.2 Projections of lines inclined to both the planes		
3. Projections of Planes	03 Hours	10%
3.1 Projection of plane		
3.2 Auxiliary projection method		
4. Projections & Section of Solid	04 Hours	13%
4.1 Projection of solids		
4.2 Sectional view		
4.3 True shape of Sections		
4.4 Auxiliary Inclined Plane (AIP), Auxiliary Vertical Plane (AVP)		
5. Orthographic Projection	05 Hours	17%
5.1 Principle projection		
5.2 Methods of first and third angle projection with examples / problems		
6. Isometric Projections	04 Hours	13%
6.1 Terminology, Isometric scale		
6.2 Isometric view and Isometric projection with examples / problems		
7. Computer Graphics	03 Hours	10%
7.1 Introduce computer graphics		
7.2 Importance of computer graphics		
7.3 Demonstration of CAD Modeling software		
7.4 Training of Fusion 360 software		
8. Development of Lateral Surfaces	03 Hours	10%
8.1 Method of Development		
8.2 Developments of cylinder, cone, prism, pyramid		

D. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.

- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Drawing sheets/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory, discipline and behavior will be observed strictly.

E. Course Outcomes (COs):

At the end of the course, the students will be able to:

CO1	Realize and interpret various engineering drawings.
CO2	Understand the concept and application of engineering scale and engineering curve.
CO3	Visualize three-dimensional drawing of engineering components through orthographic, sectional orthographic and isometric drawing.
CO4	Explore the basic principle of projections and development of lateral surface of solid.
CO5	Use the computer for geometric modeling.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	1	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Text Books:

1. N. D. Bhatt & V. M. Panchal, “Engineering Drawing”, Charotar Publishing House Pvt. Ltd.
2. P. J. Shah, “Engineering Graphics”, S. Chand Publishing & Co.

Reference Books:

1. P.B. Patel & P.D. Patel, “Engineering Graphics”, Mahajan Publishing House.
2. Arunoday Kumar, “Engineering Graphics”, Tech-Max Publication.
3. Gopal Krishna K.L., “Engineering Drawing”, Subhas Publications
4. Venugopal, K., “Engineering Drawing made Easy”, Wiley Eastern Ltd.
5. M.L. Agrawal & R.K. Garg, “Engineering Drawing”, Vol. I, Dhanpatrai & Co.
6. T.E. French, C.J. Vierck & R. J. Foster, “Graphic Science and Design”, McGraw Hill.
7. W. J. Luzadder & J. M. Duff, “Fundamentals of Engineering Drawing”, Prentice Hall.
8. K. Venugopal, “Engineering Drawing and Graphics”, New Age international Pry. Ltd.

Reading Materials, web materials with full citations:

1. <http://nptel.ac.in/courses/112103019/>
2. <http://nptel.ac.in/downloads/112105125/>
3. <http://nptel.ac.in/syllabus/syllabus.php?subjectId=105104101>
4. <http://nptel.ac.in/courses/105107122/>
5. <https://law.resource.org/pub/in/bis/S01/is.sp.46.2003.pdf>

**FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICS**

PY142: ENGINEERING PHYSICS-I

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	50 (Internal) + 50 (External) =100	-	100	

B. Outline of the Course:

Unit No.	Title of the Unit	Minimum number of Hours
Unit 1	Mechanics Lab	16
Unit 2	Thermodynamics Lab	14
		Total Hours (Practical): 30

C. Detailed Syllabus:

1	Mechanics: Uncertainties in the measurements: Uncertainties in Measurements: Sources and estimation of errors, accuracy and precision, systematic error, random error, Significant figure and round off, error propagation. Laws of Motion: Frames of reference. Newton's Laws of motion, Dynamics of a system of particles, Centre of Mass, Projectile motion. Collisions: Elastic and inelastic collisions between particles. Momentum and Energy: Conservation of momentum, Work and energy, Conservation of energy. Motion of rockets. Rotational Motion: Angular velocity and angular momentum. Torque. Conservation of angular momentum. Elasticity: Hooke's law - Simple Stress and Strain: Introduction, Normal and Shear stresses, Stress- Strain Diagrams for ductile and brittle material, Elastic Constants	16 hours
	• Understanding Errors and Uncertainties in the measurements • Conservation of Energy, Hook's law • Young modulus and elasticity • To determine g by Bar Pendulum and Kater's Pendulum.	
2	Thermodynamics: Zeroth Law of thermodynamics and temperature. First law, second, third law and internal energy, conversion of heat into	14 hours

	work, Various Thermo dynamical Processes, Enthalpy, Gibbs, Helmholtz and Internal Energy functions, Transport Phenomena: Viscosity, Conduction and Diffusion, applications to specific heat of gases and metals, Blackbody radiation, Spectral distribution, Derivation of Planck's law	
	• Specific Heat of Metals • Thermal conductivity of materials by Searl's apparatus • Heat Transfer and Newton's Law of Cooling • Radiation from a black body: Stefan-Boltzmann Law	

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Section wise Online Quiz will be taken.
- Lab manual: Student will be required to read the lab material prior to the start of class.
A way to ensure this is by lab quizzes and assignments.
- Lab Reports: Student has to write lab reports and submit them hardcopy/electronically. The purpose of this exercise is both to demonstrate your work in lab and to guide you to think a bit more deeply about what you are doing. The act of technical writing also helps improve your communication skills, which are broadly relevant far beyond the physics lab

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Students will be able to apply and demonstrate the concepts of mechanics to practical engineering problems.
CO2	Understand the basic concepts of thermodynamics such as internal energy, thermodynamic properties, transport phenomena, blackbody radiation.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	1	1	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

❖ Text Books:

1. Physics for Scientists and Engineers by Randall D. Knight, 4th Edition, Pearson
2. University Physics by Hugh D. Young, Roger A. Freedman and A. Lewis Ford, 13th Edition, Pearson
3. Physics by John D. Cutnell & Kenneth W. Johnson, 8th Edition, John Wiley & Sons, Inc.

❖ Reference Books:

1. An introduction to mechanics, D. Kleppner, R.J. Kolenkow, 1973, McGraw-Hill.
2. University Physics, Ronald Lane Reese, 2003, Thomson Brooks/Cole.
3. Physics, Resnick, Halliday and Walker 8/e. 2008, Wiley.
4. Physics for scientists and Engineers with Modern Phys., J.W. Jewett, R.A. Serway, 2010, Cengage Learning
5. Heat and Thermodynamics, M.W. Zemansky, Richard Dittman, 1981, McGraw-Hill.
6. Concepts in Thermal Physics, S.J. Blundell and K.M. Blundell, 2nd Ed., 2012, Oxford University Press

❖ Web Materials:

1. Uncertainty and error in measurement - <https://www.youtube.com/watch?v=pWEflsClyTk>
2. Vernier Callipers principle and description (Introduction) - <https://www.youtube.com/watch?v=ySRN3yuZUT0>
3. Hooke's law - <https://www.youtube.com/watch?v=PWQm4ynYVSE>
4. Law Of Conservation Of energy in a Simple Pendulum - https://www.youtube.com/watch?v=51RCyBr_nGk,
<https://www.youtube.com/watch?v=QlViWmQcwpQ>
5. Young's modulus of the material of a beam by method of bending of beam - <https://www.youtube.com/watch?v=iUhfstf10rk>
6. Bar pendulum - https://www.youtube.com/watch?v=3uZ_Boyt_AI
7. Kater's Pendulum - <https://www.youtube.com/watch?v=TxbDyv17Jfs>
8. Specific Heat of Metals - <https://www.youtube.com/watch?v=8gHFAL2990U>

9. Thermal conductivity of materials by Searl's apparatus -

<https://www.youtube.com/watch?v=qKhhcrqhPfY>

10. Newton's Law of Cooling - <https://www.youtube.com/watch?v=IC9o6ikJlR8>

11. Radiation from a black body: Stefan-Boltzmann Law -

<https://www.youtube.com/watch?v=riRsMfNmicM>

FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI SHIVABHAI PATEL DEPARTMENT OF CIVIL ENGINEERING

CL144.01A: ENVIRONMENTAL SCIENCES

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit	Evaluation Scheme				
					Theory		Practical		Total
Hours/week	0	2	2	2	Internal	External	Internal	External	
Marks	0	100	100		--	--	30	70	100

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Multidisciplinary nature of environmental	03
2	Environmental Pollution	07
3	Ecology & Ecosystems	05
4	Biodiversity and Conservation	04
5	Natural Resources	04
6	Human Communities and the Environment	07

Total Hours (Theory): 00

Total Hours (Lab): 30

Total Hours: 30

C. Detailed Syllabus:

1.	Multidisciplinary nature of environmental studies	03 hrs.	11%
	a) Definition, scope and importance		
	b) Earths-Evolution and Systems		
	c) Components of the Environment: Biotic, Abiotic, Atmosphere, Lithosphere, Hydrosphere, Biosphere		
	d) Concept of sustainable development		

2.	Environmental Pollution a) Environmental pollution: types, causes, effects and controls; Air, water, soil and noise pollution b) Nuclear hazards and human health risks c) Solid waste management: Control measures of urban and industrial waste d) Pollution case studies – Ganga/Yamuna River, Bhopal Gas Tragedy, Delhi Air Pollution, Effect of Pandemics on the Environment	07 hrs.	23%
3.	Ecology & Ecosystems a) Concept of an ecosystem, Structure and function of an ecosystem b) Producers, consumers and decomposers c) Energy flow in the ecosystem, Food chains, food webs and ecological pyramids d) Case studies of the following ecosystems: Forest ecosystem, Grassland ecosystem Aquatic ecosystems (ponds, streams, lakes, rivers)	05 hrs.	17%
4.	Biodiversity and Conservation a) Introduction – Definition: genetic, species and ecosystem diversity b) Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values c) Hot-spots of biodiversity, Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. d) Endangered and endemic species of India, Conservation of biodiversity e) Biodiversity Act 2002/ BD Rule 2004: Mandate & Functions of National Biodiversity Authority (NBA), Role of State Biodiversity Board (SBB) and Biodiversity Management Committees(BMC).	04 hrs.	13%
5.	Natural Resources a) Renewable and non-renewable resources b) Recyclable and Non-recyclable resources c) Energy resources: Growing energy needs, use of alternate energy sources. Case studies. d) Role of an individual in conservation of natural resources	04 hrs.	13%
6.	Human Communities and the Environment a) Pandemics: Causes, Effects, Impact on the Environment: Positive & Negative, Lessons to learn b) Floods, Cyclones, Earthquakes, Landslides & Forest Fires	07 hrs.	23%

- | | | |
|--|--|--|
| c) Human population growth: Impacts on environment, human health and welfare. | | |
| d) Case Studies: Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents, etc. | | |

D. **Instructional Method and Pedagogy:**

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

E. **EVALUATION:**

There will be end semester university examinations based on Practical/ Viva. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students and distribute the marks (out of 30 as Internal and 40 out of 70 as External). Rest 30 marks (30 out of 70 of external) will be given based on the performance in the Practical/Viva) examination.

Evaluation Scheme

The students' performance in the course will be evaluated through the following components:

Component		Marks
Internal	Attendance	05
	Assignment/ Individual Activity Participation/ Quiz, etc.	25
External	Group Activity Participation, Project Report, Case Study with Report, Online Certification Course, MOOCS, Field Work with report, Presentation etc.	40
	Viva/ Oral Examination	30
Total		100

F. **Course Outcomes (COs):**

On the successful completion of the course, the students will be able

CO1	To perceive the elementary knowledge about natural environment and its relation with science.
CO2	To identify and analyze human impacts on the environment.
CO3	To understand the facts and concepts of natural and energy resources thereby applying them to lessen the environmental degradation.
CO4	To Initiate new and innovative environmental friendly practices.
CO5	To communicate on recent environmental problems thereby creating awareness among society.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	3
CO2	-	-	-	-	-	-	3	-	-	-	-	-	-	3
CO3	-	-	-	-	-	2	3	-	-	-	-	-	-	3
CO4	-	-	2	-	-	-	3	-	-	-	-	-	2	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Recommended Study Material:

Text Books:

1. Varandani, N.S., Basics of Environmental Studies
2. Sharma, J. P., Basics of Environmental Studies

Reference Books:

1. Shah Shefali & Goyal Rupali, Basics of Environmental Studies
2. Agrawal, K.C., Environmental Pollution : Causes, Effects & Control
3. Dameja, S. K., Environmental Engineering & Management
4. Rajagopalan, R., Environmental Studies, Oxford University Press
5. Wright Richard T. & Nebel Bernard J., Environmental Science

6. Shah, S.G., Shah, S.G. & Shah, G. N., Basics of Environmental Studies, Superior Publications, Vadodara

Web Materials:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITDelhi/Environmental%20Air%20Pollution/index.htm>
- <http://nptel.iitm.ac.in/video.php?subjectId=105104099>
- http://apollo.lsc.vsc.edu/classes/met130/notes/chapter1/vert_temp_all.html
- <http://www.epa.gov>
- <http://www.globalwarming.org.in>
- <http://nopr.niscair.res.in>
- <http://www.indiaenvironmentportal.org.in>

**FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI SHIVABHAI PATEL DEPARTMENT OF CIVIL
ENGINEERING**

CL143 ENGINEERING MECHANICS

A. Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	1	2	6	4
Marks	100	-	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	04
2	Fundamental of Statics	20
3	Friction	05
4	Truss	05
5	Centroid and Centre of Gravity	06
6	Fundamentals of Kinematics and Kinetics of Particles	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

C. Detailed Syllabus:

- 1 Introduction** **04 hrs 08%**
Introduction of Mechanics, Fundamental concepts and idealization of mechanics, Fundamental principles & Laws of mechanics, Scalar and Vector Quantities,
Components, unit vectors and position vector, Composition and resolution of vector, System of Units

- | | | |
|------------|--|-----------------------|
| 2 | Fundamental of Statics | 20 Hrs 45% |
| 2.1 | Coplanar Concurrent Force system | |
| | Introduction of Force, Effect of force and Characteristics of force, Types of force, Type of force systems, Principle of Transmissibility, Resultant of force systems, Resolution of a single force, Composition and Resolution of force system, Resolution method for coplanar concurrent force system | |
| 2.2 | Moments and Couples | |
| | Moment of a force, Principle of moments, Coplanar applications, Parallel force system, Couples, Equivalent couples, Operations with couples, Equivalent system of forces | |
| 2.3 | Coplanar Non-Concurrent Force system | |
| | Introduction, Resultant of coplanar non-concurrent force system, Concentrated and distributed loads | |
| 2.4 | Equilibrium of Rigid bodies | |
| | Equilibrium, Resultant & Equilibrant, Principle of action and reaction, Free body diagram & Lami's theorem, Tensions of strings, condition of equilibrium for Coplanar concurrent forces & Coplanar non-concurrent forces, Equilibrium of Coplanar concurrent forces.
Equilibrium of Coplanar non-concurrent forces | |
| 3 | Friction | 5 Hours 11% |
| | Friction and its applications, Types of friction and Laws of dry friction, Angle of friction, Angle of repose, Coefficient of friction, Block Friction, Ladder friction, Wedge friction | |
| 4 | Truss | 5 Hours 11% |
| | Classification of Truss, Perfect & Imperfect truss, Analysis of pin jointed perfect truss using method of joints and Method of section. | |
| 5 | Centroid and Centre of Gravity | 6 Hours 14% |
| | Introduction, basic definitions and their understanding, Concept of centre of gravity, Centroids of Linear elements & Planar elements, Determination of centroids by integrations, Centroids of Composite sections (1D & 2D). | |

6 Fundamentals of Kinematics and Kinetics of Particles 5 Hours
11%

Rectilinear motion, Curvilinear motion, Motion of rigid bodies, Velocity and acceleration, Newton's law of motion, Energy and momentum.

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Two internal exams will be conducted as a part of continuous theory evaluation.
- Assignments/ Tutorial based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar/ presentation/ model making will be carried out as a part of continuous theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments related to course content will be carried out in the laboratory.

E. Course Outcomes (COs):

On the completion of the course one should be able to:

CO1	Understand laws of mechanics and their application to engineering problems.
CO2	Understand the fundamentals of statics and apply them to simple structural problems
CO3	Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple practical problems.
CO4	Compute centroid of composite sections.
CO5	Recognize different types of friction and its effects on an object.
CO6	Classify different types of truss and analyze the pin jointed perfect truss.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	1	-	-	-	-	-	-	-	1	-
CO3	1	-	-	-	-	-	-	-	-	-	-	-	1	-
CO4	2	1	1	-	-	-	-	-	-	-	-	-	1	-
CO5	1	1	-	-	-	-	-	-	-	-	-	-	1	-
CO6	1	1	2	-	1	-	-	-	-	-	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

- Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I & II, Charotar Publishing House
- Junnarkar, S. B. & Shah, H. J., Applied Mechanics, Charotar Publishing House
- Beer and Johnston, Engineering Mechanics (Statics & Dynamics)

Reference Books:

- Beer and Johnston, Mechanics of Materials
- Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
- Hibbler, R.C., Engineering Mechanics, Pearson Education
- Popov, E.P., Engineering Mechanics of Solids, Prentice Hall of India, New Delhi
- Meriam, J. L. & Kraige, L. G., Engineering Mechanics Statics, John Wiley & Son, Singapore
- A K Tayal, Engineering Mechanics (Statics & Dynamics), Umesh Publications

Web Materials:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITDelhi/Mechanics%20Of%20Solids/index.htm>
- <http://nptel.iitm.ac.in/video.php?subjectId=105106116>

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL
ENGINEERING

ME142: WORKSHOP PRACTICES

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	1
Marks	-	50	-	50	

B. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to workshop facility.	02
2.	Carpentry shop	06
3.	Fitting shop.	08
4.	Different Metal Joining Processes.	04
5.	Smithy shop.	02
6.	Sheet metal working.	02
7.	Plumbing shop.	02
8.	Introduction to machine tools.	02
9.	Injection molding process.	02

Total hours (Lab): 30

Total hours: 30

C. Detailed Syllabus:

- | | | |
|--|-----------------|------------|
| 1. Introduction to workshop facility. | 02 hours | 07% |
| 1.1 Familiarization with workshop facility. | | |
| 1.2 Introduction to different shops of the workshop. | | |
| 2. Carpentry Shop. | 06 hours | 20% |
| 2.1 Introduction to different tools of carpentry shop. | | |

2.2 Making of drawing of the job to be made.

2.3 Making of finished job as per drawing out of the given raw material of wood.

Identification on the job for traceability.

3. Fitting Shop. **08 hours** **27%**

3.1 Introduction to different tools of fitting shop.

3.2 Making of drawing of the job to be made.

3.3 Making of finished job as per drawing out of the given raw material. Identification on the job for traceability.

4. Different Metal Joining Processes. **04 hours** **14%**

4.1 Introduction to different tools of welding shop.

4.2 Making of drawing of the job to be made.

4.3 Making or demonstration of finished job as per drawing.

4.4 Introduction to Soldering and brazing of metal joining process.

4.5 Joining of two metal sheet or plate by Riveting.

4.6 Making of drawing of the job to be made by riveting.

4.7 Making or demonstration of finished job as per drawing.

5. Smithy Shop. **02 hours** **07%**

5.1 Introduction to different tools of smithy shop.

5.2 Making of drawing of the job to be made for Cold smithy.

5.3 Making or demonstration of finished job as per drawing.

5.4 Making of drawing of the job to be made for Hot smithy.

5.5 Making or demonstration of finished job as per drawing.

6. Sheet Metal Working. **02 hours** **06%**

6.1 Introduction to different tools of sheet metal working shop.

6.2 Making of drawing of the job to be made from sheet metal.

6.3 Making or demonstration of finished job as per drawing.

7. Plumbing Shop. **02 hours** **06%**

7.1 Introduction to all plumbing tools.

7.2 Demonstration of plumbing on the piping model.

8. Introduction to Machine Tools. **02 hours** **07%**

8.1 Detailed introduction to Lathe machine, Shaping machine, Drilling machine, Grinding machine, Milling machine, Bending machine, Mechanical press.

9. Injection molding process.

02 hours 06%

9.1 Introduction and demonstration to Injection Molding Process for making job out of plastic material.

D. Instructional Method and Pedagogy:

- Attendance is compulsory in laboratory session.
- Journal writing based on above course content and practical work in form of performance practical's by preparing job at the workshop floor.
- In the laboratory discipline and behavior will be observed strictly.
- All the students must follow code of conduct during working at the shop floor.
- Journal should be submitted to the respective course teacher within the given time limit.

E. Course Outcome (COs):

On the completion of the course one should be able to:

1. Recognize essential tools and process of carpentry.
2. Understand the joining process like welding, soldering and brazing.
3. Recognize essential tools and process of fitting.
4. Recognize various forging and forming processes with the aid of smithy process.
5. Understand different types of sheet metal joints which are useful in working.
6. Recognize different types of fittings and plumbing tools.
7. Identify different machines in workshops along with use.
8. Recognize plastic molding processes.

Course Articulation Matrix:

	P O1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	1	-	-	-	-	-	-	-	-	-	-	-	-	-

CO6	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO7	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO8	1	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

1. Anderson, James, and Earl E. Tatro. "Shop Theory. 5th ed." (1968).
2. Bawa, H. S. Workshop Technology. India: Tata McGraw-Hill, (1995).
3. Choudhury, Hajra. "Elements of Workshop Technology, Vol. I & II." Media Promoters Pvt Ltd (2009).
4. Raghuwanshi, B. S. Course in Workshop Technology. Dhanpat Rai and Company (P) Limited, (2009).

Reference Books:

1. Chapman, W. A. J. "Workshop Technology Part 1-3." (1998).
2. Tejwani V.K., “Basic Machine Shop Practical Vol. I, II”, Tata McGraw Hill Pub. Co., New Delhi, (1989).
3. Arora B.D., “Workshop Technology Vol. I, II”, Satya Prakashan, New Delhi, (1981).

Other materials:

Workshop book and manual.

Web sites: <http://nptel.ac.in/courses/112107145/>

FSI01A: FOUNDATION COURSE ON MATHEMATICS AND PHYSICS
1st Semester and 1st Year (Level I)
B. Tech. (All Branches)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Scientific phenomena related to Mathematics and discussion, presentation, and its explanation	15
2	Scientific phenomena related to Physics and discussion, presentation, and its explanation	15

Total hours (Theory) : -- Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 30 Hrs

C. Detailed Syllabus:

Scientific Problems in Mathematics

Sr. No	
1	Applications of Mathematics in nature through Fibonacci sequence.
2	Applications of Mathematics in science (Geometry, Graphs, Heat and Light, Solar System and Planets, Solar System and Stars, Space Science, Sun).
3	Applications of Mathematics in growth or decay model of bacteria/animals.
4	Applications of Mathematics in cryptography.
5	Applications of Mathematics in data science (data security, AI, encryptions of code etc.).
6	Applications of calculus in day to day life.
7	Applications of geometry in real life (Birds flying patterns, geometry of Bridges, plantations, Surveying etc.).
8	Applications of Graph theory in real life (Labelling Problems, colors problems, Shortest path, tree etc.).
9	Applications of Mathematics in Bioscience (COVID-19 model, infectious disease dynamics, transmissions models etc.)
10	Applications of Graph theory in networking (social networks, biological networks, internet, neuroscience etc.
11	Applications of differential equations in engineering (Electric Circuits, Problems of Vibrations, Beam-Columns Problems etc).

12	Applications of mathematics in road Traffic control problems.
13	Applications of mathematics in Weather forecasting.
14	Application of Matrices in Real life and computer network.
15	Application of Mathematics in Business.
16	Real life statistical problems e.g. counting of birds or fishes in the pond.
17	Measuring of area of an irregular figures using mathematical concepts.

Scientific Problems in Physics

Sr. No.	
1	Basic Mechanics of Rocket Launching.
2	Quantum Computation.
3	Average heights of Human in reaction of G.
4	Planetary motion.
5	Solar cells—a window to the future?
6	Big discoveries about tiny particles
7	Know about Earth's marvellous magnetism
8	How big is nano
9	Why Sky is blue? – Raman effect.
10	Communication & Solar flare.
11	Geo Satellite & GPS System
12	Tsunami and wave
13	Motion of electrons (Electricity) at our Home!
14	Laws of attraction and rejection (Gravitational, electric, magnetic, nuclear etc force)
15	Pressure of life in deep sea.
16	Why the violin sings
17	why winds never blow in a straight line
18	The challenge and promise of studying burning plasmas
19	How atom emit lights:
20	A vacuum is not exactly nothing

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Acquire the fundamental knowledge of Mathematics and Physics.
CO2:	Able to apply foundational knowledge to solve daily life problems.
CO3:	Solve basic phenomena of sciences by a scientific approach.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	2	-	-	-	1	-	-	1	-	-	1	-	-
CO3	2	2	2	-	-	-	-	-	1	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Labs will be conducted with the aid of multi-media projector, black board, OHP, models etc.
- Attendance is compulsory in labs which carries a 10% component of the overall evaluation.
- Course content presentation should be based on experimental demonstration which can induce scientific inquiry among students.
- Identify the audio-visual resources/materials related to the topic which could be played during lab session to improve understanding of theory.
- Problem solving activities should be done in lab along with specific practical exercises to inculcate prediction on final result/outcome.
- Critical thinking activities should be incorporated after course content presentation so that students can develop ability to reach generalization and apply them to solve everyday problems.
- Include a range of learning experiences, such as investigations, practical activities, field trials, field trips, speakers, and group and individual activities. On field activities and excursion(s) must be done.
- After the presentation of course content, active involvement of student must be solicited in form of either pair/group work or presentation to motivate and engage the students. Furthermore, these activities will promote higher level skills and attributes such as communication, teamwork and perseverance in the students.
- Adaptive course contents based on recent events with its background, current trends should be added in course.
- More real life exposure should be included.
- More comprehensive practical must be included to fulfil the requirement of learning while doing.
- Use of models and modelling techniques in teaching to strengthening the understanding.
- Practical should be conducted in such a way that the development of process skills like observation, classification, measurement etc. will become strong in the students.

F. Recommended Study Material:

Text Book:

Reference Book:

1. The Science Book, DK Publishing
2. How It Works Book 2: Amazing Physics Imagine Publishing
3. How Science Works: The Facts Visually Explained. DK Publishing
4. The Physics Book: Big Ideas Simply Explained. DK Publishing
5. The Math Book: Big Ideas Simply Explained. DK Publishing

Web Material:

<https://www.mathsisfun.com/>

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – II)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS201.02A : PAINTING

I. Credits and Schemes

Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				Total
			Contact Hours/Week	Theory		Practical		
				Internal	External	Internal	External	
HS201.02A	Painting	02	02	--	--	30	70	100

II. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Painting	02
2	Drawing from Nature and Object	04
3	Colour Design and Colour Value	06
4	Composition and Perspective	06
5	Figure Drawing and Proportion	04
6	Sketching	04
7	Contemporary Issues in Painting	04

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

III. Detailed Syllabus:

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Painting <i>An Introduction to Painting</i> <i>Principles of Composition</i> <i>Medium and Techniques of Painting</i> <i>History of Painting: Folk Indian Painting / Western Painting</i> <i>2D and 3D Painting</i>	2

2	Drawing from Nature and Object • <i>Objects of Drawing: Nature and Manmade / Artificial Objects</i> • <i>Drawing Still / Live Objects</i> • <i>Drawing from Memory</i> • <i>Drawing from Life</i>	4
3	Colour Design and Colour Value • Color Theory: <i>Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)</i> • Color Contrast & Attributes: <i>Interaction, harmony, psychology/mood, culture & expression</i> • Media Characteristics & Surfaces: <i>Acrylic, oil, paper, wood & canvas (primed/unprimed)</i>	6
4	Composition and Perspective • Composition: <i>Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting</i> • Perspective: <i>An approximate reproduction</i> • Types of Perspectives: <i>Linear Perspective, One-point Perspective, Twopoint Perspective, Three-point Perspective, Four-Point Perspective</i>	6
5	Figure Drawing and Proportion • <i>Proportions of the Human Body</i> • <i>Three views – Anterior (front), Lateral (side) and Posterior (back)</i> • <i>Fundamental Proportion – The Big Three</i>	4
6	Sketching • <i>Sketching and Freehand</i> • <i>Sketching Techniques</i> • <i>Sketch and Drawing Medium</i>	4
7	Contemporary Issues in Painting • <i>Contemporary Indian Art</i> • <i>Pioneers of Contemporary Indian Art</i> • <i>Contemporary Issues in Painting</i>	4
Total Hours		30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

VI. Course Outcome (COs):

At the end of the course, a student will

After completion of the course, the student would:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art history, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in painting.
CO4	have better cognizance and association of meaning of colors, shapes, and composition.
CO5	be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
CO6	have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	3	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS202.02A : PHOTOGRAPHY

I. Credits and Schemes

Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
			Contact Hours/Week	Theory		Practical		Total
				Internal	External	Internal	External	
HS202.02A	Photography	02	02	--	--	30	70	100

II. Course Outline

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Photography	03
2	Camera and Operating System	05
3	Light and Shade	10
4	Composition	09
5	Contemporary Issues in Photography	03

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

III. Detailed Syllabus:

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Photography • <i>Art, Design and Visualization</i> • <i>Basics of Photography and Various Types of Photography</i> • <i>Basics of Post Production</i> • <i>A Brief History of Photography: Early Experiments and Later Developments</i>	03

2	Camera and Operating System • <i>Role of Camera in the Photography</i> • <i>Types of Camera</i> <i>Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, sub miniature and instant camera</i> • <i>Principal Parts of Photographic Camera</i> <i>Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer</i> • <i>Types of Lenses</i> <i>Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses</i> • <i>Different Models of Camera, their Features and Operating Systems</i> • <i>Camera and Size of the Image, Speed and Power of Lens</i>	05
3	Light and Shade • <i>Reflection and refraction of light</i> • <i>Dispersion of light through a glass prism, lenses</i> • <i>Colour Filters:</i> <i>Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter</i> • <i>Photographic Light Sources:</i> <i>Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions</i> • <i>Artificial light sources:</i> <i>Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands</i> • <i>Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table</i>	10
4	Composition • <i>Different kinds of image formations</i> • <i>Principal focus and focal length of the lens</i> • <i>Depth of field, angle of view and perspective</i> • <i>Perspective and composition</i> • <i>Rules of composition</i>	09
5	Contemporary Issues in Photography • <i>Present Day Photography</i> • <i>Contemporary Photographers and their Contributions</i> • <i>Major Issues in Contemporary Photography</i>	03
Total Hours		30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

VI. Course Outcomes (COs):

After completion of the course, the student would:

CO1	Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
CO2	Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects
CO3	Become independent thinkers who will contribute inventively and critically to culture through the making of art photography
CO4	Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art
CO5	Experiment and Represent the cultivated sense and skills in Photography to the mass
CO6	Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
--	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------

CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	2	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	3	-	-
CO6	-	-	-	-	-	-	-	-	-	2	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS205.02A : MEDIA AND GRAPHIC DESIGN

I. Credits and Schemes

Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Theory		Practical		Total
			Contact Hours/Week	Internal	External	Internal	External	
HS205.02A	Media and Graphic Design	02	02	--	--	30	70	100

II. Course Outline

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Media and Graphic Design	03
2	Layout and Design	07
3	Form and Space	06
4	Computer Graphics	04
5	Fonts	04
6	Basic Print Media	03
7	Contemporary Issues in Graphic Design	03

Total hours (Theory): - Hrs.
Practical Hours: 30 Hrs.
Total hours: 30 Hrs.

III. Detailed Syllabus:

1.	An Introduction to Media and Graphic Design	03 Hours	10%
	• <i>Creating Art, Art in Context and Art as Inquiry</i> • <i>History of Graphic Design</i> • <i>Constructional, Representational, and Simplification Drawing</i>		
2.	Layout and Design	07 Hours	23%
	• <i>Layout, Design and Aesthetics</i> • <i>Elements of Design</i> • <i>Principles of Design:</i> <i>Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition</i>		

	<i>and Proportion</i> • <i>Impact/function of Design</i> • <i>Indigenous design practices</i> • <i>Role of design in the changing social scenario</i>		
3.	Form and Space	06 Hours	20%
	• <i>Types of Forms: Man-made, Nature</i> • <i>Types of Space: Negative and Positive</i> • <i>Composition of Form and Space to create Layout</i> • <i>Exploring Creativity</i>		
4.	Computer Graphics	04 Hours	12%
	• <i>An Introduction to Graphic Software</i> • <i>Flash, Coreldraw, Illustrator and Photoshop</i> • <i>Pre-press Process</i>		
5.	Fonts	04 Hours	12%
	• <i>Construction of Type</i> • <i>Anatomy of Type</i> • <i>Visual Language</i> • <i>Creating Logo and Symbol</i>		
6.	Basic Print Media	03 Hours	10%
	• <i>An Introduction to Press and its Development Phases</i> • <i>Types of Press</i> • <i>Types of Printing Technologies</i> • <i>Post-press Processes</i>		
7.	Contemporary Issues in Graphic Design	03 Hours	10%
	• <i>Present Day Graphic Designs</i> • <i>Contemporary Designers and their Contribution</i> • <i>Major Contemporary Issues in Graphic Design</i>		

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
	Total			30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
	Total			70

VI. Course Outcomes (COs):

After completion of the course, the student would:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art and designs, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in graphic design.
CO4	have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
CO5	be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	2	-	-
CO5	-	-	-	-	3	-	-	-	-	1	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), No correlation -

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS209.02A || DRAMATICS

I. Credits and Schemes

Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
			Contact Hours / Week	Theory		Practical		Total
				Internal	External	Internal	External	
HS209.02A	Dramatics	02	02	--	--	30	70	100

II. Course Outline

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Drama	06
2	History of Drama and Contemporary Theatre	06
3	Theatre Design and Techniques	06
4	Technicalities of Stage Performance	08
5	Contemporary Trends in Drama	04

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

III. Detailed Syllabus:

Module	TITLE / TOPIC	Classroom Contact Sessions
1	Introduction to Drama • Introduction to performing arts • Drama - An art, a socializing activity, & a way of learning • Form of Drama • Elements of Drama • Types of Drama	06

2	History of Drama and Contemporary Theatre • Important world dramatists & drama—from Greek to modern • Evolution of contemporary theatre in the context of developments in Indian theatre • Major Movements in Drama	06
3	Theatre Design and Techniques • Theatre Architecture • Stage craft: Set, light, costume, make up, sound, props • Theatre techniques: from selection of script to final performance	06
4	Technicalities of Stage Performance • Selection of plot and character • Improvisation • Movement • Voice, Speech, Imagination • Character Development • Scene Enactment	08
5	Contemporary Trends in Drama • New Tendencies in theatre • Drama and Society • Using drama for Social Change and Education	04

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation:

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

VI. Course Outcome (COs):

After completion of the course, the student would:

CO1	be aware about the concept of performing art and its nuances.
CO2	display a working knowledge of historic of drama, its development and current trends in dramatics.
CO3	demonstrate skills in the technical/design preparation and execution of a theatre performance.
CO4	demonstrate the ability to work collaboratively.
CO5	develop essential transferable skills in various relevant areas of the theatre.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	2	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS210.02A || CONTEMPORARY DANCE

I. Credits and Schemes

Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
			Contact Hours/Week	Theory		Practical		Total
				Internal	External	Internal	External	
HS210.02A	Contemporary Dance	02	02	--	--	30	70	100

II. Course Outline

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to dance	04
2	Types of Dance	06
3	Basic Elements of Dance	04
4	Technical Skills in Professional Contemporary Dance	06
5	Contemporary Trends in Dance	10

Total hours (Theory): - Hrs.

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

III. Detailed Syllabus:

1.	Introduction to dance	04 Hours	14%
	- Dance as a Performing Art - Dance as a Medium of Expression - History and Development of Dance		
2.	Types of Dance	06 Hours	20%
	- Western dance and classical dance - Salsa, rumba, hip hop, tap dance, belly dance, etc. - Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc. - Other Regional dance forms in India		
3.	Basic Elements of Dance	04 Hours	14%

	• Movements of different parts of a body for Expression • Concepts of: Nritya, Laya and Taal		
4.	Technical Skills in Professional Contemporary Dance	06 Hours	20%
	• Dance technique: alignment, balance, co-ordination, flexibility and control • Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc. • Skills and processes of rehearsal and production: • physical energy, stamina and athleticism		
5.	Contemporary Trends in Dance	10 Hours	32%
	• Prevalent trends and techniques in contemporary dance • Future trends in contemporary dance form • On Stage Performance		

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sr. No	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

VI. Course Outcome (COs):

After completion of the course, the student would:

CO1	be able to develop ability to express through the form of dance.
CO2	have enhanced aesthetic sensitivity.
CO3	have improved concentration, mental alertness, quick reflex action, and physical agility.
CO4	be able to express a natural way human feelings and expressions by creating harmony.
CO5	be able to deliver contemporary dance performance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	3	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
MA144: ENGINEERING MATHEMATICS –II

A. Credits and Hours:

Teaching Scheme	Theory	Tutorial	Total	Credit
Hours/week	4	1	5	4
Marks	100	-	100	

B. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	First order and First degree Ordinary Differential Equations	08
2	Higher Order Ordinary Linear Differential Equations	12
3	Partial Differential Equations and Applications	10
4	Matrix Algebra –II	10
5	Improper and Multiple Integrals	10
6	Probability and Statistics	10
	Total hours	60

D. Detailed Syllabus:

		Hours
1.	First order and First degree Ordinary Differential Equations:	08
1.1	Modeling of real world problems in terms of first order ODE	
1.2	Concept of general and particular solutions	
1.3	Initial and boundary value problems	
1.4	Existence and Uniqueness of solutions by illustrations	
1.5	Solutions of first order and first degree differential equations	
1.6	Linear, Bernoulli, Exact and non-exact differential equations	
2.	Higher Order Ordinary Linear Differential Equations:	12
2.1	Model of real world problems of higher order LDE	

2.2	General Solution of Higher Order Ordinary Linear Differential Equations with Constant coefficients	
2.3	Methods for finding particular integrals viz. variation of parameters and undetermined coefficients	
2.4	LDE of higher order with variable coefficients: Legendre's Equations (Special case: Cauchy-Euler equation)	
2.5	System of simultaneous first order linear differential equations	
3.	Partial Differential Equations and Applications:	10
3.1	Boundary valued conditions	
3.2	Methods of solutions of first order PDE	
3.3	Lagrange's Linear Partial Differential Equations.	
3.4	Special types of Nonlinear PDE of the first order	
3.5	Solutions of Heat, Wave and Laplace equations using separation of variables.	
3.6	Modeling of real world problem in terms of PDE	
4.	Matrix Algebra –II:	10
4.1	Revision of matrices, determinant.	
4.2	Eigen values and Eigen vectors of matrices	
4.3	Eigen values and Eigen vectors of special matrices	
4.4	Cayley-Hamilton's Theorem and its applications.	
4.5	LU decomposition	
5.	Improper and Multiple Integrals:	10
5.1	Improper integrals and their convergence	
5.2	Definitions, properties and examples of Beta, Gamma and error functions	
5.3	Evaluation of double and triple integrals	
5.4	Change of order of double integration	
5.5	Transformation to polar and cylindrical coordinates	
5.6	Applications of double and triple integrals	
6.	Probability and Statistics:	10
6.1	Mean, median, mode and standard deviation	
6.2	Combinatorial probability	
6.3	Joint and Conditional probability and Bayes theorem	

6.4	Random variables, probability distribution functions - Binomial, Poisson, exponential and normal.	
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E. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/tutorials which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Quiz (surprise test) /Oral tests/ Viva/Assignment will be conducted which carries 10% component of the overall evaluation.

F. Student Learning Outcomes (Course Outcome)

At the end of the course, the students will be able to

CO1	Formulate models of natural phenomena using differential equations and find its solution using standard methods.
CO2	Identify, analyze and subsequently solve physical problems analytically whose behaviour can be described by linear and nonlinear differential equations.
CO3	Find and explain significant of Eigenvalues and Eigenvectors of a square matrix, use Cayley-Hamilton's theorem to find inverse and power of a square matrix, construct LU decomposition of a square matrix.
CO4	Use advanced techniques to evaluate improper integrals, apply multiple integrals to find area, volume and mass in engineering field.
CO5	Recognize the difference between different measure of central tendency, summarize and interpret data.
CO6	Understand and solve the problems using probability axioms, rules and Bayes theorem, use distributions such as Binomial, Poisson, Exponential and Normal to solve real world problems.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	-	3	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	3	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	3	-
CO4	3	1	-	-	-	-	-	-	-	-	-	-	2	-
CO5	2	2	-	1	1	-	-	-	-	-	-	-	2	1

CO6	2	2	1	2	1	1	1	-	-	-	-	-	2	1
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Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

G. Recommended Study Material:

❖ Text Books:

1. Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
2. H. K. Dass and Rajnish Verma, Higher Engineering Mathematics, S Chand & Co Pvt. Ltd.
3. Ross, Sheldon. A first course in probability. Pearson, 2014.

❖ Reference Books:

1. M. D. Weir *et al*: Thomas' Calculus, 11th Ed., Pearson Education, 2008.
2. Stewart James: Calculus Early Transcendental, 5th Ed., Thomson India, 2007
3. Wylie, C. R., and L. C. Barrett. Advanced Engineering Mathematics. 1982, McGraw-Hill Book Company.
4. Greenberg, Michael D. Advanced engineering mathematics. Prentice-Hall, 1988.
5. Hogg, R V, E A Tanis and D L Zimmerman, Probability and Statistical Inference, 9th edition, Prentice Hall, 2015.

❖ Web Materials:

1. <http://nptel.ac.in/courses/122107037/>
2. <http://nptel.ac.in/courses/111107108/>
3. <http://nptel.ac.in/courses/122103012/>
4. <http://nptel.ac.in/courses/122104018/>
5. <http://nptel.ac.in/courses/111106100/>
6. <http://nptel.ac.in/courses/122101003/>
7. <https://ocw.mit.edu/courses/mathematics/18-02-multivariable-calculus-fall2007/lecture-notes/>
8. <https://nptel.ac.in/courses/111105041/>

FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY

IT143: FUNDAMENTALS OF COMPUTER PROGRAMMING

A. Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	02	02	04	03
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Computation and Algorithm	03
2	Introduction to Programming and 'C' language	02
3	Constants, Variables & Data Types in 'C'	03
4	Operators and Expression in 'C'	03
5	Managing Input & Output Operations	02
6	Conditional Statements ,Branching & Looping	04
7	Arrays	04
8	Character Arrays	03
9	User-Defined Functions in 'C'	03
10	Structures and Basics of pointer	03

Total Hours (Theory): 30

Total Hours (Lab): 30

Total Hours: 60

C. Detailed syllabus

Following contents will be delivered to the students during laboratory sessions.

- | | | | |
|-----|--|-----------------|------------|
| 1 | Introduction to Computation and Algorithm | 03 Hours | 9% |
| 1.1 | Various number systems: Decimal, Binary, Octal, Hexadecimal, conversion from one number system to another | | |
| 1.2 | What is computer, Algorithms, Flow-charts | | |
| 1.3 | Solve Various types of algorithms like Exchanging values of two variables, (using 3 variables & 2 variables), Arrange numbers in ascending order, Evaluate various series e.g.: $\sin x$, $1^2-2^2+3^2-\dots$, $1+2/2!+3/3!+\dots$, | | |
| 2 | Introduction to Programming and 'C' language | 02 Hours | 7% |
| 2.1 | What is program & programming, programming languages and it's types, compiler, and interpreter, History of C, Characteristics of C, Basic structure of a program, Compiling process of C a Program | | |
| 3 | Constants, Variables & Data Types in 'C' | 03 Hours | 9% |
| 3.1 | Character set, C tokens, Keyword, Constants, and Variables | | |
| 3.2 | Data types – declaration & initialization, User-defined type declaration - typedef, enum, Symbolic constant (#define) | | |
| 4 | Operators and Expression in 'C' | 03 Hours | 13% |
| 4.1 | Classification of operators: arithmetic, relational, logical, assignment, increment / decrement, bitwise, special operators. Unary, binary and ternary operators | | |
| 4.2 | Arithmetic expression, evaluation, type conversion: implicit & explicit, precedence and associativity, use of math.h file | | |
| 5 | Managing Input & Output Operations | 02 Hours | 6% |
| 5.1 | Basic input and output operations, Input a character, introduction to ASCII code, various library functions from ctype.h | | |
| 5.2 | Formatted input using scanf(), formatted output of integer and real data using printf () | | |
| 6 | Conditional Statements, Branching & Looping | 04 Hours | 14% |
| 6.1 | Decision making using if, if...else statement, nesting of if...else, else...if Ladder | | |

- 6.2 *switch*, use of if...else instead of conditional operator, *goto* statement
- 6.3 Need of looping, entry-controlled loop: while, for, exit-controlled loop: do...while, difference
- 6.4 Nesting of looping statements, use of break and continue, use of if, if...else in loop

7 Arrays 04 Hours 10%

- 7.1 Need of array, declaration & initialization 1D array, various programs of 1D
- 7.2 2D array and their memory allocation, 2D array basic programs and matrix operations

8 Character Arrays 03 Hours 8%

- 8.1 Difference of character array with numeric array and importance of NULL character
- 8.2 Declaration, initialization and various input and output methods of string.

9 User-Defined Functions in 'C' 03 Hours 12%

- 9.1 Need of modularization, advantages, introduction to user-defined function, form of C functions, function prototype, function call, function body
- 9.2 Call by value, actual & formal arguments, use of return, nesting of functions, recursion, Array as function arguments
- 9.3 storage classes: scope, life of a variable in C

10 Structures and Basics of pointer 03 Hour 12%

- 10.1 Need of user-defined data type, structure definition, declaration and initialization of variables
- 10.2 Background of memory, variable, value, address of variable, introduction to pointer, declaration & initialization, access value using pointer, indirection (*) operator

D. Instructional method and pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed. Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory and tutorial session respectively.

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify situations where computational methods would be apply and develop various problem solving skills.
CO2	Define, select, and compare data types, operators, and functions to solve mathematical as well as scientific problems.
CO3	Describe, design and develop modular programs using control structures.
CO4	Illustrate the use of different data structures with practical aspect.
CO5	Validate the logic building and code formulation by designing code capable of passing various test cases.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	1	1	1	-	-	-	-	-	-	-	-	-	1	-
CO3	2	-	-	-	1	-	-	-	-	-	-	-	1	-
CO4	-	-	-	1	-	-	-	-	-	-	-	-	1	1
CO5	2	1	2	-	-	-	-	-	-	-	-	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

1. able to apply and practice logical ability to solve the real world problems.

F. Recommended study material:

❖ Text book:

1. Programming in ANSI C, E.Balagurusamy, Tata McGraw Hill.

❖ **Reference books:**

1. C Programming Language (2nd Edition), Brian W. Kernighan, Dennis M. Ritchie, Prentice Hall (PHI)
2. C: The Complete Reference, Herbert Schildt, McGrawHill
3. Let us C: Yashwant Kanetkar, BPB publications new Delhi
4. Computer programming and utilization: M.T.Savaliya, Atul Prakashan
5. Computer concepts and Programming , Vikas Gupta, DreamTech
6. Computer fundamentals and Programming in C, Pradip dey and Manas Ghosh, Oxford.

❖ **URL Links:**

1. <http://www.tutorials4u.com/c/>
2. <http://www.cprogramming.com/tutorial.html>
3. <http://www.howstuffworks.com/c.htm>
4. <http://www.programmingtutorials.com/c.aspx>
5. http://www.physics.drexel.edu/courses/Comp_Phys/General/C_basics/

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL
ENGINEERING

ME147: BASICS OF CIVIL AND MECHANICAL ENGINEERING

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction of Mechanical Engineering	06
2.	Steam and Steam Generator	05
3.	Internal Combustion Engines	06
4.	Pumps and Compressors	03
5.	Refrigeration and Air Conditioning Systems	05
6.	Power and Motion Transmission	05
7.	Scope of Civil Engineering	02
8.	Introduction to Surveying	06
9.	Elements of building Construction	07

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

C. Detailed Syllabus:

PART A:

1. Introduction of Mechanical Engineering 06 Hours 13%

1.1 Prime movers and its types, Sources of energy

1.2 Basic terminology: Force and mass, Pressure, Work, Power, Energy, Heat, Temperature, Units of heat, Specific heat capacity, Interchange of heat, Change of state, Internal energy, Enthalpy, Entropy, Efficiency

- 1.3 Zeroth Law and First Law of Thermodynamic, Boyle's law, Charle's law and Combined gas law, Relation between C_p and C_v
 1.4 Constant volume process, Constant pressure process, Isothermal process, poly-tropic process, adiabatic process
 1.5 Numerical

2. Steam and Steam Generator **05 Hours** **11%**

- 2.1 Introduction to steam formation and its types
 2.2 Introduction to steam table
 2.3 Calorimeter and its types
 2.4 Boiler definition and its classification
 2.5 Cochran boiler, Babcock and Wilcox boiler and its mountings and accessories
 2.6 Efficiencies of boiler

3. Internal Combustion Engines **06 Hours** **13%**

- 3.1 Introduction
 3.2 Basic terminology of I.C. engine
 3.3 Types of I. C. engines
 3.4 Efficiencies of an engine
 3.5 Numerical Practice

4. Pumps and Compressors **03 Hours** **7%**

- 4.1 Introduction
 4.2 Classification and application of pumps and compressors,
 4.3 Working of Reciprocating and Centrifugal Pump

5. Refrigeration and Air Conditioning Systems **05 Hours** **11%**

- 5.1 Introduction to refrigeration and air conditioning
 5.2 Basic terminology, Principal and application of refrigeration
 5.3 Vapour compression refrigeration system
 5.4 Domestic refrigerator
 5.5 Window and split air conditioning systems

6. Power and Motion Transmission **05 Hours** **11%**

- 6.1 Introduction
 6.2 Types of couplings, brakes and clutches.
 6.3 Belt drive and its types
 6.4 Gear drives and its types, Gear trains, Chain drives

PART B:

7. Scope of Civil Engineering **02 Hours** **05%**

- 7.1 Scope of Civil Engineering, Branches of civil engineering, Role of civil engineer

8. Introduction to Surveying **06 Hours** **13%**

- 8.1 Definition of surveying, Objects of surveying, Uses of surveying,
 8.2 Primary divisions of surveying, Principles of surveying,
 8.3 List of classification of surveying, Definition: Plan and Map,
 8.4 Introduction to linear and angular measurement

9. Elements of building Construction **7Hours** **16%**

- 9.1 Types of building,
- 9.2 Design loads
- 9.3 Building components (super structure and substructure)
- 9.4 Principles of Planning
- 9.5 Basics Requirements of a building Planning
- 9.6 Types of Residential Building

D. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

E. Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand fundamental principles of mechanical engineering
CO2	Learn the formation of steam and different types of the boilers and its mounting and accessories
CO3	Explore the basics of internal combustion engine and pumps & compressors
CO4	Learn basics of refrigeration and air conditioning system and power & motion transmission systems
CO5	Learn importance and application of civil engineering
CO6	Understand the fundamentals of surveying and building planning and Identify

	different building components
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Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	1	-	-	-	-	-	-	-
CO5	2	-	-	-	-	1	-	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	1	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Text Books:

1. S. M. Bhatt, H. G. Katariya, J. P. Hadiya, “Elements of Mechanical Engineering”, Books India Publication, Ahmedabad.
2. P. S. Desai, S. B. Soni, “Elements of Mechanical Engineering”, Atul Prakashan, Ahmedabad

Reference Book:

1. Dr. Sadhu Singh, “Elements of Mechanical Engineering”, S.CHAND Publication, New Delhi
2. V. K. Manglik, “Elements of Mechanical Engineering”, PHI Learning, Delhi.
3. Khasia R.B. and Shukla R.N., “Elements of Civil Engineering”, Mahajan Publication.
4. Kandyia Anurag, “Elements of Civil Engineering”, Charotar Publishing House Pvt. Ltd.
5. Punamia B.C., “Surveying”, Vol. I & II.

Reading Materials, web materials with full citations:

1. <http://nptel.ac.in/downloads/112105125/>
2. http://www.slideshare.net/allsaintsscience/7th-grade-ch-2-sec-3-behavior-ofgases?qid=75b08741-fb53-4413-b434-5982afe602bf&v=&b=&from_search=12
3. http://www.slideshare.net/Arjun_Dedaniya/properties-of-steam-62226458?qid=fa8777fd-b5434128-813c-cf3af3b86579&v=&b=&from_search=2
4. http://www.slideshare.net/shanus1/i-c-engines-a-study?qid=69826356-b9ed-4618-9c77b2d5a3eac2e3&v=&b=&from_search=8
5. <http://nptel.ac.in/syllabus/syllabus.php?subjectId=105104101>
6. <http://nptel.ac.in/courses/105107122/>

**FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICS**

PY143: ENGINEERING PHYSICS-II

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	2
Marks	-	50 (Internal) + 50 (External) =100	-	100	

B. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of Hours
1.	Electricity and magnetism Lab	12
2.	Light and Optics Lab	8
3.	Modern Physics Lab	10
		Total Hours:30

C. Detailed Syllabus:

1	Electricity and magnetism: Charge, Conductors and Insulators, Coulomb's law, The electric field, Principle of Superposition, Electric potential energy, Electric potential, capacitance and capacitors, The electric potential inside a parallel plate capacitor, electron current, conductivity and resistivity, resistance and Ohm's law, Kirchhoff's laws and basic circuit, energy and power Magnetic field, The magnetic field of a current, magnetic dipoles, Ampere's law and solenoids, Magnetic forces on current-carrying wires, Magnetic properties of matter, Induced currents, Lenz's law, Faraday's law, Induced currents: Three Applications, Inductors, LC Circuits, LR Circuits, LRC Circuits	12 hours
	• Measurement of capacitance by the bridge method • Induction and LR, LC, and LRC Circuits • Magnetic field along the axis of a coil • Time Constant of RC Circuit • Measurement of susceptibility of paramagnetic solution (Quinck's Tube Method)	
2	Wave and Optics: Classification of waves: Plane and Spherical Waves. Longitudinal and	08 hours

	Transverse Waves. Plane Progressive (Travelling) Waves, Pressure of a Longitudinal Wave. Energy Transport. Intensity of Wave. Standing (Stationary) Waves in a String: Fixed and Free Ends, Longitudinal Standing Waves and Normal Modes, Superposition of Waves, Propagation of electromagnetic waves, Diffraction and Interference, reflection, refraction, refractive index, Basics of LASER Physics, Total Internal reflection, Basics of optical fiber, Acceptance angle and Numerical aperture	
	• The wavelength of light, LASER, and Diffraction • Numerical Aperture and Bending Losses in Optical fiber • Standing (Transverse) Waves and resonance Using Vibrating Strings. Melde's Experiment	
3	Modern Physics: Planck's quantum theory, Planck's constant and light as a collection of photons; Photo-electric effect and Compton scattering. De Broglie wavelength and matter waves; Davisson- Germer experiment, Wave-particle duality, Heisenberg uncertainty principle-impossibility of a particle following a trajectory, Two slit interference experiment with photons, atoms and particles; Radioactivity: stability of nucleus; Law of radioactive decay	10 hours
	• The Photoelectric Effect; photo current versus intensity and wavelength of light; maximum energy of photo-electrons versus frequency of light • Frank hertz experiment; To determine the ionization potential of mercury • To determine value of Planck's constant using LEDs of at least 4 different colours	

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Section wise Online Quiz will be taken.
- Lab manual: Student will be required to read the lab material prior to the start of class. A way to ensure this is by lab quizzes and assignments.
- Lab Reports: Student has to write lab reports and submit them hardcopy/electronically. The purpose of this exercise is both to demonstrate your work in lab and to guide you to think a bit more deeply about what you are doing. The act of technical writing also helps improve your communication skills, which are broadly relevant far beyond the physics lab

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Students would be able to describe the static and dynamic electric and magnetic fields for technologically important structures.
CO2	Ability to identify and illustrate physical concepts and terminology used in optics and other wave phenomena.
CO3	Students would be able to appreciate the need for quantum mechanics, wave particle duality, uncertainty principle etc. and their applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	1	1	-	-	-	-	-	-	-	-	-	-
CO3	3	3	-	2	1	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Text Books:

1. Physics for Scientists and Engineers by Randall D. Knight, 4th Edition, Pearson
2. University Physics by Hugh D. Young, Roger A. Freedman and A. Lewis Ford, 13th Edition, Pearson
3. Physics by John D. Cutnell & Kenneth W. Johnson, 8th Edition, John Wiley & Sons, Inc.

Reference Books:

1. Electricity and Magnetism, Edward M. Purcell, 1986 McGraw-Hill Education
2. Waves: Berkeley Physics Course, vol. 3, Francis Crawford, 2007, Tata McGraw-Hill.
3. Fundamentals of Optics, F.A. Jenkins and H.E. White, 1981, McGraw-Hill
4. Principles of Optics, Max Born and Emil Wolf, 7th Edn. 1999, Pergamon Press.
5. Optics, Ajoy Ghatak, 2008, Tata McGraw Hill
6. Concepts of Modern Physics, Arthur Beiser, 2002, McGraw-Hill.
7. Introduction to Modern Physics, Rich Meyer, Kennard, Coop, 2002, Tata McGraw Hill

8. Introduction to Quantum Mechanics, David J. Griffith, 2005, Pearson Education.

Web Materials:

1. Measurement of capacitance by the bridge method -
<https://www.youtube.com/watch?v=loZC-2A3LLg>
2. Phasor Diagram of RL, RC and RLC Circuits -
<https://www.youtube.com/watch?v=HaFrY0qQ-NU>
3. Magnetic field along the axis of a coil - <https://www.youtube.com/watch?v=S0N4eVg7I3Y>
4. Quinck's Tube Method - <https://www.youtube.com/watch?v=yzgdq8uUfO4>
5. The wavelength of light, LASER - <https://www.youtube.com/watch?v=desLn3tMLcc>
6. Numerical Aperture and Bending Losses in Optical fiber -
<https://www.youtube.com/watch?v=b7dLcINlvwE>,
<https://www.youtube.com/watch?v=Wh9knsYSodI>
7. Melde's Experiment - <https://www.youtube.com/watch?v=pvX5y95Sye0>,
<https://www.youtube.com/watch?v=1CyFsGk-14>
8. The Photoelectric Effect - <https://www.youtube.com/watch?v=6VqNz4oT0ng>,
<https://www.youtube.com/watch?v=kcSYV8bJox8>
9. Frank hertz experiment - <https://www.youtube.com/watch?v=aFLnOglBxDk>
10. To determine value of Planck's constant using LEDs -
<https://www.youtube.com/watch?v=fmbSTt8dDWs>

FACULTY OF TECHNOLOGY & ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE145: BASICS OF ELECTRONICS & ELECTRICAL ENGINEERING

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

B. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic Electrical Terms and Units	04
2	Electrical Circuit Analysis	07
3	Electrostatic	08
4	Electromagnetism	05
5	AC Fundamentals	05
6	Single Phase AC Series Circuits	05
7	Polyphase Circuits	04
8	Basics of Electronics	07

Total hours (Theory) : 45 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 75 Hrs

C. Detailed Syllabus

1	Basic Electrical Terms and Units	04 Hours	08%
1.1	Ohm's law, resistor and its coding, properties, temperature co-efficient of resistance, resistance variation with temperature, examples		
2	Electrical Circuit Analysis	07 Hours	15%
2.1	Kirchhoff's current and voltage law, mesh and nodal analysis, Examples		
2.2	Series parallel circuits, star-delta transformation		
3	Electrostatic	08 Hours	18%
3.1	Capacitors, charge and voltage, capacitance, electric fields, electric field strength and electric flux density, relative permittivity, dielectric strength, Examples		

3.2	Capacitors in parallel and series, Calculation of capacitance of parallel plate and multi plate capacitor, examples		
4	Electromagnetism	05 Hours	12%
4.1	Magnetic field, its direction and characteristics, magnetic flux and flux density, magneto motive force and magnetic field strength, examples		
4.2	Faraday's law of electromagnetic induction, Fleming's left hand and right hand rule, Lenz law, force on a current carrying conductor, examples		
4.3	Self and mutual inductance		
5	AC Fundamentals	05 Hours	12%
5.1	AC Waveform and definition of its terms, relation between speed and frequency		
5.2	Average and RMS value and its determination for sinusoidal wave shapes, examples		
6	Single Phase AC Series Circuits	05 Hours	12%
6.1	R-L and R-C series circuit, power in ac circuits, examples		
6.2	R-L-C series circuit, resonance in R-L-C series circuit, relevant examples		
7	Polyphase Circuits	04 Hours	08%
7.1	Phase sequence, voltage and current relations in star and delta connected system		
8	Basics of Electronics	07 Hours	15%
8.1	Electronic Systems: Basic amplifier, voltage, current and power gain, Basic attenuators, CRO		
8.2	Transmission and Signals: Analog and digital signals, bandwidth,		
8.3	Forward and reverse bias of PN junction diode, Zener diode		
8.4	Rectifiers: Half Wave, Full Wave - Centre Tap, Bridge		
8.4	Transistor: Bipolar junction transistor, construction and biasing, configuration		

D. Course Outcomes (COs):

On the completion of the course, one should be able to:

CO1:	Describe resistors, capacitors and inductors properties, readings and calculation
CO2:	State the basic electrical laws and apply these laws to solve electrical network.
CO3:	Identify the property of magnetic materials and understand the laws of emf generation
CO4:	Solve the series and parallel DC circuits and AC circuits for single and poly-phase networks
CO5:	Develop skill and design AC-DC rectification circuits, operate basic electrical and electronics instruments

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO2:	2	3	-	-	-	-	-	-	-	-	-	-	1	-
CO3:	1	3	-	-	-	-	-	-	-	-	-	-	1	-
CO4:	1	2	-	2	-	-	-	-	-	-	-	-	3	-
CO5:	1	2	3	-	3	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), No correlation “-”

E. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, pre-requisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, blackboard, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

• Recommended Study Material:

❖ Text Books:

1. Elements of Electrical Engineering and Electronics by U.A.Patel and R. P. Ajwalia
2. A Text Book of Electrical Technology by B. L. Thareja, S. Chand

3. Principles of Electrical Engineering and Electronics by V. K. Mehta, S. Chand

❖ Reference Books:

4. Electrical Technology by Hughes, Pearson Education.
5. Electrical Engineering Fundamentals by Vincent Del Toro, Pearson Education.

❖ Web Material

1. <https://www.electronics-tutorials.ws/>

FSI02A: FOUNDATION COURSE ON CHEMISTRY AND BIOLOGY
2nd Semester and 1st Year (Level I)
B. Tech. (All Branches)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Scientific phenomena related to Chemistry and discussion, presentation, and its explanation	15
2	Scientific phenomena related to Biology and discussion, presentation, and its explanation	15

Total hours (Theory) : -- Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 30 Hrs

C. Detailed Syllabus:

Scientific Problems in Chemistry

Sr. No.	
1	Acidic Seas: How Carbon Dioxide is Changing the Oceans.
2	Why Paper Thrives in a Digital World? Chemistry of Paper.
3	A solar future. Chemistry of solar cell.
4	Smartphones: Smart Chemistry
5	The Chemistry of Sleep.
6	Cheese chemistry
7	Why does your coffee taste and smell delicious?
8	Chemistry of Compost.
9	How chemistry changed world war-1?
10	What's in the smell of first rain?
11	What's that stuff? (FIREWORKS, FOOD PRESERVATIVES, FOOD COLORING, PENCILS & PENCIL LEAD, SUNSCREENS, ARTIFICIAL SWEETENERS etc.)
12	Significant facts of colours and lights:
13	What is a battery and how it works?

14	How corrosion occurs and can it be prevented?
15	How does detergents/soap clean dirty clothes?
16	Importance of pH in drinking water.
17	Chemistry of insect bites
18	Role of metal ions in life.
19	Chemistry of Cosmetics.
20	Acidity and its treatment
21	chemistry of aging in humans

Scientific Problems in Biology

Sr. No.	
1	What is life? How did life begin and evolve on Earth?
2	Are all cells alike?
3	Is it possible to reverse tissue type?
4	What are the building blocks of cells?
5	What is genetic material?
6	What are micro-organisms? How does our body fight against the disease causing bacteria?
7	Can we create life in the lab?
8	Can human organs and organ systems inspire new technologies?
9	Physics tools in biology (ECG, EEG, CT Scan, MRI, X Ray, respirometer, ventilators etc.)

D. Course Outcomes (Learning Outcomes):

Upon successful completion of this course, a student will be able to

CO1:	Acquire the fundamental knowledge of Mathematics and Physics.
CO2:	Able to apply foundational knowledge to solve daily life problems.
CO3:	Solve basic phenomena of sciences by a scientific approach.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	2	-	-	-	1	-	-	1	-	-	1	-	-
CO3	2	2	2	-	-	-	-	-	1	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Labs will be conducted with the aid of multi-media projector, black board, OHP, models etc.
- Attendance is compulsory in labs which carries a 10% component of the overall evaluation.
- Course content presentation should be based on experimental demonstration which can induce scientific inquiry among students.
- Identify the audio-visual resources/materials related to the topic which could be played during lab session to improve understanding of theory.
- Problem solving activities should be done in lab along with specific practical exercises to inculcate prediction on final result/outcome.
- Critical thinking activities should be incorporated after course content presentation so that students can develop ability to reach generalization and apply them to solve everyday problems.
- Include a range of learning experiences, such as investigations, practical activities, field trials, field trips, speakers, and group and individual activities. On field activities and excursion(s) must be done.
- After the presentation of course content, active involvement of student must be solicited in form of either pair/group work or presentation to motivate and engage the students. Furthermore, these activities will promote higher level skills and attributes such as communication, teamwork and perseverance in the students.
- Adaptive course contents based on recent events with its background, current trends should be added in course.
- More real life exposure should be included.
- More comprehensive practical must be included to fulfil the requirement of learning while doing.
- Use of models and modelling techniques in teaching to strengthening the understanding.
- Practical should be conducted in such a way that the development of process skills like observation, classification, measurement etc. will become strong in the students.

F. Recommended Study Material:

Text Book:

Reference Book:

1. The Science Book, DK Publishing
2. How Science Works: The Facts Visually Explained. DK Publishing

Web Material:

<https://wonderopolis.org/wonders?tag=biology>

<https://www.schoolandcollegelisting.com/XX/Unknown/1399327706947869/Chemistry-is-Fun>

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY												
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING												
Sem	Course Code	Course Title	Teaching Scheme					Examination Scheme				
			Contact Hours				Credit	Theory		Practical		Total
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External	
SY Sem-III	XXXXX	University Elective I	0	2	0	2	2			30	70	100
	MA252	Advanced Engineering Mathematics	4	0	0	4	4	30	70	0	0	100
	EE251	Analog and Digital Electronics	3	2	0	5	4	30	70	25	25	150
	EE252	Circuit Theory	3	2	0	5	4	30	70	25	25	150
	EE243	Electrical Measurement and Industrial Instrumentation	4	2	0	6	5	30	70	25	25	150
	HS121.02A	Creativity, Problem Solving and Innovation		2		2	2	0	0	30	70	100
			14	10	0	24	21	150	350	105	145	750
SY Sem-IV	XXXXX	University Elective II	0	2	0	2	2			30	70	100
	EE254	Electrical Power Generation and Its Economics	3	0	0	3	3	30	70	0	0	100
	EE246	Microprocessor and Microcontrollers	4	2	0	6	5	30	70	25	25	150
	EE247	Transformers and Induction Machines	4	4	0	8	6	30	70	50	50	200
	EE255	Control Systems	3	2	0	5	4	30	70	25	25	150
	HS111.02A	Human Values and Professional Ethics		2		2	2	0	0	30	70	100
			14	12	0	26	22	120	280	160	240	800

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 3rd Semester Undergraduate Programme
(Effective from Academic year 2022-23)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC281.01: Introduction to MATLAB Programming	EC / FTE
2	CE281.01: Art of Programming	CE / FTE
3	CL281.01: Environmental Sustainability and Climate Change	CL / FTE
4	EE284: Python Programming	EE/FTE
5	IT283: Web Designing & UI/UX	IT/FTE
6	ME281.01: Engineering Drawing	ME/FTE
7	PH233.01: Fundamentals of Packaging	RPCP/FPH
8	PD260.01: Basic Laboratory Techniques	PDPIAS/FAS
9	NR251.01: First Aid & Life Support	NURSING / FMD
10	PT191.01: Health Promotion and Fitness	ARIP / FMD
11	CA224: Introduction to Web Designing	CMPICA / FCA
12	BM231: Banking and Insurance	I2IM / FMS
13	PD261: Astrophysics, Space and Cosmos-I	PDPIAS/FAS
14	CL283: SDG Handprint Laboratory	CL / FTE

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 4th Semester Undergraduate Programme
(Effective from Academic year 2022-23)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC282.01: Prototyping Electronics with Arduino	EC / FTE
2	CE282.01: Web Designing	CE / FTE
3	CL282.01: Basics of Environmental Impact Assessment	CL / FTE
4	EE287: MATLAB Programming	EE/FTE
5	EE288: Maintenance of Household Apparatus	EE/FTE
6	IT284: Data Visualization	IT/FTE
7	ME282.01: Material Science	ME/FTE
8	PH238.01: Cosmetics in daily life	RPCP/FPH
9	NR261.01: Life Style Diseases & Management	NURSING / FMD
10	PT192.01: Occupational Health & Ergonomics	ARIP / FMD
11	CA225: Programming the Internet	CMPICA / FCA
12	BM241: Health Care Management	I2IM / FMS
13	PD262: Astrophysics, Space and Cosmos-II	PDPIAS/FAS

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – III)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE284: PYTHON PROGRAMMING (UE)

3rd Semester and 2nd Year (Level II)
 B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	---	2	2	2
Marks	---	100	100	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Variables and type	05
2	Function, Basic recursion	05
3	Control flow: Branching and repetition	05
4	Introduction to objects: Strings and lists	05
5	Python modules, debugging programs	05
6	Introduction to data structures : Dictionaries	05

Total hours (Theory) : 00Hrs
 Total hours (Lab) : 30Hrs
 Total hours : 30Hrs

C. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Write program for real world problems.
CO2	Describe the Math functions, Strings, List, Tuples and Dictionaries in Python.
CO3	Implement solution for mathematical problems, processing data, and handling strings.
CO4	Debug and Troubleshoot python code.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
-----------------	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------

CO1	1	1	2	-	3	-	-	-	-	-	-	-	3	1
CO2	3	2	-	-	-	-	-	-	-	-	-	-	-	-
CO3	1	1	3	1	-	-	-	-	-	-	-	-	2	1
CO4	-	3	1	2	1	-	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Laboratories will be conducted with the aid of multi-media projector.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- Attendance is compulsory in laboratory, which carries five marks of the overall evaluation.
- Two viva voce will be conducted during the semester and average of two will be considered as a part of overall evaluation.

E. Recommended Study Material:

Text Book:

1. Think Python – How to Think like a Computer Scientist, Allen Downey, Green Tea Press.

Reference Book:

1. Python Programming: A Complete Guide for Beginners to Master, Python Programming Language, Brian Draper
2. Python: The Complete Reference, Martin C. Brown, Tata McGraw Hill

Web Material:

- <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016>
- <https://www.python.org/about/gettingstarted>
- https://onlinecourses.nptel.ac.in/noc18_cs21/preview

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF SCIENCE
MATHEMATICAL SCIENCES
MA252: ADVANCE ENGINEERING MATHEMATICS
B. TECH. 3rd SEMESTER (EC/EE)

Credits and Hours				
Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	-	4	4
Marks	100	-	100	

A. Outline of the course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Laplace Transforms and Its Applications	15
2.	Fourier Series	10
3.	Linear Algebra	07
4.	Vector Differential Calculus	08
5.	Z – transforms	10
6.	Fourier Transforms	10
		60

B. Detailed Syllabus:

1.	Laplace Transforms and Its Applications:	15 Hours 25%
1.1	Laplace transforms as an improper integral and its existence.	
1.2	Laplace transforms of elementary functions and its properties.	
1.3	Inverse Laplace transforms and its properties.	
1.4	First and second shifting theorems. Laplace transforms of derivatives and integrals.	
1.5	Convolution theorem and its application to obtain inverse Laplace transform.	
1.6	Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function).	
1.7	Solving differential equations using Laplace transform.	
1.8	Applications of Laplace transform to ODE: Electric circuits (RLC and RL)	
2.	Fourier Series:	10 Hours 17%
2.1	Periodic functions, Dirichlet's conditions, Trigonometric series.	
2.2	Euler formulae, Fourier series of periodic functions of period p .	
2.3	Fourier series of Discontinuous functions.	

2.4	Fourier series of Even and odd functions; half range Fourier series.	
3.	Linear Algebra:	07 Hours 11%
3.1	Introduction of Vector Space and its examples.	
3.2	Linear dependence and independence and its examples.	
3.3	Basis and its examples.	
4.	Vector Differential Calculus:	08 Hours 13%
4.1	Revision of concepts of Vector algebra, Scalar and Vector fields.	
4.2	Gradient of a scalar functions, Directional derivatives.	
4.3	Divergence and Curl of a vector field and their properties.	
4.4	Physical interpretations of gradient, divergence and curl: Irrotational and solenoidal vector fields.	
4.5	Scalar potential function.	
5.	Z – transforms:	10 Hours 17%
5.1	Introduction to Z – transform. Representation of sequences.	
5.2	Z – transforms of basic sequences.	
5.3	Unit impulse and Unit step sequence. Basic operations on sequence.	
5.4	Properties of Z – transforms: Damping rule and shifting property.	
5.5	Introduction to inverse Z – transforms. Inverse Z – transform using partial fraction and geometric series.	
6.	Fourier Transforms:	10 Hours 17%
6.1	Introduction to Fourier integral: sine and cosine integrals.	
6.2	Introduction to Fourier Transform and inverse Fourier Transform.	
6.3	Properties and examples of Fourier transform.	
6.4	Fourier sine and cosine transforms.	

D. Instructional Method and Pedagogy:

- At the starting of the course, the course delivery pattern, prerequisite of the subject must be discussed.
- Lectures may be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 5% component of the overall evaluation.
- Minimum two internal tests/ unit tests will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Quizzes (surprise tests) /Oral tests/ Viva/Assignment will be conducted which carries 10% component of the overall evaluation.

E. Course Outcome:(COs)

At the end of the course, the students will be able to

CO1	Analyze the Fourier series of periodic functions and determine the solution of complex engineering problems. Solutions of heat, wave and Laplace's equations using it.
CO2	Classify Laplace transforms and inverse Laplace transforms and its techniques to

	solve ordinary differential equations of real world problems.
CO3	Z transform and inverse Z transforms of elementary sequences and use it to solve compound engineering problems.
CO4	Fundamental mathematical properties of the Fourier transform and inverse Fourier transform including linearity, shift, symmetry, scaling, modulation and convolution.
CO5	Dot product, cross product, length of vectors, partial derivatives, derivatives of vector-valued functions, gradient functions and these tools are employed successfully in different branches of engineering.
CO6	Infer important concepts of linear algebra and apply them to solve problems in the engineering domain related to Linear Algebra.

• Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO2	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO3	3	2	-	1	1	-	-	-	-	1	-	1	2	2
CO4	3	2	-	1	1	-	-	-	-	1	-	1	2	2
CO5	3	2	2	1	1	-	1	-	-	1	-	1	2	2
CO6	2	2	1	-	-	-	-	-	-	1	-	1	1	1

- Correlation levels 1, 2 or 3 as defined below:
- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. Recommended Study Material:

❖ Text Books:

1. Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
2. Debnath, Lokenath, and Dambaru Bhatta. Integral transforms and their applications. CRC press, 2014.
3. K. A. Stroud and D. J. Booth, Advanced Engineering Mathematics, Palgrave Macmillan, 2006.

❖ Reference Books:

1. Howard Anton and Chris Rorres, Elementary linear algebra. John Wiley & Sons, 2010.
2. H. K. Dass, Advanced engineering mathematics, S. Chand, 2008.
3. M. D. Greenberg: Advanced Engineering Mathematics, 2nd ed., Pearson, 1998.
4. B. S. Grewal, Higher engineering mathematics. 43rd Ed., Khanna Publishers, New Delhi, 2014.
5. M. D. Weir *et al*: Thomas' Calculus, 11th Ed., Pearson Education, 2008.

6. C. Ray Wylie and L. C. Barrett: Advanced Engineering Mathematics, McGraw Hill pub., 1982.

❖ **URL link:**

1. <http://nptel.ac.in/courses/111107098/>
2. <http://www1.maths.leeds.ac.uk/~frank/math2420/notes.pdf>

EE251: ANALOG AND DIGITAL ELECTRONICS
3rd Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	BJT and FET	04
2	Operational Amplifiers	04
3	Applications of op-amp	09
4	Specialized ICs	05
5	Introduction to Digital Electronics and Numbering Systems	04
6	Digital Logic families and Design	07
7	Combinational Logic	06
8	Flip-flops, Timers and Counters	06

Total hours (Theory) : 45 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 75 Hrs

C. Detailed Syllabus:

1	BJT and FET	10%	04 Hrs
	BJT: V-I Characteristics, Transistor DC load line and Operating point, Transistor as a switch and an amplifier, Faithful Amplification, Transistor biasing and its Methods, Transistor Stability and Stability Factor, Testing and Transistor data-sheet. FET: Types of FET, Construction and working principle of JFET, Difference between BJT and JFET, V-I characteristics, Pinch-off Voltage, Shorted-gate Drain Current, Parameters of JFET, Biasing of JFET, Working of MOSFET and V-I Characteristics, Difference between JFET and MOSFET.		

2	Operational Amplifiers	10%	04 Hrs
	Introduction to Integrated Circuits, IC classifications, packages and symbols, Pin diagram of 741 Op-amp, Internal block diagram of a Typical op-amp, Ideal op-amp and its equivalent circuit, Specification and Datasheet parameters of op-amp (Differential Input Resistance (R_i), Output Resistance(R_o), Common Mode Rejection Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio (PSRR), Input offset Voltage, Input offset Current, Input bias Current, Output offset Voltage), Feedback Configurations, An op-amp with and without Feedback, Closed-loop Voltage Gain.		
3	Applications of op-amp	20%	09 Hrs
	Linear applications: DC and AC Amplifier, Voltage follower, Peaking, Summing, Scaling, Averaging Amplifier, Differential Amplifiers and Instrumentation Amplifier, V to I converter, I to V converter, Integrator, Differentiator Active filters & Oscillators: First order Low-Pass and High-Pass Filters, second order Low-Pass and High-Pass Filters, Band-Pass and Band-Reject filters, All-Pass filters, Phase Shift, Wien Bridge Oscillators; Square, Triangular, Saw-tooth Wave Generators. Comparators & Converters: Basic Comparator, Zero crossing Detector, Schmitt Trigger, Basic Positive and Negative Clippers and Clampers, Absolute-Value output Circuit, Peak Detector, ADC and DAC using op-amp.		
4	Specialized ICs	10%	05 Hrs
	555 Timer IC, 555 timer IC as Monostable, bistable and astable Multivibrator, Three terminal regulator ICs (Fixed Voltage Regulator ICs, Adjustable output voltage regulator ICs, their use and basic design considerations for designing regulated power supplies), Analog Multiplier MPY634.		
5	Introduction to Digital Electronics and Numbering Systems	8.34%	04 Hrs
	Digital circuits, Digital signals, Binary number system, Signed & unsigned binary number system, 1's complement and 2's complement, Binary arithmetic, octal, hexadecimal and decimal Number systems and their inter conversion, BCD numbers, ASCII and EBCDIC codes.		
6	Digital Logic families and Design	15%	07 Hrs
	Boolean Algebra: Basic Logic gates and their truth tables, Universal Gates, Laws of Boolean algebra, De-Morgan's theorem, Min term, Max term, POS, SOP, Karnaugh		

	Maps, Minimization of logical function, Don't care condition. Logic families and Design: Introduction, Resistor transistor logic, Direct coupled transistor logic, Integrated injection logic, DTL logic, TTL logic, CMOS Logic.		
7	Combinational Logic	13.33%	06 Hrs
	Half adder, Full adder, Binary subtraction, Arithmetic logic unit, Digital comparators, Priority encoder, decoders, Multiplexer & demultiplexer.		
8	Flip-flops, Timers and Counters	13.33%	06 Hrs
	S-R flip-flop, D- flip-flop, T-flip-flop, J-K flip-flop, Clocked flip-flop design, Registers, Asynchronous counter, Synchronous counter.		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Develop & Enhance their skills in basics of analog and digital circuits.
CO2	Describe the performance and classification of ICs.
CO3	State linear and non-linear applications of operational amplifiers.
CO4	Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates, timer and counters.
CO5	Analyze characteristics of popular MOS and bipolar logic families, with emphasis on CMOS and TTL technologies.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	2	-
CO3	1	3	-	-	-	-	-	-	-	-	-	-	2	-
CO4	1	2	2	3	3	-	-	-	-	-	-	-	3	1
CO5	1	1	3	3	-	-	-	-	-	-	-	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Principles of Electronics by V. K. Mehta S. Chand & Company Ltd.
2. Op-Amp and Linear Integrated Circuit technology- Ramakant A Gayakwad, PHI Publication
3. Analog and Digital Electronics by J. S. Katre, Techmax Publications
4. Modern digital electronics, Second edition, R.P.Jain
5. Fundamental of digital circuits by A. ANANDKUMAR, PHI Publication

Reference Book:

1. Operational Amplifier and Linear integrated Circuits By Robert F.Coughlin, Frederick F. Driscoll
2. Operational Amplifier and Linear Integrated Circuits by K.LAL kishore. Pearsons
3. Electronics Devices and Circuits an Introduction by A. Mottershed, PHI Publication.
4. Digital Fundamentals by Morris and Mano, PHI Publication

Web Material:

- <https://www.facstaff.bucknell.edu/mastascu/eLessonsHTML/EEIndex.html>
- <http://www.electronics-tutorials.ws/>
- <http://nptel.ac.in/courses/108106068/>
(Videos and Notes by Prof. K. Radhakrishna Rao, IIT Madras)
- <http://nptel.ac.in/courses/108106069/>
(Videos and Notes by Dr. Amitava Dasgupta, IIT Madras)

EE252: CIRCUIT THEORY
3rd Semester and 2nd Year (Level II)
B.Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1.	Circuit elements and energy sources	2
2	Network Topology	4
3.	Network Equations	4
4	Network theorems and Impedance function	9
5	Two-Port Network Analysis	8
6	Laplace Transformation, its Application and Special Signal Waveforms	12
7	Initial conditions in networks	6

Total hours (Theory) :45 Hrs
 Total hours (Lab) :30 Hrs
 Total hours : 75 Hrs

C. Detailed Syllabus:

- 1 **Circuit elements and energy sources** **4.45% 02 Hrs**
 Circuit Elements And Their Characteristics, Energy Sources, Source Transformation.
- 2 **Network Topology** **8.89% 04 Hrs**
 Concept Of Network Graph, Terminology Used In Network Graph, Number Of Trees In A Graph, Tie-Set Matrix, Fundamental Tie-Set Matrix, Fundamental Cut-Set, Cut-Set Matrix, The Dot Convention For Coupled Circuit, Principle Of Duality

- 3 **Network Equations** 8.89% 04 Hrs
Kirchhoff's laws, Current division, Voltage division in series circuits, Nodal and mesh analysis of electric circuits in DC and AC excitations.
- 4 **Network theorems and Impedance function** 20% 09 Hrs
Superposition theorem, Thevenin theorem, Norton Theorem, Reciprocity and Maximum power transfer theorem in Electrical Circuits with DC and AC excitations, Transform impedance and transform circuits
- 5 **Two-Port Network Analysis** 17.78% 08 Hrs
Network Elements, Classification of Network, Network Configuration, Recurrent Network, Z-parameters, Y-parameters, ABCD Parameters, Inverse ABCD Parameters, Interrelationship between parameters, Concept of Reciprocity & Symmetricity for network parameters
- 6 **Laplace Transformation, its Application and Special Signal Waveforms** 26.66% 12 Hrs
Basic Types Of Special Signals, Laplace Transformation Of Special Signal Waveforms, Step Response of R-L, R-C, R-L-C Series & Parallel Circuit, Transient Response of Series R-L, R-C, R-L-C With DC Excitation, Transient Response of Series R-L, R-C, R-L-C With AC Excitation.
- 7 **Initial conditions in networks** 13.33% 06 Hrs
Initial conditions in elements, Geometrical Interpretation of Derivatives, A Procedure of Evaluating Initial conditions, R-L, R-C & R-L-C Circuits analysis using Initial Conditions.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	To understand basics of network elements and their behaviour.
CO2	To understand basics of network topologies, tie set and cutset schedules.
CO3	To apply concept of mesh, nodal analysis technique, various network theorems such as Superposition, Thevenin, Norton, Reciprocity, and Maximum Power Transfer.
CO4	To calculate two port parameters such as Z, Y, ABCD, etc. for the given two-port network.
CO5	Develop the skills related to analyse circuit taking into account of initial conditions and solving the DC and AC circuits with differential equations with the help of Laplace approach.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	3	-	2	-	3	1	3	3	-
CO2	3	3	-	-	-	-	-	-	-	1	-	-	3	-
CO3	2	1	1	3	2	1	-	-	-	1	-	2	2	3
CO4	3	1	1	2	2	-	2	2	3	-	2	2	2	2
CO5	3	1	1	3	3	-	2	2	3	-	2	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Circuit Theory by A. Chakrabarti , Dhanpat Rai Publications
2. Network Analysis by Van Valkenburg , Prentice Hall College
3. Network Analysis and Synthesis by Ravish R Singh, McGraw Hill Education Pvt. Ltd

Reference Book:

1. U. A. Patel, Network analysis and Synthesis, Mahajan Publisher

2. G. K. Mithal, Network analysis, Khanna publications
3. F. F. Kuo, Network Analysis, John Wiley
4. Electric Circuits and Networks :- By K. S. Suresh Kumar – Pearson Education
5. Linear Circuits Analysis 2nd edition :-By DeCarlo/ Lin – Oxford University Press(Indian edition)

Web Material:

- <http://nptel.ac.in/courses/108102042/9>
- <http://freevideolectures.com/Course/2350/Networks-Signals-and-Systems>

EE243: Electrical Measurements & Industrial Instrumentation
3rd Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction of Instrumentations and Measurement	04
2	Temperature and Strain Measurement	08
3	Flow and Pressure Measurement	08
4	Measurement of torque, velocity and vibration	04
5	Miscellaneous Instruments in Industrial, Biomedical and Environmental Applications	08
6	Recorders	03
7	Instrument Transformer	09
8	Analog and Digital instruments	10
9	Recent Development in instrumentation and measurements	06

Total hours (Theory) :60Hrs

Total hours (Lab) :30 Hrs

Total hours : 90Hrs

C. Detailed Syllabus:

1	Introduction of Instrumentations and Measurement	06%	04 Hrs
	History to the Instrumentation and Measurements, Application of Instrument Systems, Functional Elements of Measurement system, classification of Instruments, Microprocessor based Instrument, Static and Dynamic Performance Characteristics of Instruments, Standards and Calibration.		
2	Temperature and Strain Measurement	13%	08Hrs
	Temperature Measurement: Temperature Scales, International Practical Temperature Scale, Measurement of Temperature, Non-Electrical Methods,		

	Electrical Methods and Radiation Methods (Pyrometer) Strain Measurement: Construction, Operation, Theory and Factors Affecting Strain Measurements, Applications of Strain Gauges, Load Cells, Advantages, Disadvantages.		
3	Flow and Pressure Measurement	13%	08Hrs
	Flow Measurement Devices: Venture Tube, Orifice Plate, Flow Nozzle, Rotameter, Pitot Tube, Ultrasonic Flow Detection, Turbine Flow Meter, Electromagnetic Flow Meter, Hot Wire Anemometer Pressure Measurement Devices: Construction, Operation, Advantages, Disadvantages and Applications of Elastic Transducers (Diaphragm, Bellow, Bourdon Tube), Variable Inductance and Capacitance Transducers, Vacuum Pressure Measuring Devices (Pirani Gauge, Ionization Gauge, McLeod Gauge., Piezo-Electric Transducers		
4	Measurement of torque, velocity and vibration	06%	04Hrs
	Torque measurement: Strain Gauge Torque Meters and Magneto-Strictive Transducers Vibration measurement: Nature of Vibrations, Quantities Involved in Vibration, Piezoelectric Transducer (Accelerometer) Velocity measurement: Moving Magnet Type, Moving Coil Type Velocity Transducer, Tachometers		
5	Miscellaneous Instruments in Industrial, Biomedical and Environmental Applications	16%	10Hrs
	Specific gravity measurement: Bubbler System, Hydrometer Method, Totally Immersed Float Method, Nuclear Absorption Method, Fixed Volume Method Liquid Level measurement: <u>Direct method</u> : Dip-Stick Method, Sight Glass Method, Hook Gauge, Float Gauge <u>Indirect method</u> : Hydrostatic Pressure Level Measurement Device, Bubbler or Purge Technique, Resistive, Inductive and Capacitive Transducers, Ultrasonic Level Gauge and Radiation Method. Viscosity measurement: Capillary Tube Viscometer, Efflux Type Viscometer, Rotation Concentric Cylinder Viscometer, Variable Area Viscometer. Humidity and Moisture measurement: Dew Point Meter, Hair Hygrometer, Psychrometer, Resistive and Capacitive Hygrometer. Measurement of PH value & Conductivity: Ph meter& Conductivity meter Biomedical measurement: Blood Pressure Measurement, Measurement of Bioelectric Potentials Measurement of Environmental air pollution parameters: Orsat Apparatus For Exhaust Gas Analysis, Gas Chromatography, Non-Dispersive Infrared and Non-Dispersive Ultraviolet Gas Analyzers, and Smoke Density Measurement.		
6	Recorders	05%	03Hrs

	Strip Chart Recorders, X-Y Recorders, Circular Chart recorders, Digital Recorders, Data Acquisition System, Data Display and Storage.		
7	Instrument Transformer	15%	09Hrs
	Application of instrument transformers, CT, PT, testing of CT and PT Current Transformer: Relationship Between Various Quantities in CT, Errors in CT, Characteristics of CT, types of CT Potential Transformer: Relationship Between Various Quantities in PT, Errors on PT, Characteristics of PT, Types of PT		
8	Analog and Digital instruments	16%	10Hrs
	Analog instruments: Electrodynamic and Induction Type Instruments, Power Factor Meter, Analog Frequency Meter, Synchroscope, Phase Sequence Indicator, Extension of Instrument Ranges and Multipliers. Digital Instruments: Introduction, Digital Voltmeter, Digital Multimeter, Digital Frequency Meter and Analyzer, Signal Generator, Digital Signal Oscilloscope, Harmonic Analyzer		
9	Recent Development in instrumentation and measurements	10%	06Hrs
	Computer-Aided measurement: Introduction, Microcomputers for Data Acquisition Purposes, Data Transfer and Communication, Data Acquisition System Fiber optic transducer: Introduction, types of Configuration, Applications. Advantages and Disadvantages, Micro Sensors, Smart Sensors, Virtual Instrumentation		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Recognize different sensors, transducers and various measuring instruments
CO2	Understanding the constructional and operational aspects of various sensors, transducers and measuring instruments used for industrial, biomedical and environmental applications
CO3	Aware with the performance characteristics of instrument transformer used for measurement of currents and voltages
CO4	Aware about the recent developments in measuring instruments and sensors

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
-----------------	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------

CO1	3	-	-	-	-	-	-	-	2	-	-	3	2	-
CO2	3	-	2	2	-	-	-	-	2	-	-	-	2	-
CO3	3	-	-	-	-	-	-	-	-	-	-	3	3	-
CO4	2	2	-	3	3	-	-	-	-	-	-	2	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. B.C. Nakra & K.K. Chaudhry, Instrumentation, Measurement and Analysis, TATA McGraw Hill.
2. A. K. Shawhney, Electrical and Electronics Measurement and Instrumentation, Dhanpat Rai Publishers

Reference Book:

- i. Rangan, Sharma, Mani, Industrial Instrumentation, Tata McGraw Hill
- ii. Murthy, Transducers and Instrumentation, Tata McGraw Hill
- iii. Patranabis, ‘Principles of Industrial Instrumentation’, Tata McGraw Hill Publishing Company Ltd, Revised edition
- iv. R. K. Rajput, Electrical and Electronics Measurement and Instrumentation, S. Chand Limited.
- v. S. K. Singh, Industrial Instrumentation & Control, Tata McGraw Hill.R.S.

Khandpur, Handbook of Analytical instruments, Tata McGraw Hill.

Web Material:

- www.isa.org
- <http://nptel.iitm.ac.in/video.php?courseId=1062>
- http://www.mywbut.com/syllabus.php?mode=SM&paper_id=57&dept_id=6

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS121.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
III	HS121.02 A	Creativity, Problem Solving and Innovation	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

III. Detailed Syllabus:

1.	Introduction to Creativity, Problem Solving and Innovation	06 Hours	20%
	Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills		
2.	Questioning, Learning and Visualization	6 Hours	20%
	Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3.	Creative Thinking and Problem Solving	6 Hours	20%

	Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		
4.	Logic, Language and Reasoning	6 Hours	20%
	Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5.	Contemporary Issues and Practices in Creativity and Problem Solving	6 Hours	20%
	Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

IV. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

V. Evaluation:

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them. There may also be a Computer Based Test at the end.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Attendance	100 %	--	20
2	Individual Activity Participation	As stipulated by the Resource Person(s) in the Training		20
3	Group Activity Participation			20
4	Presentation			30
5	Feedback on Improvement			10
Total				100

VI. Course Outcomes (COs)

At the end of the course, learners will be able to:

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	2	-	-
CO2	-	-	-	2	-	-	-	-	-	2	-	3	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	2	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO6	-	-	-	3	-	-	-	-	-	2	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VII. Books / Reference Books / Reading

Text Books

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference Books

3. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
4. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
5. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
6. Tony Buzan, How to Mind Map, Thorsons Publication 2002
7. Scott Berkun, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
8. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
9. Ira Flatow, The all Laughed, Harper Publication 1992

10. Paul Sloane, Des Mac Hale & M.A. Di Spezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

11. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
12. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
13. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
14. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
15. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
16. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
17. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
18. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
19. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
20. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
21. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
22. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
23. Howard Gardner, Creating minds, Basic Books Publication 1993
24. Mihaly Csikszentmihalyi, Creativity–Flow and Psychology of Discovery and Invention, Harper Publication 1996
25. Martin Gerdner, W. H., Aha! Insight, Freeman Publication 1978
26. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication 1994
27. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

- Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – IV)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE287: MATLAB PROGRAMMING (UE-II)
4th Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	---	2	2	2
Marks	---	100	100	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	System of Linear Equations	08
2	Interpolation	04
3	Nonlinear Equations	06
4	Ordinary Differential Equations	08
5	Matrices and Eigenvalues	04

Total hours (Theory) :00Hrs

Total hours (Lab) :30Hrs

Total hours : 30Hrs

C. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Write program for real world problems.
CO2	Understand the use of MATLAB for numerical methods.
CO3	Implement solution for mathematical problems.
CO4	Debug and Troubleshoot MATLAB code.
CO5	Enhance the programming skills in MATLAB.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	-	2	-	-	-	-	-	-	-	3	2
CO2	3	3	-	2	2	-	-	-	1	-	-	-	3	2

CO3	3	2	2	2	-	-	-	-	-	-	-	-	3	3
CO4	-	3	3	2	2	-	-	-	-	-	-	-	2	-
CO5	-	-	3	2	2	-	-	-	-	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Laboratories will be conducted with the aid of multi-media projector.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- Attendance is compulsory in laboratory, which carries five marks of the overall evaluation.
- Two viva voce will be conducted during the semester and average of two will be considered as a part of overall evaluation.

E. Recommended Study Material:

Text Book:

1. Applied Numerical Methods Using Matlab, Won Young Yang, Wenwu Cao
Tae-Sang Chung, John Morris, JOHN WILEY & SONS, INC.

Reference Book:

1. Computer Oriented Numerical Methods, V. Rajaraman, PHI Publication
2. Engineering Optimization: Theory and Practice, S S Rao, JOHN WILEY & SONS, INC.

Web Material:

- <http://www.nptel.ac.in/courses/122104018/node109.html>
- http://www.geos.ed.ac.uk/~yliu23/docs/lect_spline.pdf
- https://mat.iitm.ac.in/home/sryedida/public_html/caimna/interpolation/lagrange.html

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE288: MAINTENANCE OF HOUSEHOLD APPARATUS (UE-II)
4th Semester and 2nd Year (Level II)
B.Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	---	2	2	2
Marks	---	100	100	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Electricity and Electrical Safety Rules:	04
2	Household Wiring	10
3	Soldering and Desoldering Electronic Components	04
4	Maintenance of Household Electrical Appliances	10
5	Introduction to Batteries	02

Total hours (Theory) : 00Hrs
Total hours (Lab) : 30Hrs
Total hours : 30Hrs

C. Detailed Syllabus:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Electricity and Electrical Safety Rules: Introduction to Electricity. Electrical Safety Rules, Introduction to voltage, current, frequency, phase difference, power, energy, and its measurement.	04
2	Household Wiring: Wiring of a tube light, fan, Extension board (Power Socket), Staircase wiring (Two-way wiring), Fuse, MCB, ELCB; Importance of Earthling; Know Your Bill; Preparing the drawing for wiring a newly built room without any electrical wiring along with a bill of materials and specifications.	10
3	Soldering and Desoldering Electronic Components:	04

	Assembling Electronic Circuit on bread board, Solder and De-solder Electronic Component on PCB.	
4	Maintenance of Household Electrical Appliances Maintenance and Troubleshooting of Tube-light, Fan, LED light, Pump, Water Purifier (RO), Air-conditioner(AC); Power Consumption of Household appliances	10
5	Introduction to Batteries	2

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Safely operate Electrical/Electronic devices and safeguard their self from Electrical Shock.
CO2	Install and maintain/repair domesticated wiring.
CO3	Maintain and repair/troubleshoot house-hold electrical appliances.
CO4	Develop small electronic project.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	1	-	3	-	-	-	-	-	-	2	-
CO2	2	3	1	1	-	1	-	-	1	-	-	-	2	-
CO3	2	3	1	1	-	1	-	-	-	-	-	-	1	-
CO4	1	1	3	1	1	-	-	-	2	-	1	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- A student has to prepare a laboratory term work as per instruction given by lab instructor.
- Attendance is compulsory in laboratory, which carries five marks of the overall evaluation.
- Two viva voce will be conducted during the semester and average of two will be considered as a part of overall evaluation.

F. Recommended Study Material:

Text Book:

1. Electrical Design Estimation and Costing, Raina K.B. & Bhattacharya S.K., Willet Estern Ltd.
2. India Electrical Rules 1956 Hand book, Chudley R., Butterworth –London, New Delhi.
3. A Textbook of Electrical Technology in S.I. Units Volume III, B L Thareja, S Chand

Reference Book:

1. National Building code of India Group 1 and Group 4, Bureau of Indian standard, New Delhi, Book no. 1604, Latest Edition
2. Troubleshooting and Repairing Major Appliances Hardcover, Eric Kleinert, McGraw Hill

Web Material:

- <https://www.instructables.com/id/How-to-solder/>
- <https://www.udemy.com/course/electronic-electronics-maintenance-electronic-devices-maintenance/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE254: ELECTRICAL POWER GENERATION AND ITS ECONOMICS
4th Semester and 2nd Year (Level II)
B.Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction	01
2	Steam Power Plants	08
3	Hydro Electric Power Plant	04
4	Nuclear Power Plant	04
5	Combined cycle power plant	03
6	Wind Power Generation	04
7	Solar Power Generation	03
8	Other Renewable Energy Sources	04
9	Insulated Cables	05
10	Components of Overhead Lines	10
11	Mechanical Design of Transmission Lines (Sag-Tension Relationship)	03
12	Economics of Power System	07
13	Power Factor Improvement in Power System	04

Total hours (Theory) :45 Hrs
 Total hours (Lab) :0 Hrs
 Total hours : 45 Hrs

C. Detailed Syllabus:

- | | | |
|---|---|---------------|
| 1 | Introduction
Power and Energy, sources of energy such as coal, water, nuclear, wind, solar, tidal etc. | 01% 1 Hrs |
| 2 | Steam Power Plants
Classification of steam power plants, layout of steam power plant, Component of steam power plants, Essential requirement for steam power plant, selection of site, Choice of steam condition, fuel handling, Combustion equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers, steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment, advantages and disadvantages of steam power plant. | 14% 06 Hrs |
| 3 | Hydro Electric Power Plant
Application, advantages and disadvantages of hydro power plant, Site selection for hydro power plant, layout and elements of hydro power plant, Classification of hydro power plant, Hydraulic turbines. | 07% 03 Hrs |
| 4 | Nuclear Power Plant
General aspect of nuclear engineering, layout of nuclear power plant, Nuclear reactors, Main components of nuclear power plants, advantages and disadvantages of nuclear power plant. | 07% 03 Hrs |
| 5 | Combined Cycle Power Plant:
Various combined cycle power plant, working, advantages and disadvantages, comparison of various power plant. | 04% 02 Hrs |
| 6 | Wind Power Generation
Applications, Merits and demerits of wind energy, nature and origin of wind, variables in wind energy conversion system, wind velocities and height from ground and site selection, Types of wind energy system, wind turbine generator unit with battery storage facilities, solar wind hybrid, Wind farm siting. | 06% 03 Hrs |
| 7 | Solar Power Plant
Solar energy routes and prospects, merits and limitations of solar energy conversion and utilization, Types of solar thermal collectors, comparison between conventional and solar thermal power plant, ratings of solar power plant heat transfer fluids, solar pond and binary cycle solar thermal power plant. | 04% 02 Hrs |
| 8 | Other Renewable Energy Sources
Fuel cell based power plant: Introduction, concept, types, Electrochemical | 07% 03 Hrs |

Reactions, Hydrogen, Oxygen Fuel cells.

MHD Power Generation: Principles, MHD Systems, Advantages of MHD Systems, Electrical conditions.

Geothermal energy: Applications, utilization of geothermal energy, origin of geothermal resources, liquid dominated geothermal electric power plant.

Ocean and tidal energy: Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC. Tidal energy conversion : High and low tides, tidal energy conversion.

9 Insulated Cables 09% 04 Hrs

Requirement of insulated cables, properties of conductor and insulating material for cables, construction of cables, types of cables, types of insulating material for cable, Insulation resistance of a single core cable, capacitance of single and three core cable, Examples, Electrostatic stresses in a single core cable, most economical conductor size in cable, examples, Capacitance grading and inter sheath grading, examples, Calculation of losses in cable and factors affecting it, current rating of a cable.

10 Components of Overhead Lines 17% 8 Hrs

Basic power system structure, generation, transmission, substation, sub-transmission, distribution, Complex power, single phase transmission, three phase transmission. Conductors and line supports, Types of Insulators, potential distribution over a string of suspension insulators, String efficiency and methods to improve it, Examples on string efficiency.

11 Mechanical Design of Transmission Lines (Sag-Tension Relationship) 05% 02 Hrs

Sag-tension relation and its example with different configurations.

12 Economics of Power System 12% 05 Hrs

Loads in power system, effects of variable load on power system, load curves and load duration curves, Important terms and factors related to power system economics of power system, importance of those factors, Examples on load curves, Load curves and selection of generating units, Examples, Cost of electrical energy and its expression, methods of determining the depreciation, examples, Types of Tariffs, Examples

13 Power Factor Improvement in Power System: 07% 03 Hrs

Causes of low power factor, disadvantages of low power factor, power factor

improvement, Power factor improvement equipments, calculations of power factor correction, most economical power factor, examples.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Acquire the knowledge regarding the basic functioning of conventional and non-conventional power plants.
CO2	Describe the working of various power plants and various equipment of power plants.
CO3	Develop skill and Identify the key point to improve the performance of the power plants.
CO4	Aware with the new era of energy generation technology like wind, solar and fuel cell etc.
CO5	Analyze and select the proper cable for particular application.
CO6	Determine the basic structure of the tariff and means to reduce the electricity charges by improving the power factor.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	3	1	-	-	-	-	-	-	-	-	-	2	-
CO3	-	3	-	-	-	2	3	1	1	-	-	-	3	2
CO4	-	2	-	-	3	-	-	-	-	-	-	-	2	2
CO5	1	3	3	-	-	-	-	-	-	-	-	-	3	-
CO6	1	3	3	-	2	-	-	-		-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of power plant will clear the concept of power generation and student will realise the actual size of generating equipments.

F. Recommended Study Material:

Text Book:

1. Energy Technology by S. Rao & Dr. B.B.Parulekar, Khanna Publication, New Delhi.
2. Principles of Power system by V.K. Mehta, Publisher: S. Chand & Company Ltd.
3. Renewable energy sources and conversion technology by N.K. Bansal, Tata McGraw Hill, New Delhi.
4. A Course in electrical power by Soni & Bhatnagar, Dhanpat Rai & Sons.
5. Energy sources, by G.D. Rai, Khanna Publication, New Delhi

Reference Book:

1. Power Generation Technologies by Paul Breeze, Newnes
2. Renewable energy in Power system by Leon Freris, David Infield, John Wiley Inc.
3. Renewable Energy Resources by John Twidell and Tony Weir, Taylor & Francis

Web Material:

- <http://nptel.ac.in/courses/108105058/>
- <http://nptel.ac.in/courses/108108078/>
- <https://www.youtube.com/watch?v=u0jAVR-j60o>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE255: CONTROL SYSTEMS
4th Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to control Systems	2
2	Mathematical Modeling of Control systems	8
3	Time response analysis	8
4	Stability of Control Systems	6
5	Frequency response analysis	8
6	Compensation Technique	7
7	State Variable Analysis of Control Systems	6

Total hours (Theory) :45 Hrs

Total hours (Lab) :30 Hrs

Total hours :75 Hrs

C. Detailed Syllabus:

- 1 Introduction to control System** 4.44% 02 Hrs
 Introduction, Classification of control systems: open loop, closed loop, continuous, discrete, linear & non linear control system, Examples of control system application, Effect of feedback on control system performance, Feed forward control system.
- 2 Mathematical modeling of control systems** 17.78% 08 Hrs
 Transfer function representation, Block diagram algebra, Block diagram reduction technique, Signal flow graph, Mason's gain formula, Modeling of DC motors,

Servomechanism

- 3 Time response analysis 17.78% 08 Hrs
Introduction, Standard test signals, Time response of first order and second order system, Damping ratio, natural un damped frequency, Transient response specifications, Steady state error and error constant, Effect of integral and derivative control action on system performance.
- 4 Stability of Control Systems 13.33% 06 Hrs
Concept of stability, Necessary condition for stability, Absolute and relative stability, Routh Hurwitz stability criteria, Root locus concept, Construction of root loci.
- 5 Frequency response analysis 17.78% 08 Hrs
Bode plot, Procedure for drawing bode plot and determination of gain margin, phase margin and stability, Polar plot.
- 6 Compensation Technique 15.56% 07 Hrs
Phase lead compensation, Phase lag compensation, Phase lead lag compensation, Selection of Compensator, Design of Phase Lag, Phase Lead and Phase Lag-Lead Compensator.
- 7 State Variable Analysis of Control Systems 13.33% 06 Hrs
State space representation of dynamic system, Concept of state, state variable and state model, Concept of controllability and Observability, State –Transition Matrix ,Computation of State Transition Matrix, Eigen Values and Eigen Vectors.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Demonstrate an understanding of the fundamentals of control systems
CO2	Acquire the skill related to development and usage of models for physical which are suitable for use in the analysis and design of control systems.
CO3	Express and solve system equations in state-variable form (state variable models).
CO4	Determine the (absolute) stability of a closed-loop control system.
CO5	Determine the time and frequency-domain responses of first and second-order systems to step and sinusoidal (and to some extent, ramp) inputs with controller designs.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	2	-	-	-	-	-	2	3	-
CO2	3	3	2	2	1	-	-	-	-	-	-	-	-	1
CO3	3	2	2	3	3	1	-	-	-	-	-	2	1	3
CO4	3	2	2	3	2	2	-	-	-	2	2	2	2	-
CO5	3	2	3	3	3	-	-	3	-	2	1	1	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Automatic control system by B.C. Kuo
2. Modern control engineering by K. Ogata
3. Control System Engineering by J Nagrath & M Gopal New Age International Publishers

Reference Book:

1. Control Systems by A. Anand Kumar, PHI Learning Pvt. Ltd.
2. Automatic Control Systems by S. Hasan Saeed, Katson Books.
3. Control Systems by Ashfaq Hussain, Haroon Ashfaq, Dhanpat Rai & Co.

Web Material:

- http://nptel.iitg.ernet.in/courses/Elec_Engg/IIT%20Delhi/Control%20Systems.htm
- <http://nptel.ac.in/courses/108101037/>
- <http://nptel.ac.in/courses/108102043/>
- <http://nptel.ac.in/courses/108103007/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE246: MICROPROCESSOR AND MICROCONTROLLERS
4th Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of 8085	10
2	Introduction to Microcontroller	8
3	8051 Programming	12
4	On chip Hardware Modules/Peripherals in 8051	15
5	Interfacing Microprocessor/Microcontroller & Applications	15

Total hours (Theory) :60 Hrs
Total hours (Lab) :30 Hrs
Total hours :90 Hrs

C. Detailed Syllabus:

- 1 **Basic of 8085** 17% 10 Hrs
Introduction to Microprocessor, Block Diagram of 8085: Various Registers, stack pointer, stack memory, Program Counter, Instruction decoder, Bus organization; Pin diagram and Pin functions, Multiplexing/Demultiplexing of address/data bus, Types of Memory (i.e. various type of RAM and ROM memory), External Memory Interfacing and required control signal interfacing.
- 2 **Introduction to Microcontroller** 13% 08 Hrs
Comparison between Microprocessor and Microcontroller, Introduction to 8051

Family (MCS-51) microcontrollers and salient features, Architectural block Diagram of INTEL 8051, Pin diagram and Pin Functions, General Purpose and Special Function Registers, Memory organization, Crystal Frequency & Crystal Connection, Calculation of Speed of Execution, 6/12 clock mode operation, PORT Structure.

3 8051 Programming. 20% 12 Hrs

Software & Hardware Required for Programming: Concept of Assembler, Editor, Linker, Loader, Debugger in IDE, Hardware Loader Circuit for ISP featured 8051.

Programming Language: Assembly Instruction Sets: Data Transfer & Addressing modes, Logical, Branching, Arithmetic & Bit wise Operation, Comparison of Assembly and C Programming, Data type in C, Keywords in C, Various Operator in C, Control loops, Functions and Arrays, Special Keywords & Interrupt function in Keil Microvision IDE.

4 On chip Hardware Modules/Peripherals in 8051 25% 15 Hrs

Internal Architecture of Timer/Counter and Related C Programs with all modes, Basics of Interrupt: Mask-able/ Non mask-able Interrupt, Enabling the mask-able Interrupt, Interrupt vector table, interrupt latency time and interrupt priority, External Interrupt and Related C Programs, Basics of Serial communication Protocol, Difference Between Synchronous and Asynchronous Communication, Internal Architecture of Serial Communication module in 8051 (UART) and related C programs with all modes (using SBUF register and printf library function).

5 Interfacing Microcontroller & Applications 25% 15 Hrs

Interfacing: Hardware interfacing diagram for driving a relay by ULN 2803/Transistor ,solid state relay(162341), Controlling a Lamp using Opto-isolator(ILD74),External Memory Interfacing (RAM and ROM), Keyboard Interfacing (Key Debouncing Concept, Simple Key interfacing Hardware & Software program, Matrix key board), Displays Interfacing (LED, 7-Segment display and LCD display), ADC & DAC interfacing (Serial and parallel output ADC/DAC).

Applications: Temperature measurement(LM35) and display, Counting Cars passing through passage/bottles on belt, Unidirectional and Bidirectional DC Motor Control with PWM using L293 and ILD74, Stepper Motor Drive with ULN2003.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Understand the basic building blocks, architecture of a microcontroller/microprocessor, its salient feature and difference between various microcontrollers of MCS-51 family with basic overview of other microcontrollers.
CO2	Work on Integrated development Environment(IDE)/Compiler to Analyze and develop programs using assembly code and embedded 'c'.
CO3	Gain hands-on experience in doing experiments on microcontroller by using hardware kit in the laboratory and present the report.
CO4	Interface memory and I/O devices with 8 bit microcontroller.
CO5	Develop small embedded system and applications.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO2	1	2	3	-	3	-	-	-	-	-	-	-	2	-
CO3	1	1	2	3	-	-	-	-	-	1	-	-	2	-
CO4	1	1	1	3	-	-	-	-	-	-	-	-	2	-
CO5	1	-	3	2	3	-	-	-	-	-	-	-	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.

- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Microprocessor Architecture, Programming, and Applications with the 8085, By Ramesh Gaonkar, Penram International Publishing (India) Ltd.
2. Exploring C for Microcontrollers by Jivan Parab, Vinod Shelake, Rajanish Kamat, Gourish Naik, Springer
3. The 8051 Microcontroller Architecture by Kenneth Ayala
4. The 8051 Microcontroller and Embedded Systems Using Assembly And C, by M A Mazidi, Janice Mazidi, Rolin Kinlay, Pearson Publication
5. Microcontrollers and PLCs by Sanjay Attri, Dhanpatrai & Co.

Reference Book:

1. Programming and Customizing the 8051 Microcontroller, 1999, by Myke Predko, Tata McGraw Hill
2. 8051 Microcontroller – Internals, Instructions, Programming and Interfacing, 2010, by Subrata Ghoshal, Pearson
3. 8051 Microcontrollers MCS 51 family and its variants, 2010, by S. Shah, Oxford Higher Education
4. User guide for keil(c51) and Related document from: www.keil.com
5. Datasheet Intel8051 Microcontroller, ADC 0809 (Parallel ADC), MAX 1112 (Serial ADC), DAC 0808 (Parallel DAC), ULN2003, ILD74, L293
6. Programming in ANSI C by E. Balagurusamy, McGraw Hill
7. MCS@51 Microcontroller Family User's Manual by INTEL

Web Material:

- A web tutorial on 8051:
http://old.zntu.edu.ua/base/lection/rpf/lib/zhzh03/8051_tutorial.pdf

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE247: TRANSFORMERS & INDUCTION MACHINES
4th Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	4	8	6
Marks	100	100	200	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Electromechanical energy conversion	04
2	Fundamentals of single phase transformer	09
3	Testing and performance analysis of single phase transformers	06
4	Fundamentals of three phase transformers.	07
5	Operation and performance of three phase transformers.	05
6	Construction and operation of polyphase induction machines.	09
7	Testing and analysis of three phase induction machines.	08
8	Starting and speed control of three phase induction machines.	04
9	Single phase induction motor.	08

Total hours (Theory) :60Hrs
Total hours (Lab) :60 Hrs
Total hours : 120Hrs

C. Detailed Syllabus:

- 1 Electromechanical energy conversion 06% 04 Hrs
Basic principles of electro-mechanical energy conversion, Force and torque in magnetic field systems, energy balance, Singly, multiply excited magnetic field system, Energy flow in electro mechanical devices, Force and torque in permanent

magnet system, Energy conversion via electrical field, Dynamical equations of Electromechanical energy systems.

2 Fundamentals of single phase transformer

15% 09Hrs

Introduction; Basics of magnetic circuits, Ampere's law, linear and nonlinear magnetic circuits; Faraday's law of electromagnetic induction; Concept of an ideal transformer, assumptions; ideal transformer at no load and on load, phasor diagram, voltage current and power relations; Transformer on DC; Emf equation, Voltage and current transformation ratio, Effects of voltage and frequency variations, No load current wave shape; Basic construction of a practical single phase transformer and its types, conceptual differences with an ideal transformer; Constructional details of a practical single phase transformer, core, winding, insulation and cooling methods; Circuit model of a practical transformer incorporating an ideal transformer; Analysis of a practical single phase transformer, exact and approximate equivalent circuits, phasor diagram, relation between referred variables and parameters, per unit system of representation; Examples.

3 Testing and performance analysis of single phase transformers. 10% 06Hrs

Open circuit and short circuit tests, per unit parameters; Experimental determination of single phase transformer equivalent circuit parameters, example; Percentage regulation of a single phase transformer, phasor diagram, derivation, per unit representation; example; Efficiency of a single phase transformer, maximum efficiency, all day efficiency; Sumpner's test for experimental determination of a single phase transformer efficiency; Auto transformer; Examples.

4 Fundamentals of three phase transformers.

12% 07Hrs

Bank of three single phase transformers, voltage, current and kVA rating with different connections, open Δ connection; example; Three phase transformer as a single unit; constructional features, three limb, five limb, core type and shell type constructions; comparison with bank of three single phase transformers; Three phase transformer connections, polarity and terminal convention, vector groups; Y/Y and Δ/Δ connections, connection and phasor diagram, per phase equivalent circuit based analysis; Y/ Δ and Δ /Y connections. Connection and phasor diagrams, per phase equivalent circuit based analysis; examples. Y/Z and Δ /Z connections. Connection

and phasor diagrams, voltage, current relations, Phase conversion: 3 to 6 phase and 3 to 2 phase conversions (Scott Connections).

- 5 Operation and performance of three phase transformers. 08% 05Hrs**
Tap changing transformer and voltage control; Parallel operation of single and three phase transformers, conditions and connection diagrams, per phase equivalent circuit based analysis with equal and unequal no load voltages, load sharing; Harmonics and switching transients in transformers, effect of transformer connections, inrush current; Examples
- 6 Construction and operation of polyphase induction machines. 15% 09Hrs**
Introduction, Classification of three phase induction motor, Advantages, Disadvantages and Application of Induction Motor, Production of three phase rotating magnetic field, Locking of stator and rotor field, Construction, Principle of operation, slip, Frequency of rotor current, Speed of rotor field, Rotor EMF, Rotor current, Rotor reactance under starting and running condition, Relation between torque and rotor power factor, Torque equation, Torque under running condition, Torque-Speed curves and effect of change in rotor resistance, Operating region, Starting torque, Full load torque, Breakdown torque, Condition for maximum torque, Effect of change in supply voltage and frequency on torque and slip, Braking of induction motor, examples. Losses in induction machines, Power flow diagram in a three phase induction machine, air gap power, slip power, mechanical power; Torque, Synchronous watt, Efficiency, examples.
- 7 Testing and analysis of three phase induction machines. 14% 08Hrs**
Induction motor as generalized transformer, Phasor diagram of induction motor, Exact and approximate per phase equivalent circuit; Induction motor no load test and blocked rotor test, determining equivalent circuit parameters; losses and efficiency, examples; Concept of circle diagram, Series circuit and current locus, Construction of circle diagram, Induction machine performance computation from circle diagram; examples. Effect of space harmonic fields, harmonics induction torques, Harmonic synchronous torques, Crawling and Cogging, Deep bar and double cage induction motor (High torque motors), Equivalent circuit, Characteristics, examples.

8 Starting and speed control of three phase induction machines. 06% 04 Hrs

Starting of squirrel cage and slip ring induction motor; Direct on line starting, stator resistance and reactance based starting, rotor resistance based starting; Y/ Δ and auto transformer based starting; examples; Speed control of induction motors by stator voltage variation and pole changing techniques, Variable frequency speed control of induction machines, v/f control method, slip power control method.

9 Single phase induction motor. 14% 08Hrs

General description of the motor and constructional features; Rotating and pulsating field, development of equivalent circuit based on double revolving field theory; starting of single phase induction motor, Determination of parameters by test, performance analysis, Types of induction motor: split phase or resistance start, Capacitor-start, Capacitor start capacitor-run, Permanent capacitor, Shaded pole motor: Construction, Working, Starting and running performance, Characteristics and Applications; selection of capacitor value for starting and running conditions, Selection capacitance values for maximum torque, examples.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Get in-depth knowledge in fundamental concepts of electromechanical conversion equipment and transformer
CO2	Acquire the knowledge about the construction details, operating principle, performance characteristics of induction machines
CO3	Acquire the knowledge of testing and applications of induction machines
CO4	Acquire the knowledge about the construction details, performance characteristics, testing and applications of transformers
CO5	Familiarize with the recent developments in transformers and induction machines to get the opportunity to work as a team member/leader in multi-disciplinary branches

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1	1	-	-	-	-	-	-	-	2	2	-
CO2	2	2	2	1	-	-	-	-	-	-	-	2	2	-
CO3	1	2	1	1	-	-	-	-	-	2	-	1	2	-
CO4	2	2	2	1	-	-	-	-	-	2	-	2	2	-
CO5	2	1	-	2	-	-	-	-	3	-	-	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Theory and performance of electrical machines by J.B. Gupta, S.K. Kataria and sons
2. Electric Machinery Fundamentals by Stephen. J. Chapman, McGraw Hill
3. Electric Machinery 6th Edition by A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, Tata McGraw Hill

4. The performance and design of alternating current machines by M.G. Say, CBS Publishers & Distributors
5. A textbook of electrical technology VOL II (AC& DC machines) by B.L.Theraja & A.K. Theraja S. Chand Publication.

Reference Book:

1. Electrical Machines by D.P. Kothari & I. J. Nagrath,
2. Electrical Machinery by Dr. P.S. Bimbhra, Khanna Publisher
3. Electrical Machines by D.P. Kothari & I. J. Nagrath, Tata McGraw Hill
4. Fundamentals of Electrical Machines by B.R. Gupta, New Age International Publishers
5. Electrical machine Drives and Power System by Wildi, 6th Edition, Pearson Publications.
6. Charles .I. Hubert, Electrical Machines (Theory, operation, application, adjustment and control) Pearson Publication, 2nd edition.
7. Theory of Alternating of Alternating Current Machinery by A.S Langsdorf.

Web Material:

- NPTEL
- www.elk.itu.edu.tr/~ozdemir/DC-notes-1
- <http://www.ece.ualberta.ca/~knight/ee332/ee332.htm>
- http://vlabs.iitb.ac.in/vlabs-dev/vlab_bootcamp/bootcamp/Sadhya/experimentlist.html
- <http://em-coep.vlabs.ac.in/List%20of%20experiments.html?domain=Electrical%20Engineering>
- <http://vem-iitg.vlabs.ac.in/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS111.02 A: HUMAN VALUES AND PROFESSIONAL ETHICS

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				
					Theory		Practical		Total
					Internal	External	Internal	External	
IV	HS111.02 A	Human Values & Professional Ethics	02	02	---	---	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

III. Detailed Syllabus:

1.	Introduction to Values and Ethics	5 Hours	17%
	Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics		
2.	Elements and Principles of Values	8 Hours	26%
	Universal & Personal Values, Social, Civic & Democratic Value		
3.	Applied Ethics	8 Hours	26%
	Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics		
4.	Value, Ethics & Global Issues	5 Hours	17%

	Cross-Cultural Issues, Role of Ethics & Values in Sustainability		
5.	Contemporary Issues in Values and Ethics	4 Hours	14%
	Case Studies, Presentations, Projects		

IV. Instruction, Method & Pedagogy:

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, critical reading, group work, independent and collaborative research, presentations, etc.

V. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University Practical Examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment	1	25	25
2	Project Work	1	25	25
3	Quiz	2	10 + 15	25
4	Presentations / Posters / Tasks	1	25	25
	Total			100*
5	Attendance and Class Participation			05
	Total			30

***To be converted to 25.**

External Evaluation

The University Practical examination will be of 70 marks and will test the value system of the students by asking them application based questions. There may also be a Computer Based Test at the end.

Sl. No.	Component	Number	Marks per incidence	Total Marks
---------	-----------	--------	---------------------	-------------

1	Practical Exam / Viva Voce	01	70	70
			Total	70

There may also be a Computer Based Test at the end.

VI. Course Outcome (COs):

At the end of the course, learners will be able to:

At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	2	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	2	2	-	-	-	-	-	-	-
CO5	-	-	-	-	-	2	-	-	-	2	-	2	-	-
CO6	-	-	-	-	-	1	-	3	2	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

VII. Reference Books / Reading:

- Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).

- Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
- Values and Ethics for 21st Century, BBVA.
(https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)
- www.ethics.org

CE291: PYTHON PROGRAMMING SKILLS

B.TECH 4TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – I

(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	2	4	6	4
Marks	100	100	200	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Basics of Python	01
2.	Variables and Data Types	04
3.	Loops and Conditional Control	06
4.	Advanced Data Structures	04
5.	Functions	02
6.	Object Oriented Programming in python	08
7.	Reading and Writing Files	03
8.	Packages and Modular Programming	02

Total Hours (Theory):

30

Total Hours (Lab): 60

Total Hours: 90

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1. | Basics of Python | 01 Hour | 03% |
| 1.1 | Introduction to Python | | |
| 1.2 | Installation and Environment Setup | | |
| 1.3 | Simple Python Program Practice | | |
| 2. | Variables and Data Types | 04 Hours | 13% |
| 2.1 | Variables, Operators, Precedence and Associativity | | |
| 2.2 | Data Types , Indentation, Comments, Reading Input, Print Output | | |
| 2.3 | Memory Optimization | | |

2.4	Type Conversion		
3.	Loops and Conditional Control	06 Hours	20%
3.1	The if... Decision Control Flow Statement, The if...else Decision Control Flow Statement, The if...elif...else Decision Control Statement		
3.2	Nested if Statement, The while Loop, The for Loop,		
4.	Advanced Data Structures	04 Hours	13%
4.1	Lists- Basic List Operators, Replacing, Inserting, Removing an Element, Searching and Sorting Lists, Tuples.		
4.2	Dictionaries- Dictionary Literals, Adding and Removing Keys, Accessing and Replacing Values; Traversing Dictionaries.		
4.3	Sets, Strings, Files and their Libraries		
5.	Functions	02 Hour	07%
5.1	Defining a Function, Calling a Function		
5.2	Pass by Reference vs Value, Function Arguments, Required and Default Arguments, Keyword Arguments and Variable-Length Arguments		
5.3	The Anonymous Functions, The Return Statement		
5.4	Scope of Variables		
6.	Object Oriented Programming in python	08 Hours	27%
6.1	Object-Orientation Introduction, Class Definition, Class Objects, Instance Objects, Method Objects		
6.2	Class and Instance Variables, Inheritance, Private Variables, The Polymorphism		
7.	Reading and Writing Files	03 Hour	10%
7.1	Reading Files, Parsing and Processing Files		
7.2	File storage, File operations (delete, copy)		
8.	Packages and Modular Programming	02 Hour	07%
8.1	Executing modules as Scripts, Import and Use of Built-in Modules, Accessing Third Party Packages and Modules		
8.2	Creating Custom Modules and Packages, Importing and Using Custom Modules		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Practicals/Tutorials related to course content will be carried out in the laboratory.

D. Course Outcomes (COs):

At the end of the course the students will be able to:

CO1	Interpret the fundamental python syntax, semantics and fluent in the use of python control flow statements. (Understand)
CO2	Determine the methods to create and manipulate python programs by utilizing the data structures like lists, dictionaries, tuples and sets. (Apply)
CO3	Express proficiency in the handling of strings and functions. (Apply)
CO4	Articulate the Object-Oriented Programming concepts such as inheritance and polymorphism as used in Python. (Create)
CO5	Identify and use the commonly used operations involving file systems, and regular expressions (Create)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	-	-	-	1	-	1	2	-	-	2	-	-
CO2	-	1	-	2	2	3	2	-	-	-	2	-	2	-
CO3	1	-	2	1	3	2	3	2	2	2	-	2	-	3
CO4	-	1	2	2	3	3	2	3	-	3	3	-	-	-
CO5	1	2	-	2	3	3	2	2	2	-	3	3	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

E. Recommended Study Material:

❖ **Text Books:**

1. Nagar, Sandeep, “Introduction to Python for Engineers and Scientists - Open Source Solutions for Numerical Computation”, Apress, 2018.
2. Kenneth Lambert, “Fundamentals of Python: First Programs”, ISBN-13: 978-1337560092, Cengage learning publishers, first edition, 2012.
3. Magnus Lie Hetland, "Beginning Python from Novice to Professional", Third Edition, Apress, 2017
4. Reema Thareja, “Python Programming using Problem Solving Approach”, ISBN-13:978-0-19- 948017-3, Oxford University Press, 2017.

❖ **Reference Books:**

1. David Beazley, Brian K. Jones, "Python Cookbook", 3rd edition, OREILLY, 2016
2. Brett Slatkin, "Effective Python: 59 Specific Ways to Write Better Python", Novatec, 2016
3. Allen Downey, "Think Python: How to Think Like a Computer Scientist", Green Tea Press, 2015
4. Mark Lutz "Learning Python", 4th Edition, O'REILLY, 2016
5. Vamsi kurama, “Python programming: A modern approach”, ISBN-978-93-325-8752- 6, Pearson, 2018.
6. W. Chun, “Core python programming”, ISBN-13: 978-0132269933, Pearson, 2nd edition, 2016.

❖ **Web Materials:**

1. <https://www.python.org/>
2. <http://www.diveintopython3.net/>
3. <https://developer.mozilla.org/en-US/docs/Learn/Server-side/Django>
4. <https://www.fullstackpython.com/django.html>
5. <https://codelabs.developers.google.com/>
6. <https://www.tutorialspoint.com/python/index.htm>

❖ **Software:**

1. Python IDLE
2. Anaconda Python
3. PyCharm

ME29I: BASICS OF ELECTRICAL VEHICLES
B.TECH 4TH SEMESTER (Electrical)
INSTITUTE ELECTIVE – I
 (For Minor Specialization in “Electric Vehicle Systems”)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Electric Vehicles	14
2.	Automotive Design	14
3.	Design of Electric and Hybrid Electric	15
4.	Regulation and Standardization of Vehicles	02

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

- | | | |
|--------------|--|------------------------------|
| 1 | Introduction to Electric Vehicles | 14Hours 31.11% |
| 1.1 | I.C Engine powered Vehicles Vs Electrical Vehicles (EVs), How EV works, Types of EVs, Hybrid Electric Drive-train | |
| 1.2 | Tractive effort in normal driving, Energy consumption Concept of Hybrid Electric Drive Trains, Architecture of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains, Parallel hybrid electric drive trains, Electric Propulsion unit. | |
|
2 |
Automotive Design |
14Hours 31.11% |
| 2.1 | Major parts of automobiles, Chassis, Frame, Body of automobile, Transmission system, Steering and Breaking systems, Suspension, Accessories. | |
|
3 |
Design of Electric and Hybrid Electric Vehicles |
15Hours 33.34% |

- 3.1 Series Hybrid Electric Drive Train Design: Operating patterns, control strategies, Sizing of major components, power rating of traction motor, power rating of engine/generator, design of PPS
- 3.2 Parallel Hybrid Electric Drive Train Design: Control strategies of parallel hybrid drive train, design of engine power capacity, design of electric motor drive capacity, transmission design, energy storage design
- 4 Regulation and Standardization of Vehicles 02Hours 4.44%
- 4.1 Motor vehicle act, registration of motor vehicles, driving license, control of traffic, insurance against third party, claims for compensation, traffic signs, central motor vehicle rules, vehicle safety standards and regulations, classification and definition of vehicles, enforcement of emission norms, duties of surveyor.

C. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand needs and working of various systems in automobiles.
CO2:	Practically identify different automotive systems and subsystems.
CO3:	Practically identify different automotive components.
CO4:	Understand market and businesses of automobile industry.
CO5:	This course will give the students some awareness to recent trends and research areas in Automobiles.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2:	3	1	1	-	-	-	-	-	1	-	-	1	-	-
CO3:	3	1	1	-	-	-	-	-	1	-	-	2	-	-
CO4:	-	-	-	-	-	-	-	-	-	-	2	1	-	-

CO5:	1	1	1	1	-	-	-	-	1	-	-	1	-	-
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1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial Visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of assignment.

G. Recommended Study Material:

Text Books:

1. Singh Kirpal, “Automobile Engineering” Volume I & II, Standard Pub.& Dist.
2. Gupta K. M. “Automobile Engineering” Volume I & II, Umesh Pub.
3. Gupta R. B. “Automobile Engineering”, Satya Prakashan.
4. Giri N. K., “Automobile Technology”, Khanna Pub.

Reference Books:

1. Crouse W., “Automotive Mechanics”, Tata Mc Graw Hill.
2. Narang G. B. S. “Automobile Engineering”, Khanna Pub.
3. John Lowry and James Larminie , “Electric Vehicle Technology Explained”, Wiley

Web Material:

1. <http://nptel.iitm.ac.in>
2. <http://www.sae.org/pubs/automotive/>
3. <https://www.springer.com/journal/12239/>

Other Material:

1. International Journal of Automotive Technology (www.ijat.net/)
2. International Journal of Automotive Technology

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Th.	Pr.	Total		Int.	Ext.	Int.	Ext.	
TY Sem– V	EE351	Electrical Power Transmission and Distribution	3		3	3	30	70			100
	EE342	Synchronous and DC Machines	4	4	8	6	30	70	50	50	200
	EE353	Power Electronics and Drives – I	3	2	5	4	30	70	25	25	150
	EE344	Minor Project I		4	4	2			50	50	100
	EE345	Electrical Product Survey		2	2	2			50	50	100
	EE350	Summer Internship I		3	3	3			75	75	150
	HS131.02A	Communication and Soft Skills		2	2	2			30	70	100
	EE371-EE375	Advanced Microcontrollers/ VLSI Technology and Design/ Optimization Techniques/ Distributed Generation and Microgrid/ Energy Conservation, Audit and Management	4	2	6	5	30	70	25	25	150
	TOTAL:		14	19	33	27	120	280	305	345	1050
TY Sem - VI	EE346	Digital Signal Processing	3	2	5	4	30	70	25	25	150
	EE347	Programmable Logic Controllers and Industrial Automation	2	4	6	4	30	70	50	50	200
	EE348	Power Electronics and Drives II	3	2	5	4	30	70	25	25	150
	EE349	Fault Analysis and Switchgear	4	2	6	5	30	70	25	25	150
	EE360	Minor Project II		4	4	2			50	50	100
	HS132.02A	Contributor Personality Development		2	2	2	0	0	30	70	100
	EE376-EE381	Special Electrical Machines and Applications/ Embedded Systems/ Power Electronics Applications in Power System/ High Voltage Engineering/ Internet of Things/ Hybrid and Electric Vehicles	3	2	5	4	30	70	25	25	150
	TOTAL:		15	18	33	25	150	350	230	270	1000

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – V)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE351: ELECTRICAL POWER TRANSMISSION AND DISTRIBUTION
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	00	3	3
Marks	100	00	100	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Transmission Line Parameters	10
2	Characteristic and Performance of Transmission Line	10
3	Design of Transmission system	08
4	Design of Power System and Substation	06
5	Design of Distribution system	08
6	Power System Planning	03

Total hours (Theory) : 45 Hrs

Total hours (Lab) : 00 Hrs

Total hours : 45 Hrs

C. Detailed Syllabus:

1 Transmission Line Parameters 22.22% 10 Hrs

Basics of Electric field.

Resistance of line, basic concepts of inductance and capacitance.

Flux linkages of an isolated current carrying conductor.

Inductance of a single phase two wire line, examples.

Flux linkages of one conductor in a group and inductance of composite conductor lines, examples.

Inductance of three-phase line with unsymmetrical spacing and symmetrical spacing, importance of transposition of tower, examples. Inductance of double circuit line, examples. Inductance of bundled conductors and importance of bundled conductors, examples. Electric field of a long straight conductor, potential difference between two conductors of a group of parallel conductors. Capacitance of two-wire line. Capacitance of three-phase line with equal and unequal spacing, examples.

- | | | | |
|----------|---|--------|--------|
| 2 | Characteristic and Performance of Transmission Line | 22.22% | 10 Hrs |
| | Voltage regulation of short transmission line, example. | | |
| | T and π representation of medium transmission line and their Phasor diagram, example. | | |
| | Performance analysis of long transmission line and its ABCD parameter representation. | | |
| | Surge impedance and surge impedance loading, Equivalent T and π representation of long line, Interpretation of long line equation, Ferranti effect. Examples. | | |
| 3 | Design of Transmission system | 17.77% | 08 Hrs |
| | Requirement of Transmission lines, selection of voltage for high-voltage transmission lines. choice of conductors , spacing of conductor, corona. | | |
| | Surge impedance loading of transmission lines. | | |
| | Electrical design of transmission lines, Earth Wires and towers. | | |
| | Transmission of Electrical Power at extra-high voltage. Design consideration of EHV lines. | | |
| | Selection & spacing of conductors for EHV line. | | |
| | Radio & television interference. | | |
| 4 | Design of Power System and Substation | 13.33% | 06 Hrs |
| | Introduction, Selection of size and location of generation stations. | | |
| | Selection of specification of transmission lines. | | |
| | Size and locations of substations. Role of vector group in designing of sub-station. | | |
| | Sub-station Grounding, Earthing, step and touch potential. | | |
| | Bus-bar arrangements in sub-station. | | |
| 5 | Design of Distribution system | 17.77% | 08 Hrs |
| | Distribution system, requirements of distribution system, design consideration of distribution system, classification of distribution system. | | |
| | Connection schemes of Distribution system. | | |
| | D.C. distribution system, Types of distribution system. | | |
| | A.C. distribution system, A.C. distribution system calculations. | | |
| 6 | Power System Planning | 6.66% | 03 Hrs |

Introduction, Methods of Power system planning.

Forecasting load and energy requirements.

Generation Planning , Transmission System planning.

Distribution System Planning.

Reliability of electrical power system, Method of measuring power system reliability.

Trends in power system Planning in India.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Calculate the different parameters of transmission line with different configuration of transmission line & bundled conductor.
CO2	Analyze & calculate the performance of short, medium & long transmission line and power transferred through transmission line.
CO3	Design of EHV transmission line with tower and Demonstrate the knowledge regarding the surge impedance loading of transmission line.
CO4	Chose the size and location of generating station and sub-station. Prepare the specification of transmission line, Construct the arrangement of different parts of sub-station and bus-bar arrangement.
CO5	Design and Connection schemes of A.C. & D.C. distribution system can create the best alternative based on economics.
CO6	Acquire the knowledge of Power system planning for Generation, Transmission & Distribution System.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	-	-	-	-	1	1
CO2	3	3	-	-	-	-	-	-	-	-	-	-	2	1
CO3	2	2	3	-	3	-	-	-	3	3	-	-	3	3
CO4	-	-	3	3	3	3	-	2	3	2	-	-	3	3
CO5	-	3	3	3	-	-	-	1	-	-	-	-	3	3

CO6	-	-	-	2	3	3	2	-	3	3	3	3	3	3
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of nearby substation will clear the concept of transmission line design and student will realise the actual size of transmission towers and spacing of conductors.
- In lab session students have hands on practical about the different configuration of transmission line.

F. Recommended Study Material:

Text Book:

1. Electrical Power System Design, by M. V. Deshpande, Tata Mcgraw Hill.
2. Principles of Power system by V. K. Mehta, Publisher: S. Chand & Company Ltd.
3. A Course in electrical power by Soni & Bhatnagar, Dhanpat Rai & Sons.
4. Sub-station Design and Equipment by P.V. Gupta and P. S. Satnam, Dhanpat Rai & Sons.

Reference Book:

1. Handbook of Electrical Power Distribution, by Ramamurthy University Press (I) Pvt. Ltd
2. Power System Analysis and Design by B.R. Gupta, S. Chand Publications
3. Power system analysis by Hadi Saadat, Tata McGraw Hill Publishing Company, New

Delhi

4. Modern Power System Analysis by D.P. Kothari & I. J. Nagrath, Tata Mcgraw Hill
5. Power System Analysis by Grainger & Stevenson, Tata Mcgraw Hill

Web Material:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPUR/powersystem/ui/TOC.htm>
- http://en.wikipedia.org/wiki/Electric_power_transmission

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE342: SYNCHRONOUS AND DC MACHINES

5th Semester and 3rd Year (Level III)

B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	4	8	6
Marks	100	100	200	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Fundamentals of DC Machines	12
2	Operation and performance Characteristics of DC Machine	12
3	Construction and Operation of Synchronous Generator	12
4	Voltage Regulation, Power Equations and Synchronization of Alternators	14
5	Synchronous Motor	10

Total hours (Theory) :60Hrs

Total hours (Lab) :60 Hrs

Total hours : 120Hrs

C. Detailed Syllabus:

1 Fundamentals of DC Machines 20% 12Hrs

Classification of rotating electrical machines, Construction of DC machines, Comparison of AC and DC Generators, Principle of DC generator and motor, Comparison of motor and generator action, Elementary DC generator, essential parts,

Types of DC machines (methods of excitation), EMF equation, Voltage build up process, Causes of its failure, Critical resistance and speed, Armature reaction and its effect, Commutation in DC machines, Reactance voltage, methods of improving commutation, compensating winding.

- 2 **Operation and performance Characteristics of DC Machine.** 20% 12Hrs
Characteristics of separately excited DC machines (motor and generator), Characteristics of shunt, series and compound machines (generators & motors), Power flow diagram of DC machines, Losses in DC machines, Efficiency, condition for maximum efficiency, Voltage regulation in DC generator, Speed regulation in DC motor, Parallel operation of DC machines, Examples, Necessity of starter, types of starter, Speed control of DC shunt and series motors, Braking of DC motors, Examples, Applications of all types of DC generators and DC motors.
- 3 **Construction and Operation of Synchronous Generator** 20% 12Hrs
Introduction, classification of synchronous machine, details of construction, advantages of rotating field type of construction, difference between salient and cylindrical alternator, damper winding, operating principle, Speed and Frequency, Synchronous speed, pitch factor, distribution factor and emf equation, emf waveform for concentrated and distributed winding, Examples, armature leakage reactance, synchronous reactance and impedance, armature reaction, Equivalent circuit, phasor diagram, Synchronous generator operating alone, operation of alternator on no load and on load, effect of variation in load on synchronous generator, Laboratory methods of determination of synchronous reactance, oc/sc test, determination of effective resistance of armature, Short circuit ratio, Losses and efficiency, power flow diagram, Examples,
- 4 **Voltage Regulation, Power Equations and Synchronization of Alternators** 23% 14Hrs
Voltage Regulation, methods of calculating voltage regulation, Direct load test, EMF method, MMF method, ZPFC method, Comparison of methods, examples, Active, Reactive and Complex Input and Output Power Equations for cylindrical synchronous generator, Maximum input and output power and torque developed,

Power Angle Characteristics, Two reactance theory for salient pole synchronous machine, Phasor Diagram, Power Equations and Torque Angle Characteristics, Determination of X_d and X_q by slip test, Capability Curves for Synchronous Generators, Prime mover characteristics, House Diagram, Frequency Power and Voltage Reactive Power relations (F-P and Q-V), Power shared by two alternators, Necessity of parallel operation, condition required for parallel operation, the general procedure for synchronization of alternators, Infinite Bus, Synchronizing current, power, torque, Time period of Oscillations, effect of increasing the driving torque of one the alternator, effect of change in excitation of one of the alternator, load sharing between two alternators, operation of synchronous generator with infinite bus bar, operation of synchronous generator with other generator, various types of alternators based on prime movers, Factors affecting the size and efficiency of alternators, Transients of the alternator

5 Synchronous Motor

17% 10Hrs

Introduction, construction, principle of operation, Equivalent circuit and phasor diagram for cylindrical and salient rotor, synchronous motor with different excitation, starting methods of synchronous motor, power flow equations for synchronous motor, examples, steady state synchronous motor operation - effect of load change on synchronous motor, effect of field current change on synchronous motor, power flow diagram, examples, Different torques of synchronous motor, stability and maximum load angle, construction of V curves and inverted V curves, O curves, Synchronous condenser, synchronous phase modifier, hunting, speed control of synchronous motor, merits, demerits and application. Comparison of three phase synchronous and induction motors

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Able to recognize the electric and magnetic circuits of AC and DC machines.
CO2	Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines.
CO3	Develop skill for the practical exposure of rotating electrical machines and understand the various testing methods of Synchronous and DC machines.

CO4	Aware with the performance characteristics of Synchronous and DC machines.
CO5	Analyze Synchronous and DC machines performance and application various conditions.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	1	-	-	-	-	-	-	-	-	-
CO2	2	3	1	1	1	-	-	1	-	-	-	-	-	-
CO3	3	2	3	2	1	-	-	-	-	-	-	2	2	1
CO4	-	3	2	2	-	-	-	-	1	-	-	-	-	-
CO5	2	3	2	3	2	-	-	-	-	-	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Electrical Machines, by Ashfaq Husain, Dhanpat Rai & Co.
2. A textbook of Electrical Machines by K R Siddhapura and D B Raval, Vikas Publishing House Pvt. Ltd.
3. Theory and performance of electrical machines by J.B.Gupta, S.K.Kataria and sons
4. A textbook of electrical technology VOL II (AC & DC machines) by B.L.Theraja & A.K.Theraja S. Chand Publication.

Reference Book:

1. Electrical Machines III by K Krishna Reddy, Scitech Publication (INDIA) Pvt. Ltd.
2. Electric Machinery Fundamentals by Stephen. J. Chapman, McGraw Hill
3. Electric Machinery 6th Edition by A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, Tata McGraw Hill
4. The performance and design of alternating current machines by M.G.Say, CBS Publishers & Distributors
5. Electrical Machinery by Dr. P.S.Bimbhra, Khanna Publisher
6. Electrical Machines by D.P. Kothari & I. J. Nagrath, Tata McGraw Hill
7. Fundamentals of Electrical Machines by B.R. Gupta, New Age International Publishers
8. Charles .I. Hubert, Electrical Machines (Theory, operation, application, adjustment and control) Pearson Publication, 2nd edition.
9. Theory of Alternating of Alternating Current Machinery by A.S Langsdorf.

Web Material:

- <http://nptel.ac.in/courses/108106072/>
(video lectures Prof.D. Kastha, IIT Kharagpur)
- http://nptel.iitm.ac.in/courses/IIT-MADRAS/Electrical_Machines_I
- www.nptel.ac.in/courses/IIT-MADRAS/Electrical_Machines_II/pdf/2_1.pdf
- <http://www.ece.ualberta.ca/~knight/ee332/ee332.htm>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE353: POWER ELECTRONICS AND DRIVES - I
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction	01.50
2	Power Semiconductor Devices	13.00
3	Triggering, Commutation and Driver Circuit	07.50
4	DC-DC Converters (Choppers)	07.50
5	AC-DC Converters (Controlled Rectifier)	07.50
6	DC Motor Drives	08.00

Total hours (Theory) : 45 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 75 Hrs

C. Detailed Syllabus:

- 1 Introduction 03.33% 1.5 Hrs

Introduction, Scope and applications, Classification of power electronic converters.

- 2 Power Semiconductor Devices 28.33% 13 Hrs

Diode: Classification and its application

SCR: V- I characteristics, Switching characteristics, Gate characteristics, Ratings, Effect of dv/dt and di/dt, Snubber circuit and its design

Thyristor Family: Triac, DIAC, LASCR, SUS, SCS

Gate commutated devices: GTO, MOSFET and IGBT

- 3 Triggering, Commutation and Driver Circuit 16.67% 7.5 Hrs

Triggering circuits: Resistance triggering circuit, Resistance capacitance trigger circuit, DIAC trigger circuit, UJT based trigger circuit; Micro-controller based Trigger circuit

Commutation circuits: Line commutation, Load commutation, Forced commutation, External pulse commutation

Driver Circuit: Optocoupler and pulse transformer based SCR driver circuit, Gate drive circuit for power MOSFET, Driver circuit for IGBT and BJT

- 4 DC-DC Converters (Choppers) 16.67% 7.5 Hrs

Introduction, Principle of chopper operation, Control strategies, Step down (buck) converter, Step up(boost) converter, Step up down converter, Cuk DC-DC converter, Chopper configuration

- 5 AC-DC Converters (Controlled Rectifier) 16.67% 7.5 Hrs

Introduction, Principle of phase control, Single phase half wave converter with different loads, Single phase full wave converter with different loads, Single phase semi converter, Three phase half wave converter with different loads, Three phase full converter, Three phase semi converter with different loads, Dual converter, Multi-pulse Rectifier

- 6 DC Motor Drives 18.33% 08 Hrs

Concept of Electric Drive, Basic DC Machine Equations and its characteristics, Control of DC separately excited motor from single phase and three phase controlled rectifier, Rectifier control of DC series motor, Chopper control of separately excited DC motor, Chopper control of DC series motor, Multi quadrant operation of, DC separately excited motor

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Enhance their skills in basics of Power Electronic devices and converters.
CO2	Know the performance and classification of Power Electronic converters.
CO3	Understand applications of converters used in industrial drives.
CO4	Design and troubleshoot hardware circuits like choppers, Inverters, Controlled rectifiers etc.
CO5	Hardware implementation of Power Electronic Converters and its control strategies.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	2	3	2	-	-	-	-	-	-	-	-	-	-	-
CO4	-	3	3	-	2	-	-	-	-	-	-	-	2	3
CO5	-	3	-	-	3	-	-	-	2	-	-	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Power Electronics by P. S. Bimbhra, Khanna Publishers
2. Power Electronics Circuits, Devices and Applications by Muhammad H. Rahid, Pearson Education
3. Power Electronics by M.D. Singh and K. D. Khanchandani, Tata McGraw-Hill Education

Reference Book:

1. Fundamentals of Electrical Drives by Gopal K. Dibey, Alpha Science.
2. Power Electronics Converters, Applications and Design by Ned Mohan, Tore M Undeland and William P. Robbins, Wiley India.
3. Power Electronics and Motor Drives by Bimal K. Bose, Academic Press.
4. Power Electronics: Devices, Drivers, Applications, and Passive Components by B. W. Williams, Tata McGraw-Hill

Web Material:

- <http://www.nptel.ac.in/courses/108105066/>
(Notes from IIT Kharagpur)
- <http://www.nptel.ac.in/courses/108101038/1>
(Videos and Notes by Prof. B.G. Fernandes, IIT Bombay)
- <http://www.nptel.ac.in/courses/108108035/1>
- <http://www.nptel.ac.in/courses/108108036/>
(Videos and Notes by Prof. L. Umanand, IIT Bangalore)
- <http://www.nptel.ac.in/courses/108102046/>
- <http://www.nptel.ac.in/courses/108108077/>
(Videos and Notes by Prof. K. Gopakumar, IISc, Bangalore)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE344: MINOR PROJECT I
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	04	4	2
Marks	00	100	100	

B. Instructional Method and Pedagogy:

- Minor project-I has 2 credit subject and 4 hours per week.
- Department/Institute will help students to find appropriate minor projects.
- The student must be find the appropriate guide for their project and make the project
- proposal and submit the project proposal to departmental co-coordinator.
- Project must be from the electrical bag round other project must be rejected.
- Student must meet at least once in week to his/her guide.
- The student shall submit two documents to the Coordinator for the evaluation of the minor project:
 - Project report
 - Project presentation
- Upon the completion of project, a hard copy of “project Report” must be submitted to the Coordinator by the end of the semester.
- The report must outline with practical knowledge which you gain during project making. Report must be verified by the guide.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the project.

C. Format of Project Report:

Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12 pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done during project, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- INTRODUCTION: In this section, give the purpose of the project, reasons for choosing the project, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the project . It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the project. Present your observations, contributions and intellectual benefits.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet

link and document number. In this manner you don't have to print it and waste tons of paper.

D. Course Outcome (COs):

After completion of the course students will able:

CO1	To apply knowledge and skills learned from project to real-world problems.
CO2	To solve engineering problems by the team work and work with other discipline.
CO3	Use experience related to professional and ethical issues in the work environment
CO4	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.
CO5	To implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE345: ELECTRICAL PRODUCT SURVEY
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	02	02	2
Marks	00	100	100	

B. Instructional Method and Pedagogy:

- Electrical product survey has 2 credit subject and 2 hours per week.
- Students can find his/her electrical product.
- The student may take the guidance from department faculty members.
- Product must be from Electrical bag round.
- The student shall submit product presentations to the Coordinator for the evaluation of the product survey.
- After the completion of product survey, a soft copy of “presentation” must be submitted to the Coordinator by the end of the semester.
- The students must make a group of 3 or 4 and submit their proposal to departmental coordinator.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision. If the revised presentation is still unsatisfactory the student shall be requested to repeat the project.

C. Course outcomes (COs):

After completion of the course students will able:

CO1	To apply knowledge and skills learned from project to real-world engineering problems.
CO2	To function in a team work with teammates from other disciplines.
CO3	To use experience related to professional and ethical issues in the work

	environment.
CO4	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context.
CO5	To demonstrates knowledge of contemporary issues related with engineering in general with the use new tools and technologies.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO2	1	-	1	1	2	-	-	-	3	-	-	-	-	1
CO3	1	-	-	-	-	1	-	2	-	-	-	-	-	1
CO4	1	-	-	-	-	-	2	-	1	-	2	-	1	1
CO5	3	1	-	1	-	-	-	-	-	-	-	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE350: SUMMER INTERNSHIP I
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week (In Summer Break)	00	30	90	3
Marks	00	150	150	

B. Instructional Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.

- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

C. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division

- Administrative tree (if available)
- Main functions of the company
- Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

D. Course Outcome (COs):

After completion of the course students will able:

CO1	To solve engineering problems by working in a team with across other disciplines.
CO2	To apply knowledge and skills learned in company/industry/organization to real-world problems.
CO3	To use experience related to professional and ethical issues in the work environment

CO4	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context
CO5	To demonstrates knowledge of contemporary issues related with engineering in general by adopting new tools and technologies

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS131.02A: COMMUNICATION AND SOFT SKILLS

I. Credits and Schemes

Sem	Course Code	Credits	Teaching Scheme	Evaluation Scheme				
			Contact Hours/Week	Theory		Practical		Total
				Internal	External	Internal	External	
V	HS131.02 A	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

III. Detailed Syllabus:

1.	An Introduction to Communication	06 Hours	20%
	Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		
2.	Cross-cultural Communication and Globalization	03 Hours	10%
	Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication		
3.	Communication for Career Building	10 Hours	33%

	Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose		
4.	Group Dynamics and Soft Skills	05 Hours	17%
	An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset		
5.	Effective Presentation Strategies	04 Hours	14%
	Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion		
6.	Contemporary Issues in Communication and Soft Skills	02 Hours	06%
	Trends and Practices in Communication, Case Studies		

IV. Pedagogy

Teaching will be facilitated by reading material, discussions, task-based learning, projects, assignments and interpersonal activities like group work, independent and collaborative study projects and presentations, etc.

V. Evaluation

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment / Tutorial	01	25	
2	Project	01	25	
3	Quiz	02	25	
4	Case Study and Presentation	01	25	100*
5	Attendance and Class Participation			05
Total				30

*** Marks will be converted into 25**

External Evaluation

The University Examination will be for 70 marks and will be based on practical (computer based) tests and a viva-voce. There may also be a Computer Based Test at the end.

Details are:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical / Viva	01	70	70
Total				70

VI. Course Outcomes (COs)

At the end of the course, the students will be able to

CO1	Gain thorough understanding and proficiency in various Professional Communication Skills.
CO2	Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
CO3	Develop business writing and presentation skills to succeed in career.
CO4	Develop soft skills to stand out and take their career to the next level.
CO5	Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	3	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

VII Preference

It will be preferred that students take globally recognized tests by Cambridge, British Council or any other equivalent agency. Equivalent marks shall be considered in the University Exam.

VIII Recommended Study Material:

❖ Text book:

1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
3. Raman, M & Singh, P. Business Communication, Oxford University Press

❖ Reference book:

1. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited

❖ Web material:

1. <https://www.coursera.org/learn/careerdevelopment>
2. <https://www.futurelearn.com/courses/writing-applications>
3. <https://www.futurelearn.com/courses/workplace-englis>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE371: ADVANCED MICROCONTROLLERS
PROGRAMME ELECTIVE II
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. rOutline of the Course:

Sr. No.	Title of Units	Number of Hours
1.	Embedded C Language Programming for Microcontroller	12
2.	Introduction to c8051f120 Microcontroller	08
3.	Timers in c8051f120 Microcontroller	12
4.	Analog Peripherals of c8051f120 Microcontroller	12
5.	PCA & Application	10
6.	Communication Protocol	06

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

C. Detailed Syllabus:

1.	Embedded C Language Programming for Microcontroller	20%	12 Hrs
	Comparison of Assembly & High Level Programming, Concept of Super loop/Infinite loop used in Embedded C Programs, Concept of Round Robin Execution in Real Time Operating System based Programming Environment. Introduction to Keil Microvision IDE & Related debugging Techniques, Basic Working of various Control Loops, Variables: Global & Local, Various types of Variables & it's Memory Scope, Mathematical & Logical Operators, Array, Structure & Union, Pointers		

	and its use in programming. Various C library and its use in programming.		
2.	Introduction to c8051f120 Microcontroller	13%	08 Hrs
	Comparison of 8051 & CIP-51 core Microcontrollers, Peripherals available in c8051f120 Microcontroller, Watchdog Timer & it's Reset codes, System Clock Configuration with & without Internal PLL, SFR & SFR Paging, Concept of Crossbar & Crossbar Decoder, Internal Structure & Configuration of Port in Open drain & Push pull mode with & without Weak pull up disabled.		
3.	Timers in c8051f120 Microcontroller	20%	12 Hrs
	Block diagram & Operation of Timer0 & 1 with Associated SFR, Calculation of input clock frequency for various SYSCLKOUT, Various Modes of Timer & Associated Programs With & Without Interrupt enabled. Block diagram & Operation of Timer2,3 & 4 with Associated SFR. Various modes of Operations & Associated Programs.		
4.	Analog Peripherals of c8051f120 Microcontroller	20%	12 Hrs
	Basic Terminology for A/D Conversions, Specifications of c8051f120 A/D Converters: ADC0 & ADC2, Block diagram & Working of 12 bit ADC0 & 8 bit ADC2 A/D Converters, Associated SFR, Start of Conversion using Various modes, ADC0 in Window Comparator Mode, Temperature sensing using On-chip Temperature Sensor, Programs Related to ADC Operation, Digital to Analog Converter in c8051f120 with various mode of conversion, Generation of waveforms using DAC, On-chip Comparator.		
5.	PCA and Application	17%	10 Hrs
	Block diagram of PCA module, PCA Timer in Capture Compare, High Speed output, Frequency Output & 8/16 Bit Pulse Width Modulation Mode.: Associated Register & Working, Associated Programs.		
6.	Communication Protocol	10%	06 Hrs
	SPI, SMBUS and I2C, UART Communication.		

D. Course Outcomes (COs):

At the end of this course, the student should:

CO1	To understand C programming fundamentals in detail, and be able to differentiate
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	between super loop based and RTOS execution, and to understand fundamentals of programming in both approaches.
CO2	Have a good understanding of the architecture and programming model of the c8051f120 microcontroller and to write C program for using various internal modules and driving external hardware.
CO3	Be aware of software/code debugging and related techniques and also to rectify it and also to think about faults in hardware and hardware design and to solve it.
CO4	To be able to access peripherals in 32 bit microcontroller using pointer, structure, union and write pseudo code with registers for modules available in unknown (Which is not directly covered in course) microcontroller.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	3	1	-	2	-	-	-	-	2	-
CO2	3	3	3	-	3	1	-	-	2	2	2	-	2	-
CO3	3	3	-	-	3	2	-	-	-	-	-	-	2	-
CO4	3	3	-	-	3	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials/ Assignments based on course content will be conducted/ given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.

- The course includes a laboratory, where students have an opportunity to build and appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

- [1] “Embedded System Design using C8051” by Han-Way Huang, Cengage Engineering, 2009. ISBN-13: 9788131512241.
- [2] “Introduction to Embedded Systems” by Shibu K V, Tata McGraw Hill.

Reference Book:

- [1] “Microprocessors & Interfacing”, Douglas Hall, Tata McGraw Hill
- [2] The 8051 Microcontroller and Embedded Systems using Assembly and C by Muhammad Ali Mazidi, Pearson
- [3] Exploring C for Microcontrollers: A Hands on Approach, Jivan S. Parab, Vinod G. Shelake, Springer

Web Material:

- [1] http://www.keil.com/dd/docs/datashts/cast/cast_c8051.pdf (Data sheet of C8051)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE372: VLSI TECHNOLOGY AND DESIGN
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction and fabrication of MOSFET	06
2	MOS Transistor	10
3	MOS Inverters: static characteristic and switching characteristic	16
4	MOS combinational circuits	08
5	MOS sequential logic circuits	08
6	MOS dynamic logic circuits	08
7	Chip input and output	04

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

C. Detailed Syllabus:

1	Introduction and fabrication of MOSFET	10%	06 Hrs
	VLSI Design Flow, Design hierarchy, Design Methodology, nMOS, pMOS, CMOS fabrication process.		
2	MOS Transistor	17%	10 Hrs
	Metal Oxide Semiconductor (MOS) structure, MOS System under external bias, Structure & Operation of MOS transistor, MOSFET Current-Voltage characteristics, MOSFET scaling & small-geometry effects, MOSFET capacitances.		

3	MOS Inverters: static characteristic and switching characteristic	27%	16 Hrs
	Introduction, Resistive load Inverter, Inverter with n-type MOSFET load (Enhancement & Depletion type MOSFET load), CMOS Inverter, Delay-time definitions, Calculation of Delay times, Inverter design with delay constraints, Estimation of Interconnect Parasitic, Calculation of interconnect delay, Switching Power Dissipation of CMOS Inverters.		
4	MOS combinational circuits	13%	08 Hrs
	Introduction, MOS logic circuits with Depletion nMOS Loads, CMOS logic circuits, Complex logic circuits, CMOS Transmission Gates (TGs).		
5	MOS sequential logic circuits	13%	08 Hrs
	Introduction, Behaviour of Bistable elements, The SR latch circuit, Clocked latch & Flip-flop circuit, CMOS D latch & Edge-triggered flip-flop.		
6	MOS dynamic logic circuits	13%	08 Hrs
	Principles of pass transistor circuits, Voltage Bootstrapping, Synchronous Dynamic Circuit Techniques, CMOS Dynamic Circuit Techniques , High performance Dynamic CMOS circuits.		
7	Chip input and output	7%	04 Hrs
	On chip Clock Generation and Distribution, Latch –Up and its Prevention		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Identify the recent trends in VLSI Technology and Design methodologies.
CO2	Understand physics of MOSFET and hence apply this in analysis of basic building block of digital integrated circuits.
CO3	Design combinational and sequential circuits using various topologies that realizes myriad digital function of given specification.
CO4	Understand dynamic logic concept and implement this to design high density and high performance digital circuits.
CO5	Learn latest CAD/EDA/Verilog & Xilinx ISE suite tools used in industry for design of digital integrated circuits.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	2	-	-	-	-	-	-	-	-	-	-	-	-
CO3	1	2	3	1	-	-	-	-	-	-	-	-	1	-
CO4	2	1	2	1	-	-	-	-	-	-	-	-	1	-
CO5	-	-	-	1	3	-	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Sung-Mo-Kang, UsufLeblebici ,CMOS digital integrated circuits: Analysis and Design, Tata McGraw-Hill Education.

Reference Book:

1. Douglas Pucknell, Basic VLSI Design, PHI, 1999.
2. Wayne Wolf , Modern VLSI Design., Person Education, 2001.
3. John Uyemura ,Introduction to VLSI circuits and systems, Wiley, 2002.

Web Material:

- <http://www.eng.auburn.edu/~strouce/elec4200.html>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE373: OPTIMIZATION TECHNIQUES (PROGRAMME ELECTIVE -1)
5th Semester and 3rd Year (Level III)
B.Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Fundamentals of Optimization	05
2	Linear Programming	12
3	Unconstrained optimization	13
4	Nonlinear Programming	15
5	Least squares based methods	15

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

C. Detailed Syllabus:

- 1 **Fundamentals of Optimization** 8.33% 05 Hrs
Introduction, Feasibility and optimality, Convexity, constraints, Rates of convergence.
- 2 **Linear Programming** 20% 12 Hrs
Introduction, Formulation of Linear programming problem, Graphical method, Simplex method, Basic solution, Basic feasible solution, Simplex algorithm, Big M method (Penalty method), Two phase method, Special cases in simplex method application. Economic load dispatch using Linear programming method.
- 3 **Unconstrained optimization** 21.66% 13 Hrs

Introduction, Optimality conditions, Newton's method for minimization, Line search methods, Steepest-Descent method, Quasi-Newton method.

4 Nonlinear Programming 25% 15 Hrs

Optimality conditions for constrained problems, Kuhn Tucker conditions, Penalty function method, Barrier method, The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, Computing the Lagrange multipliers, Sequential quadratic programming, Interior point method. Active and reactive power optimization using NLP.

5 Least squares based methods 25% 15 Hrs

Introduction, Integral LSQ fit, Basic assumptions, Determination of unknown coefficients, Power series LSQ fit, Assumptions, Shifted waveform, Approximating the shifted waveform by a power series, Multi-variable series LSQ technique, Derivation of the multi variable series.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	At the end of course, the students will understand basics and engineering applications of optimization.
CO2	Develop skill for solving any problems using classical optimization techniques.
CO3	The students will understand different methods of unconstrained and least squares based optimization for solving real world problem.
CO4	The students will be able to solve nonlinear optimization problems.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	1	-	-	-	-	-	2	-	-	-	-
CO2	3	2	3	-	2	-	-	-	1	-	-	1	1	3
CO3	3	3	2	-	2	-	-	-	1	-	-	1	1	3
CO4	3	3	2	-	2	-	-	-	1	-	-	1	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the end of the course, the students will acquire the knowledge regarding various classical optimization methods.
- They will be able to solve various real world optimization problems and they can apply those methods to solve optimization problems of Electrical power systems.

F. Recommended Study Material:

Text Book:

1. S.S.Rao, Engineering optimization: Theory & Practice, 3rd Ed., John Wiley & Sons, Inc.
2. Stephen G. Nash and Aridla Sofer, Linear and nonlinear programming, McGraw Hill
3. K. Deb, Optimization for engineering design, PHI

Reference Book:

1. G.K. Ranganath, A textbook of computer oriented numerical methods and linear programming, S Chand.
2. A. Ravindran, K. M. Ragsdell, G. V. Reklaitis, Engineering Optimization: Methods and Applications, 2nd Ed., John Wiley & Sons, Inc.
3. M. Avriel, Ron S. Dembo, Engineering Optimization
4. Narayan Rao, Optimization Principles, John Wiley & Sons.

Web Material:

1. <https://nptel.ac.in/courses/111/105/111105100/>
2. <https://nptel.ac.in/courses/105/108/105108127/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE374: DISTRIBUTED GENERATION AND MICROGRID (PE-1)
5th Semester and 3rd Year (Level III)
B.Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction	08
2	Basic of Distributed Generation	12
3	Impact of Grid Integration	16
4	Basic of Microgrid	08
5	Control and Operation of Microgrid	16

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

C. Detailed Syllabus:

1 Introduction 23.33% 08 Hrs

Conventional power generation: advantages and disadvantages, Energy crises, Non-conventional energy (NCE) resources: review of Solar PV, Wind Energy systems, Fuel Cells, micro turbines, biomass, and tidal sources.

2 Basic of Distributed Generation (DG) 20% 12 Hrs

Concept of distributed generations, topologies, selection of sources, regulatory standards/framework, Standards for interconnecting Distributed resources to electric power systems: IEEE 1547. DG installation classes, security issues in DG implementations. Energy storage elements: Batteries, ultra-capacitors, flywheels.

Captive power plants.

- 3 **Impact of Grid Integration** 23.33% 16 Hrs
Requirements for grid interconnection, limits on operational parameters: voltage, frequency, THD, response to grid abnormal operating conditions, islanding issues. Impact of grid integration with NCE sources on existing power system: reliability, stability and power quality issues..
- 4 **Basic of Microgrid** 13.33% 08 Hrs
Concept and definition of microgrid, microgrid drivers and benefits, review of sources of microgrids, typical structure and configuration of a microgrid, AC and DC microgrids, Power Electronics interfaces in DC and AC microgrids.
- 5 **Control and Operation of Microgrid** 20% 16 Hrs
Modes of operation and control of microgrid: grid connected and islanded mode, Active and reactive power control, protection issues, anti-islanding schemes: passive, active and communication based techniques, microgrid communication infrastructure, Power quality issues in microgrids, regulatory standards, Microgrid economics, Introduction to smart microgrids.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Review the conventional power generation.
CO2	Analyze the concept of distributed generation and installation.
CO3	Design the grid integration system with conventional and non-conventional energy sources.
CO4	Get the knowledge regarding design of the DC and AC micro grid.
CO5	Analyze power quality issues and control operation of micro grid.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	2	-	2	-	-	1	-	-	3	-
CO2	3	2	1	-	-	-	-	-	1	-	-	-	2	-

CO3	3	2	3	3	2	-	-	-		-	-	-	2	2
CO4	3	3	1	2	-	-	2	-	1	-	-	-	2	2
CO5	3	-	2	-	2	-	-	-	1	2	2	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. N. Hatziaargyriou, Microgrids: Architectures and Control, Wiley-IEEE Press, 1st Edition, 2014
2. Loi Lei Lai, Tze Fun Chan, Distributed Generation- Induction and Permanent Magnet Generators, IEEE Press, John Wiley & Sons, Ltd., England. 2007.
3. Amirnaser Yezdani, and Reza Iravani, Voltage Source Converters in Power Systems: Modeling, Control and Applications, IEEE John Wiley Publications, 2009.

Reference Book:

1. N. Twidell & A. D. Weir, Renewable Energy Sources, University press ,Cambridge, 2001.
2. James Larminie , Andrew Dicks , Fuel Cell Systems, John Wiley & Sons Ltd, 2000

3. J. F. Manwell , J. G. McGowan, A. L. Rogers , Wind Energy Explained, John Wiley & Sons Ltd 2009.

Web Material:

3. <https://nptel.ac.in/courses/108/107/108107143/>
4. <https://nptel.ac.in/courses/108/101/108101005/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE375: ENERGY CONSERVATION, AUDIT AND MANAGEMENT (PE IV)
5th Semester and 3rd Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Electrical Energy Conservation	13
2	Energy Management	15
3	Energy storage system	13
4	Energy Audit	11
5	Energy Efficient Technologies In Electrical Systems	08

Total hours (Theory) :60Hrs

Total hours (Lab) :30 Hrs

Total hours : 90Hrs

C. Detailed Syllabus:

1 Electrical Energy Conservation 22% 13Hrs

Introduction to energy science and energy technology, various forms of energy. Law of conservation of energy. Primary and Secondary energy, Commercial and Non-commercial energy, Renewable and Non-Renewable energy. Carbon credit.

Energy conservation act-2001 and its features, key definitions as outlined in the Act, Schemes of BEE under the energy conservation Act, 2001.

National Mission for Enhanced Energy Efficiency.

Energy scenario of India, Energy conservation and its importance.

Load Management and Maximum demand control.

Potential energy conservation opportunities in:

- (1) HVAC System
- (2) Lighting systems
- (3) Motors and Transformers

2 Energy Management 25% 15Hrs

Definition and objective of energy management, Responsibilities and Duties of energy manager, Management tools for effective Implementation like 5S, Kaizen, Small group Activity, TPM, TQM & ISO 50001.

Electrical System Optimization: Electricity rate Tariff, Key to reduction in electrical energy consumption, methods to improve plant power factor, Load Management, Conduction Loss, Switching Loss, Magnetic Loss.

Cogeneration system

Intelligent buildings

Energy planning, Energy staffing, Energy Organization, Energy Requirement, Energy Costing, Energy Budgeting, Energy Monitoring, Energy consciousness, Energy Management Professionals, Environment pollution due to energy use.

3 Energy Storage system 22% 13Hrs

Energy storage: Demand for energy storage, Classification of Energy storage systems, Mechanical energy storage: Pumped hydro, Compressed Air, Flywheels. Electrochemical energy storage: secondary batteries, Flow batteries. Chemical energy storage: Hydrogen, Synthetic natural gas. Electrical and magnetic energy storage systems.

4 Energy Audit 18% 11Hrs

Introduction, Definition, Need for Energy Audit, Types and Procedures for energy audit. Preliminary energy audit, detailed energy audit, Industrial Audit approaches. Procedure for energy audit and equipment's required. Energy Audit Report, Benchmarking,

Electricity Billing for LT and HT consumer.

5 Energy Efficient technology In Electrical system 13% 08Hrs

Automatic Power Factor Controllers, Energy Efficient Motors, Variable Speed Drives and variable frequency Drives. Soft Starter, Energy Efficient Transformers,

Electronic Ballasts, Energy Efficient Lighting. Electrical Furnaces.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Understand and implement of Energy conservation Technics as per efficient technology available in Market also suggest the energy conservation technique that will be most suitable for required application.
CO2	The students will be well aware with the minimization of losses, standard benchmarking, combine generation of heat and power system .Also can able to calculate the payback period of any system installed.
CO3	Student will understand about different kind of storage system and their process.
CO4	Develop skill to carry out Procedure for energy audit of any industry and Analysis of Electrical System.
CO5	Student can apply knowledge on Energy efficient technology for different appliance available in Market.
CO6	Acquire the knowledge regarding the energy management, audit & conservation in any electric system.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	3	2	3	-	2	-	3	3	2	2	3	2
CO2	2	3	-	-	-	-	-	-	-	-	-	3	3	-
CO3	1	-	1	-	-	2	-	-	-	-	-	-	1	-
CO4	3	-	3	-	3	-	-	3	3	3	3	3	3	3
CO5	2	-	-	3	-	2	-	-	-	3	-	3	3	3
CO6	2	-	1	-	-	-	2	-	-	-	-	3	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures, which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room, chiller plant and nearby substation will clear the concept of load calculation and student will realise the actual Audit process.
- The course shall include a project work, where students have an opportunity to find out loop holes of wastage of energy and can suggest best remedies for the same.
- Industrial visit to be arranged for practical knowledge and exposure.
- In lab session, the student will understand about practical audit of system and energy management and conservation technology in industries.

F. Recommended Study Material:

Text Book:

1. Amlan Chakrabarti, Energy engineering and management, PHI Learning Private Limited.
2. K. Nagabhusan Raju, Industrial Energy Conservation Techniques, Atlantic Publishers & Distributors (P) Ltd.

Reference Book:

1. Renewable energy sources and conservation technology By- N. K. Bansal, Kleemann and Meliss
2. Non – conventional energy sources by G. D. Rai
3. Energy technology by S. Rao.

4. A guide to energy management by Barney L Capehart, William J Kennedy, Wayne C Turner.

Web Material:

1. www.bee-india.gov.in
2. www.energymanagertraining.com

CE39I: PYTHON FOR DATA ANALYTICS

B.TECH. 5TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – II

(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	NumPy	10
2	Pandas	12
3	Matplotlib	10
4	Statistics for data analytics	13

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

C. Detailed Syllabus:

1	NumPy	10 Hours	22%
1.1	NumPy Introduction		
1.2	Create NumPy Arrays		
1.3	Data Type Object, type		
1.4	Numerical Operations on NumPy Arrays		
1.5	Changing the Dimensions of Arrays		
1.6	Python, NumPy and Probability		
1.7	Creation of Synthetic Test Data		
1.8	Boolean Masking of Arrays		
1.9	Matrix Arithmetic		

1.10	Reading and Writing and arrays		
2	Pandas	12 Hours	27%
2.1	Pandas Data Frames		
2.2	Replacing Values in Data Frames and Series		
2.3	Pandas Data Files		
2.4	Dealing with NaN		
2.5	Binning Data		
2.6	Pandas Tutorial Continuation: multi-level indexing		
2.7	Data Visualization with Pandas and Python		
2.8	Python, Pandas and Timeseries		
2.9	Estimating the number of Corona Cases with Python and Pandas		
3	Matplotlib	10Hours	22%
3.1	Pyplot API		
3.2	Simple Plot		
3.3	Figure Class		
3.4	Axes Class		
3.5	Multiplot		
3.6	Subplots () Function		
3.7	Formatting Axes		
3.8	Setting Ticks and Tick Labels		
3.9	Bar Plot		
3.10	Histogram		
3.11	Pie Chart		
3.12	Scatter Plot		
3.13	Three-dimensional Plotting		
4	Statistics for data analytics	13Hours	29%
4.1	Mean, Median, Mode		
4.2	Standard Deviation, Variance		
4.3	Variable Distribution		
4.4	Normalization		
4.5	Statistical Modeling using Auto Regression and Auto		
4.6	Correlation		

D. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted as per pedagogy as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

E. Course Outcome (COs):

On the completion of the course the students will be able to:

CO1	Understand mechanisms for missing data and analytic implications (Understand)
CO2	Build and code a Graphical User Interface (GUI) to run a program
CO3	Understand and use python data science libraries as a tool for data analytics (Apply)
CO4	Implement numerical programming, data handling and visualization through NumPy, Pandas and Matplot-lib modules. (Create)
CO5	Analyze the significance of python program development environment by working on real world examples (Create)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	1	-		-	1	-	-	2	-	-	-	-	-
CO2	-	1	-	2	2	3	2	-	-	-	-	-	-	-

CO3	1	-	2		2	2	3	2	2	2	2	2	-	-
CO4	-	1	2	2	2	3	3	2	-	3	3	3	2	-
CO5	1	2	-	2	2	3	3	2	2	-	3	3	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Text Books:

- 1) William McKinney, “ Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Ipython” O’Reilly Media, Inc.,
- 2) Nagar, Sandeep, “Introduction to Python for Engineers and Scientists - Open Source Solutions for Numerical Computation”, Apress, 2018.
- 3) Jesus Rogel-Salazar, “Data Science and Analytics with Python”, CRC Press, Taylor & Francis Group, LLC

Reference Books:

- 1) Magnus Lie Hetland, "Beginning Python from Novice to Professional", Third Edition, Apress, 2017
- 2) Mark Lutz "Learning Python", 4th Edition, O’REILLY, 2016

Web Materials:

- 1) <https://nptel.ac.in/courses/106/106/106106212/>
- 2) <https://matplotlib.org/gallery/index.html>
- 3) <https://realpython.com/python-matplotlib-guide/>
- 4) <https://www.tutorialspoint.com/python/index.htm>

EE391: CONTROL OF ELECTRIC MOTORS FOR VEHICULAR APPLICATIONS

5th Semester and 3rd Year (Level III)

B. Tech. (Electrical Engineering)

Minor Specialization Subject-2

INSTITUTE ELECTIVE – II

(For Minor Specialization in “Electrical Vehicle Systems”)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Electrical Machines for Vehicular Applications	15
2	DC Motor Dynamics & Control	10
3	Induction Motor Dynamics & Control	11
4	Special Electric Motor Dynamics & Control	24

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

C. Detailed Syllabus:

- 1 Electric Machines for Vehicular Applications 15 Hours 25 %
Overview of conventional ac and dc motors, special electric machines such as permanent magnet machines, switched reluctance machines, linear motors, machine and engine rating, requirements, torque – speed characteristics and speed control, cooling of electric machines, testing of machines and testing standards
- 2 DC Motor Dynamics & Control 10 Hours 16.66 %

Various schemes of closed loop control using sensors, Sensor less operation of drive, DC motor drive using PWM rectifiers.

- | | | |
|---|--|--------------------|
| 3 | Induction Motor Dynamics & Control | 11 Hours 18.34% |
| | Principles of soft starting, Loss minimization, Various schemes of closed loop control, Direct and indirect field oriented control of induction motor, Direct torque control. | |
| 4 | Special Electrical Motors Dynamics & Control | 24 Hours 40 % |
| | <ul style="list-style-type: none">• Switched Reluctance Motor Drive: Structure of SR Machines, Principle of SR Machines, Converter configurations, Modified converter topology, Control strategies, Sensor and sensor less control of SRM, Reduction of torque pulsation.
• Stator-PM Motor Drives : Doubly-Salient PM Motor Drives, Flux-Reversal PM Motor Drives, Flux-Switching PM Motor Drives, Hybrid-Excited PM Motor Drives Flux-Mnemonic PM Motor Drives, Design Criteria of Stator-PM Motor Drives for EVs, Application, Advantages and Limitations for EVs.
• Brushless Motor Drive: Fundamentals of permanent magnet motors, Control strategies, Vector control of the Permanent Magnet Motor drive, Direct torque control of Permanent Magnet Motor drive, Sensor less control, Reduction of torque pulsations, Parameter sensitivity of drive.
• Advanced Magnet less Motor Drives: Introduction of Advanced Magnet less technology, Synchronous Reluctance Motor Drives, Doubly-Salient DC Motor Drives, Flux-Switching DC Motor Drives | |

D. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- In lab session, the student will be going through the various motors drive control strategies and simulations which will help to make some demo drives of the various motors used for electrical vehicles.

E. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand the different electrical machines used for vehicle application.
CO2:	Design the DC motor control with sensor as well as without sensor.
CO3:	Understand the control of induction motor with respect to electric vehicle
CO4:	Understand the control strategies for PM motors
CO5:	Understand the brushless motor drive and magnet less motor drive

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	-	-	-	-	-	-	-	-	1	-	-
CO2:	3	2	3	1	2	2	-	-	1	-	-	2	1	-
CO3:	3	2	1	1	1	-	-	-	-	-	-	1	1	-
CO4:	3	2	1	1	1	-	-	-	-	-	-	1	1	-
CO5:	3	1	-	-	1	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

- | | |
|----|---|
| 1. | Mehrdad Ehsani, Yimin Gao and Ali Emadi, “Modern Electric, Hybrid Electric and Fuel cell vehicles: Fundamentals, Theory and Design”, CRC Press, 2004. |
| 2. | Electric Vehicle Machines and Drives Design, Analysis and Application by K. T. Chau, Wiley-IEEE Press; 1st edition |
| 3. | Advanced Electric Drive Vehicles (Energy, Power Electronics, and Machines) by Ali Emadi, CRC Press; 1st edition (9 December 2014) |

Reference Books:

- | | |
|----|---|
| 1. | Electric Vehicle Technology Explained by James Larminie & John Lowry, Wiley |
|----|---|

Web Material:

- | | |
|----|---|
| 1. | https://nptel.ac.in/courses/108/106/108106170/ |
| 2. | https://onlinecourses.nptel.ac.in/noc21_ee112/ |
| 3. | https://nptel.ac.in/courses/108/102/108102121/ |

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VI)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE346: DIGITAL SIGNAL PROCESSING
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

Outline of the Course:

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Signal Processing	10
2	Z Transform and Realization of Discrete System	10
3	Fourier Analysis	10
4	Filter Design	15

Total hours (Theory) :45Hrs

Total hours (Lab) :30Hrs

Total hours :75Hrs

C. Detailed Syllabus:

1 Introduction to Signal Processing 22% 10 Hrs

Basic block diagram of Digital processing system, Sampling and Sampling Theorem for Reconstruction of signal in Digital domain, Aliasing and Solution, Re-construction Filter, Errors and noise, Mathematical and graphical representation of various signals and basic operations over it, Various Discrete systems and their characterization, Convolution and Correlation. Introduction to DSP Processor.

2 Z Transform and Realization of Discrete System 22% 10 Hrs

Definition and ROC for Z Transform, S plane to Z plane mapping using various techniques like impulse invariant and Bilinear Transformation, Inverse Z Transform,

Realization of IIR & FIR Systems (Direct, Cascade and Parallel form Realization), Discrete Integrator and Differentiator

3 Fourier Analysis

22% 10 Hrs

Continuous time and Discrete time Fourier Series (CFS and DFS) and their properties, Discrete Time Fourier Transform (DTFT), Discrete Fourier Transform (DFT), Computational Advantage of Fast Fourier Transform (FFT) Over DFT, Bit Reversed ordering, Decimation in time (DIT) & Decimation in frequency (DIF) Algorithm for implementation of FFT.

4 Filter Design

34% 15 Hrs

Introduction to Digital Filters and their classification, Applications, Design of IIR and FIR Filters using various methods, their comparison. Introduction to adaptive Filtering concepts. Multi-rate Signal Processing: Concept and Applications, Decimation, Interpolation, Sampling rate conversion by rational factor, Multi-rate filter banks.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	To understand theory of various signal processing algorithms and systems like Fourier transform, wavelet Transform, Convolution and Correlation, Filters.
CO2	To understand fundamentals and mathematical representation of signal and system, mathematics related to signal processing and their computation using math and also with Computer programming and simulation.
CO3	To develop skill to identify the DSP algorithm and systems of blocks used in a real system, and in complex engineering problem and solution required for benefits to society.
CO4	To find application of signal processing algorithm and systems in current research and to present overall research in form report and block diagram.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	-	-	2	-	-	-	-	-	-	-	2	-
CO3	-	-	3	-	-	3	2	1	-	-	-	-	2	-
CO4	-	-	-	3	-	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- In lab session students will solve the numerical by hand computations and will also verify the results through simulations in MATLAB.

F. Recommended Study Material:

Text Book:

1. Anand Kumar “Digital Signal Processing” 2nd Edition PHI Learning Private Ltd.
2. Udayashankara "Real Time Digital Signal Processing: Fundamentals, Algorithms and Implementation using TMS Processor", Prentice-Hall Of India Pvt Limited, 2010
3. Ramesh Babu, “Digital Signal Processing”, SCITECH Publications, 4th Edition, 2009

Reference Book:

1. Alan V Oppenheim, Alan S. Willsky with Hamid Nawab, “Signals & Systems”, Pearson Education, 2nd Edition (2000).
2. John Proakis, Dimitris G Manolakis, “Digital Signal Processing Principles”, Algorithms and Application”, PHI, 3rd Edition (2000).
3. S Salivahanan , A Vallavaraj , C Gnanapriya “Digital Signal Processing” 2nd Edition, McGraw Hill

Web Material:

1. <http://nptel.iitm.ac.in/video.php?subjectid=117102060> nptel video course by prof.s.c. duttaroy, iitdelhi.
2. DSP Video Lectures by Prof. E. Ambikairajah:
http://onlinevideolecture.com/?course_id=373
3. http://www.analog.com/en/processorsdsp/processors/beginners_guide_to_dsp/fca.htm
l a beginner's guide to digital signal processing

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE347: PROGRAMMABLE LOGIC CONTROLLERS AND
INDUSTRIAL AUTOMATION
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	4	6	4
Marks	100	100	200	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of Control	02
2	Introduction to Programmable Logic Control (PLC)	02
3	Input and Output devices	03
4	PLC Input and Output Module	04
5	PLC Operation	03
6	PLC Programming	10
7	Designing Automation System	06

Total hours (Theory) :30Hrs

Total hours (Lab) :60Hrs

Total hours : 90Hrs

C. Detailed Syllabus:

1	Basics of Control	06.67 %	02Hrs
	Introduction to Sequential/ Logic Control, control strategy, control philosophy and control algorithm, Difference between Relay logic control and PLC, Difference between Analog/ Automatic and Sequential/ Logic, Control, Evolution of Control System		
2	Introduction to Programmable Logic Control (PLC)	06.67%	02Hrs

	Definition of PLC, introduction to standard (IEC61131) used for PLC and basics of process automation (along with information about standard control signals used), History and Evolution of PLC, Block Diagram of PLC, Advantages and Disadvantages of PLC		
3	Input and Output devices	10.00%	03Hrs
	A brief overview of sensors and its types, Overview of Transmitter, difference between 2 wire Transmitter and 4 wire transmitter, Brief overview of types of output devices connected to PLC, Control Valves and its classification		
4	PLC Input and Output Module	13.33 %	04Hrs
	Power supply circuit/Wiring Diagram, Selection of power supply, Function of input and output modules, Classification and block diagram of Input and output Modules, AC input module (circuit/wiring diagram), Discrete/Digital input module (circuit/wiring diagram) Special input module cards TTL Logic, Relay and Triac output module (circuit/wiring diagram), Selection of cards/modules		
5	PLC Operation	10.00%	03Hrs
	PLC memory types and its mapping, PLC Register basics , Addressing: Internal and External, PLC Scan Cycle and response time		
6	PLC Programming	33.33%	10Hrs
	Types of programming languages, Conversion of Gate Logic and process statement into ladder and vice versa, Various PLC Instruction: NO, NC and output contacts and coils, Set, Reset, Timer (along with types), Counters (its types), PID, Logical, Arithmetic, Data handling, skip, MCR, jump, Bit operations and move instructions, Programming related to above instructions		
7	Designing Automation System	20.00%	06Hrs
	Selection of PLC, Documentation: System architecture, Piping and Instrumentation Diagram (P&ID) (Flow sheet symbols as per standard ISA S5.11984(R1992)), General Arrangement (GA) Drawing, Wiring Diagram, I/O Listing and program flowchart, Debugging (Simulation), Commissioning, troubleshooting and maintenance of PLC system		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Enhance their skills in basics of PLC.
CO2	Know the performance and classification of Input and Output Devices.
CO3	Understand different Programming language used in PLC.
CO4	Design any automated control system for any machinery.
CO5	o program PLC and they can apply their knowledge in the field of automation and control.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	1	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	3	3	3	2	-	-	-	2	-	-	-	3	-
CO5	-	3	3	3	2	-	-	-	2	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Programmable Logic Controllers by John W. Webb and Ronald A. Reis, PHI Publication.
2. Programmable Logic Controllers by W. Bolton, Newnes

Reference Book:

1. Programmable Logic Controllers & their Engineering Application by A.J. Crispin, McGraw Hill.
2. Programmable Logic Controllers by Thomas A. Hughes, ISA.
3. Programmable Logic Controllers by John R. Hackworth, Pearson Education.
4. Exploring Programmable Logic Controllers with Applications by Pradeep Kumar Srivastava, BPB Publications
5. Programmable Logic Control: Principles and Applications, NIIT, PHI

Web Material:

- <http://coep.vlab.co.in/?sub=33&brch=97>
- <http://www.plcdev.com/book/export/html/9>
- <http://www.plcmanual.com/>
- <http://nptel.ac.in/courses/108105062/>
- <http://nptel.ac.in/courses/108105088/1>
- <http://nptel.ac.in/courses/108105063/>
- www.siemensplcweb.com
- <http://www.ab.com/programmablecontrol/plc/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE348: POWER ELECTRONICS AND DRIVES - II
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Inverters	15
2	Power Supplies	5
3	AC Voltage controllers	6
4	Cycloconverters	7
5	AC Drives	12

Total hours (Theory) :45Hrs

Total hours (Lab) :30Hrs

Total hours :75Hrs

C. Detailed Syllabus:

1	Inverters	33.33 %	15 Hrs
	Introduction and revision: Overview of switches, Introduction to inverters Voltage Source Inverter : Single-phase half bridge and full bridge inverter with different loads, Three-phase 180 and 120 degree mode inverter, Fourier analysis of Single-phase and Three-phase inverter, Output Voltage control in inverters, Harmonic reduction techniques in inverter output voltage Current Source Inverter: Concept of CSI, CSI with different loads Load and Self-commutated inverter: Single-phase Series Inverter and Parallel Inverter		
2	Power Supplies	11.11 %	5Hrs

	DC Power Supplies: Switched-Mode DC Power Supplies, Half bridge converter, Full bridge converter, Fly back converter, Push-pull converter AC Power Supplies: Uninterruptible Power Supplies (UPS) , Batteries used in UPS Static switches and Solid State relays		
3	AC Voltage controllers	13.33 %	6Hrs
	Voltage control techniques: Principle of voltage controllers, Phase control, Integral Cycle control, Single-phase Voltage controllers with R and RL load Sequence control of AC voltage controllers: Two-stage and Multistage sequence control of voltage controllers		
4	Cycloconverters	15.56 %	7Hrs
	Introduction: Principle and Applications of Cycloconverters, Types of Cycloconverters, Single-phase to single-phase Cycloconverters: Bridge-type Cycloconverters, Mid-point Cycloconverters, Three-phase Half-wave Cycloconverters: Three-phase to Single-phase, Three-phase to Three-phase, Output voltage Equation of a Cycloconverter.		
5	AC Drives	26.67 %	12Hrs
	Basic machine equations: Analysis and Performance of AC motors, Speed Control of three-phase Squirrel-cage Induction Motors (SCIMs): Stator Voltage Control, Stator Frequency Control, Stator Voltage and Frequency Control, Stator Current Control, Speed Control of three-phase Wound Rotor Induction Motors (WRIMs): Static Rotor Resistance Control, Slip-energy recovery Control.		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Describe the role of Power Electronics as an enabling technology in various applications such as flexible production systems, energy conservation, renewable energy, transportation etc.
CO2	Learn the concepts of inverters with different loads and its applications.
CO3	Understand, simulate and design single-phase and three-phase thyristor converters.
CO4	Identify a switching power-pole as the basic building block and to use Pulse Width Modulation to synthesize the desired output.
CO5	Learn basic magnetic concepts, analyze transformer-isolated switch-mode power supplies and design high-frequency inductors and transformers.
CO6	Learn the requirements imposed by electric drives (dc and ac) on converters and synthesize these converters using the building block approach.

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
-----------------	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------

CO1	3	-	1	-	2	-	-	-	-	1	-	-	-	-
CO2	3	2	2	1	2	-	-	-	-	-	-	-	2	-
CO3	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO4	3	3	3	1	2	-	3	-	-	-	-	-	2	-
CO5	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO6	3	1	-	-	-	-	3	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Power Electronics by Dr. P. S. Bimbhra, Khanna publishers
2. Power electronics by M D Singh and K B K Khanchandani, Tata Mcgraw-Hill Publication

Reference Book:

1. Power electronics handbook, by M.H. Rashid, Academic press
2. High power converters and AC drives by Bin Wou, IEEE press
3. Power Electronics Converters, Applications and design by Mohan, Undel and, Robbins, Wiley publication.

Web Material:

1. <http://nptel.ac.in/courses/108101038/>
2. http://cusp.umn.edu/power_electronics.php
3. <http://ecee.colorado.edu/~ecen5797/notes.html>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE349: FAULT ANALYSIS AND SWITCHGEAR
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Review of representation of power system components:	04
2	Symmetrical fault analysis	08
3	Symmetrical components	13
4	Unsymmetrical Fault Analysis	10
5	Fundamentals of Fault Clearing, Switching Phenomena and Circuit Breaker. (CB) Ratings	10
6	Arc Extinction Process in Circuit Breaker	05
7	Types of Circuit Breakers	10

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

C. Detailed Syllabus:

- 1 **Review of representation of power system components:** 07% 04 Hrs
Introduction, Single phase solution to three phase system, one line diagram, impedance and reactance diagram
Per unit system and Examples
- 2 **Symmetrical fault analysis** 13% 08 Hrs

Introduction, sudden short circuit on R-L series circuit, 3 – phase short circuit current calculation, examples.

Sub-transient, transient and steady state model of synchronous machine

3 phase short circuit analysis of power system – Examples

3 Symmetrical components 17% 10 Hrs

Introduction, Symmetrical Component transformation.

Example on symmetrical components

Sequence impedance and sequence network of transformer , Phase shift in star delta Transformers.

Sequence impedance and sequence network of synchronous machine and transmission Line.

Examples on sequence networks.

4 Unsymmetrical Fault Analysis 17% 10 Hrs

Types of unsymmetrical faults and its analysis.

Examples on unsymmetrical fault analysis

5 Fundamentals of Fault Clearing, Switching Phenomena and Circuit Breaker (CB) Ratings. 17% 10 Hrs

Current interruption in AC circuit breaker.

Transient recovery voltage, rate of rise of restriking voltage and Examples.

Effect of natural frequency, power factor on TRV- effect of reactance drop on power frequency recovery voltage- effect of armature reaction on recovery voltage- effect of first pole to clear factor- single, double frequency transient.

Interruption of low magnetizing current (current chopping), use of resistance switching for damping TRV, use of opening resistor, switching of capacitor bank, unloaded transmission lines and unloaded cables, interruption of terminal fault and short line fault (Kilometric Fault), Phase opposition switching.

Rating of AC circuit breaker.

6 Arc Extinction Process in Circuit Breaker 08% 05 Hrs.

Introduction, The matter and plasma- Ionization of gases – Deionization.

Formation of electric arc and methods of arc extinction.

Arc interruption theories.

7 Types of Circuit Breakers 22% 13 Hrs.

Air break circuit breaker, construction of air blast circuit breaker, (ABCB), arc quenching in ABCB, resistance switching in ABCB and methods of reducing overvoltage in ABCB.

Physical and dielectric properties of Sulphur Hexafluoride (SF₆), arc extinction process in

SF₆ CB, different puffer type circuit SF₆ CB, relative merits and demerits of SF₆ CB.

Electric break down in high vacuum, arc extinction in vacuum CB, construction of Vacuum CB, contact material and switching phenomena of vacuum CB.

Introduction to Direct Current Circuit Breaker (DCCB), different types of DCCB.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Able to calculate symmetrical fault current and voltage at any location of power system by applying allied theories.
CO2	Able to calculate unsymmetrical fault current and voltage at any location of power system using symmetrical components by applying allied theories.
CO3	Acquire the knowledge of the phenomena of current interruption in circuit breaker.
CO4	Acquire the knowledge of arc formation and arc interruption in circuit breaker.
CO5	Impart the knowledge of circuit breaker construction and working. Also, gain ability to select proper rating and type of circuit breaker.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	2	-	-	-	-	-	-	-	-	-
CO2	3	3	2	2	2	-	-	-	-	-	-	-	-	-
CO3	3	1	-	1	-	-	-	-	-	-	-	-	-	-
CO4	3	1	-	1	-	-	-	-	-	-	-	-	-	-
CO5	3	1	2	2	-	2	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT HT control room and nearby substation will clear the concept of circuit breaker and student will realise the actual size of Circuit breaker.
- In lab session students shall solve the numerical in computer with the software “Power World Simulator”

F. Recommended Study Material:

Text Book:

1. Power System Analysis by Hadi Saadat, Tata McGraw Hill.
2. Principles of Power system by V. K. Mehta, Publisher: S. Chand & Company Ltd.
3. Modern Power System Analysis by D.P. Kothari & I. J. Nagrath,, Tata McGraw Hill, New Delhi.
4. A Course in electrical power by Soni & Bhattnagar, Dhanpat Rai & Sons.
5. Switchgear and Protection, by S.S. Rao, Dhanpat Rai & Sons.

Reference Book:

1. Power System Analysis by Grainger & Stevenson, Tata McGraw Hill
2. Power System Analysis and Design by B R Gupta. S. Chand & Company Ltd.
3. Analysis of Faulted Power System by P.M. Anderson, IEEE Press
4. Power System Protection & Switchgear by B. Ravindranath & M. Chander, New Age International.
5. Switchgear and Protection by J B Gupta. S.K Kataria & Sons.
6. Computer Techniques In Power System Analysis By M A Pai, TMH Education Pvt. Ltd.
7. JNP Switchgear By R T Lithal.

Web Material:

- <http://courses.engr.illinois.edu/ece476/notes/?year=2010>
- <http://nptel.iitm.ac.in/video.php?courseId=1060>
- <http://freevideolectures.com/Course/2353/Power-Systems-Analysis>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE360: MINOR PROJECT II
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	04	4	2
Marks	00	100	100	

B. Instructional Method and Pedagogy:

- Minor project-II has 2 credit subject and 4 hours per week.
- Department/Institute will help students to find appropriate minor projects.
- The student must be find the appropriate guide for their project and make the project
- proposal and submit the project proposal to departmental co-coordinator.
- Project must be from the electrical bag round other project must be rejected.
- Student must meet at least once in week to his/her guide.
- The student shall submit two documents to the Coordinator for the evaluation of the minor project:
 - Project report
 - Project presentation
- Upon the completion of project, a hard copy of “project Report” must be submitted to the Coordinator by the end of the semester.
- The report must outline with practical knowledge which you gain during project making. The guide must verify report.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the project.

C. Format of Project Report:

Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12 pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done during project, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- INTRODUCTION: In this section, give the purpose of the project, reasons for choosing the project, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the project. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the project. Present your observations, contributions and intellectual benefits.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet

link and document number. In this manner you don't have to print it and waste tons of paper.

D. Course Outcomes (COs):

After completion of the course students will able:

CO1	To apply knowledge and skills learned from project to real-world problems.
CO2	To solve engineering problems by the team work and work with other discipline.
CO3	Use experience related to professional and ethical issues in the work environment.
CO4	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.
CO5	To implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT

I. Credits and Schemes:

Sem.	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				Total
					Theory		Practical		
					Internal	External	Internal	External	
VI	HS133.02 A	CONTRIBUTORY PERSONALITY DEVELOPMENT	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

III. Detailed Syllabus:

1.	Concept of Personality	06 Hours	20%
	Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling		
2.	Soft Skills and Personality Development	08 Hours	26%
	Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player		
3.	Developing Contributory Personality	06 Hours	20%

	Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities		
4.	Life skills and Personality Development	06 Hours	20%
	Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving		
5.	Contemporary Issues in CPD	04 Hours	14%
	Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations		

IV. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by personality profiling, reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies (Great Personality) , critical reading, group work, independent and collaborative research, presentations etc.

V. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The practical evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Components	Number	Marks per incidence	Total Marks
1	Assignments	2	25	
2	Project work	2	25	
3	Personality Profiling and Presentation	1	25	
4	Quiz	2	25	100*
5	Attendance and Classroom participation			05
			Total	30

*** Total Marks will be converted to 25**

External Evaluation

The University Exam will be for 70 marks and will be based on practical computer based tests and a viva-voce. It will test the contributory personality aspects and their

applications. There may also be a Computer Based Test at the end.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical Exam / Viva	01	70	70
Total				70

VI. Course Outcomes (COs)

At the end of the course, learners will be able to:

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	2	-	2	2	-	-	-	-
CO6	-	-	-	-	-	2		-	-	2	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

VII. Preference

It will be preferred that at the end of the course, students take an official Personality Profiling test like MBTI through CHARUSAT. Equivalent marks shall be considered in the University Exam.

VIII. Recommended Study Material:

❖ Text book:

4. Personality Development & Soft Skills, Oxford University Press

5. Soft Skills, Bookboon
6. Personality Development, Swami Vivekananda; Advaita Ashrama

❖ **Reference book:**

6. Contributor Personality Program Workbook (Volume 1,2),
7. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

❖ **Web material:**

4. <https://www.coursera.org/learn/wharton-success>
5. <https://www.coursera.org/learn/personality-types-at-work>
6. <https://www.coursera.org/learn/self-awareness>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE376: SPECIAL ELECTRICAL MACHINES AND APPLICATIONS
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction of Special Electrical Machines	13
2	Special Synchronous Machines	05
3	Permanent Magnet Machines	07
4	Wind Turbine Generators	06
5	Modelling & Applications	09
6	Case Study	05

Total hours (Theory) :45Hrs

Total hours (Lab) :30 Hrs

Total hours :75Hrs

C. Detailed Syllabus:

- 1 Introduction of Special Electrical Machines** 28.9% 13Hrs
Special DC Machines: Rosenberg Generator, Dynamotors, Printed Circuit Board Motor. Special Induction Machines: Synchronous Induction Motor, Induction Voltage Regulator, Linear Induction Motor. Special Transformers: Induction Heating Transformer, Welding Transformers, Pulse Transformer. Schrage Motor, Repulsion

Motor, Hysteresis Motor, Universal Motor, DC-AC Servomotor. Applications, Examples.

- 2 **Special Synchronous Machines** 11.1% 05Hrs
Stepper Motor, Switched Reluctance Motor, Linear Synchronous Motor, Applications, Examples.
- 3 **Permanent Magnet Machines** 15.6% 07Hrs
Basics of permanent magnet property, Unique features of Permanent Magnet (PM) Machines, Permanent Magnet Material & Characteristics, Permanent magnet DC motor (PMDC), Permanent magnet brushless DC motor(PMBLDC) : construction, operation, control, advantages, dis-advantages, applications, permanent magnet synchronous motor(PMSM), Applications.
- 4 **Wind Turbine Generators** 13.3% 06Hrs
Introduction, Types of induction generator (line excited and self-excited), Introduction of fixed speed induction generator, Introduction of doubly fed induction generator. Working of Induction generator: Principle of Operation, Reactive power requirement, voltage build up process, Load and Power factor control, Effect of capacitor, Characteristics of induction generator, Advantages, Applications, Examples.
- 5 **Modelling & Application** 20.0% 09Hrs
Synchronous Reference Frames: dq Axis Theory, Conversions: abc to $\alpha\beta$, $\alpha\beta$ to dq, dq to $\alpha\beta$, $\alpha\beta$ to abc. Space Vector Pulse Width Modulation: discrete and analog implementation, Applications.
- 6 **Case Study** 11.1% 05Hrs
Case Study: Grid Connected Renewable Energy Source

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Recognize & understand the performance, operation and application of various AC & DC special machines.
CO2	Define various permanent magnet material used for manufacturing of special machine.
CO3	Acquire the knowledge and provide operational instructions for permanent magnet

	motors/generators, switched reluctance motor, linear motors, hysteresis motors and stepper motors as well as make effective presentations on the working, construction, advantages & drawbacks of these machines along with the needs of improvisation in the machine.
CO4	Understand the types and operation of wind turbine generators and evaluate case studies based on renewable energy sources.
CO5	To understand the modeling of machines and reference frames for machine control.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	2	-	-	-	-	1	2	1	3	-	-
CO2	3	1	2	2	2	-	1	-	-	1	-	-	2	-
CO3	3	3	3	1	2	2	1	-	-	3	2	1	2	-
CO4	3	3	3	1	2	2	3	-	2	2	2	3	3	3
CO5	3	-	2	3	3	-	-	2	2	3	1	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.

- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Special Electrical Machines by K Venkataratnam, University Presses.
2. Special Electrical Machines by E.G. Janardanan, PHI Learning Pvt. Ltd.
3. A Text Book of Electrical Machines by K.R.Siddhapura, D.B.Raval. Vikas Publishing House Pvt. Ltd.
4. Electric Machines and Power Systems by Syed A. Nasar, Tata McGraw Hill
5. Electrical Machines III by K Krishna Reddy, Scitech Publication (INDIA) Pvt. Ltd.

Reference Book:

1. Theory and performance of electrical machines by J.B.Gupta, S.K.Kataria and sons

Web Material:

- http://www.academia.edu/9885014/special_electrical_machines_nptel_notes
- <http://www.notesnation.com/electrical-engineering/special-electrical-machines-notes/>
- http://nptel.iitm.ac.in/courses/IIT-MADRAS/Electrical_Machines_I
- www.nptel.ac.in/courses/IIT-MADRAS/Electrical_Machines_II/pdf/2_1.pdf

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE377: EMBEDDED SYSTEMS
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	ARM and FPGA architecture	06 HRS
2	Fundamental of Real Time Operating System	10 HRS
3	Expert Systems implementation on computer Onchip and ARM.	09 HRS
4	Communication Protocol and Device driver.	10 HRS
5	FPGA Programming in VHDL	10 HRS

Total hours (Theory) :45 Hrs
Total hours (Lab) :30 Hrs
Total hours :75 Hrs

C. Detailed Syllabus:

1	ARM and FPGA architecture	13.33%	06 Hrs
	Introduction to CISC, RISC, Advanced Risk Machine(ARM) , ARM assembly Instructions, Co Processor, Sharing between individual cores in multi core microcontroller, FPGA architecture. Introduction to Various Market available Embedded boards and chips.		
2	Fundamental of Real Time Operating System	22.22%	10 Hrs
	Basics of OS and RTOS: Code Execution in Operating system with simplest Round robin Scheduling, Round robin with interrupts, Types of Operating System (Real Time, Non Real Time and Soft Real time), Selection of RTOS for an Embedded system, Architectural		

	Diagram and functioning of Operating System, Kernel space and user space, Task, Thread and Process (related programming examples of POSIX thread, java thread and Windows thread), State of Task (Ready, running, suspended, deleted), Multi Processing with multi core and Muti tasking, Various Task scheduling algorithms(Various multi-tasking system), Task Communication and related programs, Task synchronization, Message Passing, mailbox, signaling, semaphore, Mutex. Problems (like Deadlock, live lock, starvation), Introduction to various markets available embedded RTOS. Experiments teaching all above OS concepts using Embedded RTOS and on computer. Free RTOS examples on ARM M4 (Put Ready programs of ADC, Port pin, PWM, LCD etc in RTOS execution)..		
3	Expert Systems implementation on computer Onchip and ARM.	20%	09 Hrs
	Introduction to several computers on chip boards, Architecture, Accessing peripherals using python, Python implementation of AI and Machine learning on Raspberry PI with examples like Gesture control using AI. Block diagrams of Market available Embedded Products and recognizing scope of Expert system in it. Expert system implementation on ARM using CMSIS.		
4	Communication Protocol and Device driver.	22.22%	10 Hrs
	Basics of Serial and Parallel bus communication protocols: I2C bus, CAN bus, USB bus, ISA bus, PCI and PCI/x bus, AMBA bus), Creating USB application on ARM with CMSIS, Device driver development Examples		
5	FPGA Programming in VHDL	22.22%	10 Hrs
	History of VHDL, Introduction to VHDL and meaning of Hardware abstraction, Entity Declaration, Architecture Body(Structural, Behavioural and Mix style of Modeling), Lexical Elements(comments, identifiers, numbers, characters, strings, bit strings), Data types and objects (Integer datatypes, Arrays, Records), Object declaration, Signal Vs Variable, Programming OR, AND and XOR Gate in VHDL, If, Case and NULL statement, Loop statements, Wait statement, Libraries, Programming exercises (Like Adder, comparator, ALU, Latch, D and JK Flip flop, Register and shift Registers, Counter, Finite state Machine), Conditional and selected signal Assignment, Writing functions and procedures, Structural modeling, Test benches, Role of RTL engineer in Embedded industries, Implementation of communication protocols on FPGA, Interfacing ADC with FPGA (related programs and diagrams).		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to get knowledge about the foundation education in Embedded C and Assembly language programming. Through lecture and out-of-class assignments, students are provided learning experiences that enable them to:

CO1	To provide knowledge of architecture of CISC, RISC, ARM, DSP architecture.
CO2	To write RTOS programs for given embedded system and Hardware.
CO3	To gain knowledge of various communication Protocols and developing device driver and its VHDL implementation.
CO4	To be able to use expert system like AI and Machine learning on Microcontroller platforms.
CO5	To understand market available embedded products, boards and systems and to recognize role of expert system in it.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	1	-	-	-	-
CO2	3	1	-	-	3	-	-	-	2	3	2	-	3	-
CO3	-	-	-	-	3	-	-	-	2	3	2	-	3	-
CO4	-	3	2	3	3	-	-	-	-	3	2	-	3	3
CO5	-	3	2	-	-	-	-	-	-	3	--	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.

- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.

F. Recommended Study Material:

Text Book:

1. Steve furber, ARM System-on-chip Architecture, Pearson
2. Embedded system by Shibu.
3. Real Time Operating Systems for ARM cortex-M Microcontrollers, by Jonathan W. Valvano.
4. VLSI Design by A. Albert Raj, T. Latha
5. Circuit Design and simulation with VHDL. By Volnei A. Pedroni

Reference Book:

1. David E. Simon, An Embedded Software Primer, Addison-Wesely.
2. An Embedded system Primer by Simon
3. Raj Kamal, Embedded Systems Architecture, programming and design (Tata – MGHill)
4. ARM Cortex M4 Cookbook by Dr. Mark Fisher
5. The Definitive Guide to ARM cortex M3 and M4 Processor, by Joseph Yiu

Video lecture:

1. NPTEL and youtube videos on Embedded system by satish kashyap.
2. <https://www.youtube.com/c/arm/videos>
2. <https://www.youtube.com/watch?v=RIVcKqoes2c>
3. <https://www.youtube.com/watch?v=frbV2fiZl7Y>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE378: POWER ELECTRONICS APPLICATIONS IN POWER SYSTEM
(PE-2)
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Unit	Min. no of hours
1	FACTS Controllers	20
2	HVDC Transmission	20
3	Power Quality problems & Standards	05

Total hours (Theory) :45 Hrs
Total hours (Lab) :30 Hrs
Total hours :75 Hrs

C. Detailed Syllabus:

- | | | | |
|---|-------------------|----------|--------|
| 1 | FACTS Controllers | 20 Hours | 44.4 % |
| <p>Need of FACTS devices, classification of FACTS devices, operating principle of SVC, V-I characteristic of SVC, advantage of slope in V-I characteristic, SVC applications for transient and voltage stability improvement, mitigation of SSR, advantages of TCSC, different mode of operation of TCSC, different modeling concepts of TCSC, Operating principle of STATCOM, SSSC and UPFC and their applications for power system performance improvement, Self-study: Power flow solution with SVC, TCSC, SSSC and UPFC</p> | | | |
| 2 | HVDC Transmission | 20 Hours | 44.4 % |

Introduction, equipment required for HVDC systems, Comparison of AC and DC Transmission, Limitations of HVDC transmission lines, reliability of HVDC systems, comparison of HVDC link with EHVAC link, HVDC system configuration and components, fundamental equations in HVDC system, HVDC links, converter theory and performance equation, valve characteristic, converter circuits, converter transformer rating, multi bridge converters, abnormal operation of HVDC system, control of HVDC system, harmonics and filters. **Self-study:** Influence of AC system strength on AC/DC system interaction, response to AC and DC system faults

3 Power Quality problems & Standards 05 Hours 11.2 %

Introduction: Definition & Reasons of Occurrence of following Voltage Dip, Brief voltage increases, Brief voltage interruption, Transients, Voltage Notches, Flickers, Distortion, Unbalance. Power Quality Indices, Limits of Harmonic Distortion according to IEEE, IEC, EN and NORSOK limits. **Self-study:** Brief Introduction of Power Quality Standards: IEC 61000-2-5, IEC 61000-2-1, IEC 1159 (Categories of Power quality variation according to IEEE 1159 standard with their relevant Spectral content, Duration of occurrence & Magnitude)

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to:

CO1	Compare the HVDC Transmission and EHVAC transmission (1,3,5,10)
CO2	Identify, analyze and understand the converter configurations used in HVDC system and its components to enhance the skill development. (1,2,3,4,5,PSO1)
CO3	Understand controllers for controlling the power flow through a dc link (1,2,3,4,5,PSO1)
CO4	Able to understand the different types of Power Quality problems & Standards. (1,2,3,4,5,7,PSO1)
CO5	Understand the role of impedance control, phase angle control and voltage control in controlling real and reactive power in transmission systems. (1,2)
CO6	Identify analyze and understand the different FACTS controller required for the power system performance improvement. (1,2,7)

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	-	1	--	2	-	-	-	-	1	--		-	-

CO2	3	2	2	1	2	-	-	-	-	-	-	-	2	-
CO3	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO4	3	3	3	1	2	-	3	-	-	-	-	-	2	-
CO5	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO6	3	1	-	-	-	-	3	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Books and References:

1. E. W. Kimbark, Direct Current Transmission, Vol 1, Wiley-Interscience, 1971
2. P. Kundur, Power System Stability and Control, McGraw Hill, 1994.
3. S. Kamakshiah and V. Kamaraju, HVDC Transmission, McGraw Hill, 2011
4. K.R. Padiyar, HVDC Power Transmission Systems: Technology and System Interactions, New Age International.
5. E. Acha, FACTS: Modeling and Simulation in Power Networks, Wiley 2004
6. T.J.E Miller, Reactive Power Control in Electric Systems, Wiley India Edition, 1982

7. R. Mohan Mathur and R. K. Varma, Thyristor-based FACTS controllers for Electrical Transmission systems, IEEE Press, 2002

Web Material:

- <https://nptel.ac.in/courses/108/107/108107114/> : For **Chapter-1: FACTS Controllers**
- <https://nptel.ac.in/courses/108/104/108104013/> : For **Chapter-2: HVDC Transmission** HVDC
- <https://nptel.ac.in/courses/108/107/108107157/> : For **Chapter-3: Power Quality problems & Standards**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE379: High Voltage Engineering (Program Elective II)
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Breakdown mechanisms in gaseous dielectrics	08
2	Breakdown Mechanisms in Solid and Liquid dielectrics	06
3	Generation of High voltages and Currents	12
4	Measurement of High voltages and Currents	09
5	Overvoltage Phenomenon and Insulation Coordination in Electric Power Systems	02
6	High Voltage testing of equipment	08

Total hours (Theory) :45Hrs

Total hours (Lab) :30 Hrs

Total hours :75Hrs

C. Detailed Syllabus:

1	Breakdown mechanisms in gaseous dielectrics	17.78%	08Hrs
	Introduction, ionization processes, Townsend's mechanism, Primary and secondary ionization co-efficient, breakdown in electronegative gases, Streamer theory, Paschen's law, breakdown under uniform and non-uniform fields, Post breakdown current-voltage characteristics, breakdown mechanisms in vacuum.		
2	Breakdown Mechanisms in Solid and Liquid dielectrics	13.33%	06Hrs
	Breakdown Mechanism in Solid Dielectrics: Introduction, solid di-electrics, intrinsic breakdown, electromechanical breakdown, thermal Breakdown, Electrochemical breakdown, treeing and tracking phenomenon of partial discharge,		

	solid dielectrics used in practice Breakdown Mechanism in Liquid Dielectrics: Introduction, classification of liquids, liquid breakdown test cell, transformer oil purification, Testing of di-electric oil as per IS 355 and IS 6792, Breakdown in pure liquids , breakdown in commercial liquids		
3	Generation of High Voltages and Currents	26.67%	12Hrs
	Introduction, Generation of high direct voltage, rectifier circuits, voltage doubler, cascaded circuits, Van de Graff generators, electrostatic generators, generation of high alternating voltages, cascade transformer, resonant transformer, Generation of high frequency alternating voltages, generation of impulse voltages, standard impulse voltage wave, Impulse generator, Marx circuit, constructional features of impulse generator, Trigatron gap, generation of impulse currents, related examples		
4	Measurement of High voltages and Currents	20%	09Hrs
	Measurement of high direct voltages, potential dividers, generating voltmeters, measurement of high alternating voltages, series voltmeters, Capacitance potential dividers and capacitance voltage transformers, electrostatic voltmeter, measurement with sphere gaps (IS 1876), Sphere gap construction and assembly, factors influencing the spark over voltage, Measurement of impulse voltages, measurement of high d.c. and a.c. currents, measurement of high frequency and impulse currents.		
5	Overvoltage Phenomenon and Insulation Coordination in Electric Power Systems	4.44%	02Hrs
	Natural Causes for Overvoltages – Lightning phenomenon, Overvoltage due to switching surges, System faults and other abnormal conditions, Insulation Coordination, BIL		
6	High Voltage testing of equipments	17.78%	08Hrs
	Introduction, Power frequency voltage test (Dry & Wet), Impulse Voltage Test (Dry & Wet), Short-circuit test and Synthetic test on circuit breakers, Tests on insulators, bushings, transformers, cables and lightning arrestors, Non-destructive tests: DC Resistivity Test, Dissolved Gas Analysis, Sweep Frequency Response Analysis, Capacitance and loss tangent measurement, Partial Discharge Test.		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Understand and remember the different types of insulations, its applications and their breakdown mechanisms when subjected to high voltage stresses to get the opportunity to work as a team leader/member in multi-disciplinary branches
CO2	Differentiate, design, simulate and analyze the performance of various types of high voltage generating and measuring techniques as per IEC, IEEE and other Indian and international standards
CO3	Demonstrate, simulate, analyze and evaluate the performance characteristics of transmission line insulators and other high voltage assets such as transformers, circuit breakers etc. by conducting the high voltage tests under different climate conditions as per IEC, IEEE, CIGRE and other Indian and international standards to get the opportunity to work as a team member/leader in multi-disciplinary branches
CO4	Apply the knowledge to predict the remaining life of high voltage assets based on condition assessment tools i.e. dissolved gas analysis, sweep frequency response analysis etc.
CO5	Describe the principles behind the insulation coordination and basic impulse insulation level

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	2	-	-	3	3	-
CO2	2	1	3	-	3	-	-	-	-	-	-	-	2	-
CO3	3	2	2	3	2	-	3	3	3	3	-	2	3	-
CO4	2	2	-	-	1	-	-	-	2	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	2	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Books:

1. High Voltage Engineering by M.S. Naidu & V. Kamaraju, TMH Limited
2. High Voltage Engineering by C.L. Wadhwa, New Age International Publisher.
3. Fundamentals of High Voltage Engineering by S. K. Singh, Dhanpat Rai & Co.

Reference Book:

1. An Introduction to High Voltage Engineering by Subir Ray.
2. High Voltage Engineering by E. Kuffel, J. Kuffel and W.S. Zaengl.
3. High Voltage Engineering by D.V. Razevig, translated by M.P. Chourasia, Khanna Publishers.
4. High Voltage Engineering & Testing (2nd Edition) by Hugh M. Rayn, The Institute of Electrical Engineers, London.
5. High Voltage Engineering Practice & Theory (2nd Edition) by Mazen Abdel-Salam, Hussein Anis, Ahdab El-Morshedy, RoshdyRadwan.
6. High Voltage Engineering Practice & Theory by Dr. J. P. Holtzhausen&Dr WL Vosloo.

Web Material:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPU/HighVoltageEngg/ui/TOC.htm>
- <http://www.mv.helsinki.fi/tpaulin/Text.hveng.htm>
- <http://ocw.mit.edu/courses/physics/8-02-electricity-and-magnetism-spring-2002/video-lectures/lecture-6-high-voltage-breakdown-and-lightning.htm>
- http://www.sayedsaad.com/High_voltage/index_solids.htm

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE380: INTERNET OF THINGS
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction To Embedded Concepts	07
2	Iot Architecture And Protocols	11
3	Cloud Architecture And Computing	09
4	Computer Networks And Internet	06
5	Internet of Things: Sensors and Actuators	06
6	Preparing Iot Projects	06

Total hours (Theory) : 45 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 75 Hrs

C. Detailed Syllabus:

1 Introduction To Embedded Concepts 15.00% 07 Hrs

Introduction to embedded systems, Application Areas, Categories of embedded systems,
 Overview of embedded system architecture, Specialties of embedded systems, recent trends in embedded systems, Architecture of embedded systems, Hardware architecture, Software architecture, Application Software, Communication Software.

2 Iot Architecture And Protocols 25.00% 11 Hrs

Overview: IoT-An Architectural Overview- Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service(XaaS), M2M and IoT Analytics, Knowledge Management
Reference Architecture: IoT Architecture-State of the Art – Introduction, State of the art, Reference Model and architecture, IoT reference Model - IoT Reference Architecture- Introduction, Functional View, Information View, Deployment and Operational View.

3 Cloud Architecture And Computing 20.00% 09 Hrs

Cloud Architecture Basics: The Cloud -Hype cycle-metaphorical interpretation-cloud architecture standards and interoperability- Cloud types; IaaS, PaaS, SaaS. Benefits and challenges of cloud computing, public, private clouds community cloud, role of virtualization in enabling the cloud.

End to End Design : Requirement analysis: strategic alignment and architecture development cycle-strategic impact-Risk impact-financial impact-Business criteria-technical criteria-cloud opportunities –evaluation criteria and weight-End to end design-content delivery networks-capacity planning-security architecture and design

Cloud Application Architectures: Development environments for service development; Amazon, Azure, Google App-cloud platform in industry

How To Move Application Into The Cloud: Web Application Design- Machine Image Design-privacy design –Database management.

4 Computer Networks And Internet 13.33% 6 Hrs

Internet - The network edge- The network core- Delay , Loss and Throughput in packet switched networks- protocol layers and their service models- TCP/IP protocol architecture- Frame relay networks- ATM networks- protocol architecture- ATM logical connections ATM cell.

Congestion Control In Data Networks And Internet: Effects of congestion – Congestion control - Traffic management - Congestion control in packet switching networks – Frame relay- Congestion control -TCP flow control and TCP congestion control.

5 Internet Of Things: Sensors and Actuators 13.33% 6 Hrs

Introduction: Internet of Things Promises-Definition- Scope-Sensors for IoT Applications- Structure of IoT- IoT Map Device

Sensors: Linear and angular displacement sensors: resistance sensor – induction displacement sensor – digital optical displacement sensor – pneumatic sensors. Speed and flow rate sensors: electromagnetic sensors – fluid flow sensor – thermal flow sensor. Force sensors: piezoelectric sensors – strain gauge sensor – magnetic flux sensor – inductive pressure sensor – capacitive pressure sensor. Temperature sensors: electrical – thermal expansion – optical.

6 Preparing Iot Projects 13.33% 6 Hrs

Creating the sensor project - Preparing Raspberry Pi - Clayster libraries - Hardware- Interacting with the hardware - Interfacing the hardware- Internal representation of sensor values - Persisting data - External representation of sensor values - Exporting sensor data - Creating the actuator project- Hardware - Interfacing the hardware - Creating a controller - Representing sensor values - Parsing sensor data - Calculating control states - Creating a camera - Hardware -Accessing the serial port on Raspberry Pi - Interfacing the hardware - Creating persistent default settings - Adding configurable properties - Persisting the settings
- Working with the current settings -Initializing the camera

D. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	Enhance their skills in basics of embedded systems.
CO2	Know the protocols of IOT and its types of protocols.
CO3	Understand applications different sensors in internet of things and in electrical engineering and programming in embedded C.
CO4	Know the all communication protocols and Design and troubleshoot hardware circuits based on different sensors.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	2	-	-	-	-	-	-	-	2	-
CO3	-	3	-	-	2	-	-	-	-	-	-	-	3	-
CO4	-	2	3	2	3	-	-	-	3	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.

- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realise the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

F. Recommended Study Material:

Reference Book:

1. Designing the Internet of Things ,by Adrian McEwen & Hakim Cassimally , John Wiley & Sons ltd.
2. From machine to machine to the Internet of Things, by Jan Holler, vlasios Tsiatsis, Catherine mulligan, Elsevier.
3. Internet of Things, Converting Technologies for Smart Environments and Integrated Ecosystems, by Ovidiu Vermesan and Peter fries, River Publishers.

Web Material:

<https://nptel.ac.in/courses/106/105/106105166/> (IIT Kharagpur)

<https://nptel.ac.in/courses/108108123/> (IISC Bangalore)

<https://nptel.ac.in/courses/106105195/#> (IIT Kharagpur)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE381: HYBRID AND ELECTRIC VEHICLES
6th Semester and 3rd Year (Level III)
B.Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction of Electric Vehicle	06
2	Electric Motor Drives for EV	16
3	EV Charging Strategy	10
4	Vehicle to Grid	10
5	EV Business Trend & Market Policy	03

Total hours (Theory) :45 Hrs
Total hours (Lab) :30 Hrs
Total hours :75 Hrs

C. Detailed Syllabus:

- 1 **Introduction to Electric Vehicle** 13.33 % 06 Hrs
History of Electric Vehicles (EVs), EV and the environment, EV challenges: pure EV, hybrid EV, grid connected EV, fuel cell EV, EV Technologies: motor drive technology, energy source technology, battery charging technology.
- 2 **Electric Motor Drives for EV** 35.55 % 16 Hrs
DC Motor Drive: system configuration, structure, operation, DC-DC converter, control of DC motor, regenerative braking. Induction Motor Drive: system configuration, structure,

operation, Inverters, control of induction motor, PMBLDC Motor Drive: PM Materials, system configuration, structure, operation, inverters, control of PMBLDC motor, Comparison of various motor drives for EV application.

- 3 **EV Batteries & Charging Strategy** 22.23% 10 Hrs
EV Batteries introduction , concept of BMS, Categorizing EV charging, EV Charging Approaches: dumb charging, smart charging coordination, PV Based PEV Charging Facilities, Control Strategies: residential photovoltaic charging, Power Demand on Grid, Voltage Profile ,.
- 4 **Vehicle to Grid** 22.23% 10 Hrs
Introduction to V2X concept, Communication And Control, Impact of EV Charging On Electrical Distribution Systems, V2G: Definition, Benefits, Risks, Grid Integration, EV Supply Equipment Design And Power Levels, Charging Station Environment, Physical Connection to the Grid, V2G in Household Smart Micro-Grid.
- 5 **EV Business Trend & Market Policy** 6.66% 03 Hrs
Introduction to centralized charging control, Electricity market, EV Aggregator, Algorithm for EV charging scheduling, data structure and requirement.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Acquire the knowledge regarding the fundamentals of Electric Vehicle
CO2	Define and recognize various parameters and terms related to Electric Vehicle.
CO3	Understand the construction, working and control of various electric motors used in EV application and analyze the performance of various electric cars.
CO4	To aware with the Vehicle to Grid concept.
CO5	Recognize, use and operation EV charging by PV source.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2

CO1	3	-	-	-	3	2	3	-	-	-	-	2	2	-
CO2	3	1	1	2	1	-	-	-	-	-	-	2	1	1
CO3	3	2	3	3	3	3	3	-	2	2	2	2	3	3
CO4	3	2	2	3	2	-	-	-	-	-	2	2	2	3
CO5	3	1	3	3	3	-	-	-	-	-	2	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Plug in Electric Vehicles in Smart Grids: Charging Strategy by Sumedha Rajakaruna, Farhad Shahnian, Arindam Ghosh, Springer.
2. Smart Energy Grid Engineering by Hossam A Gabbar, ELSEVIER.
3. Electric Vehicle Technology Explained by James Larminie & John Lowry, Wiley.
4. Electric Vehicle Machines & Drives Design, Analysis & Application by K.T.Chau, IEEE press, Wiley.

5. The Rediscovery of Synchronous Reluctance and Ferrite Permanent Magnet Motors by Gianmario Pellegrino, Thomas M. Jahns, Nicola Bianchi, Wen L. Soong and Francesco Cupertino , Springer.

Reference Book:

1. Influences of Electric Vehicles on Power System and Key Technologies of Vehicle to Grid by Canbing Li, Yijia Cao, Yonghong Kuang & Bin Zhou, Springer.
2. Advanced Hybrid and Electric Vehicle System Optimization and Vehicle Integration by Michael Nikowitz, Springer.
3. Introduction to Hybrid Vehicle System Modeling and Control by Wei Liu, Wiley.
4. Electrical Machines, Drives and Power Systems by Theodore Wildi, Prentice Hall.
5. Electric and Plug in Hybrid Vehicle Networks optimization and control by Emanuele Crisostomi, Robert Shorten, Sonja Studli and Fabian Wirth, CRC Press, Taylor & Francis Group.

Web Material:

- <https://www.edx.org/course/electric-cars-introduction> (Electric Cars: Introduction)
- <https://www.edx.org/course/electric-cars-technology-0> (Electric Cars: Technology)
- <https://www.edx.org/course/electric-cars-business> (Electric Cars: Business)
- <https://www.edx.org/course/electric-cars-policy> (Electric Cars: Policy)

CE392: MACHINE LEARNING FUNDAMENTALS
B.TECH. 6TH SEMESTER (Electrical)
INSTITUTE ELECTIVE -III
(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to AI and Machine Learning	12
2.	Supervised Learning	16
3.	Evaluation Metrics	06
4.	Unsupervised Learning	10
5.	Neural Network	16

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

B. Detailed Syllabus:

1.	Introduction to AI and Machine Learning	12 Hours	20%
1.1	Environment Setup. Introduction to Machine Learning Pipeline		
2.	Supervised Learning	16 Hours	27%
2.1	Regression, Simple Linear regression Least Square method, Simple Linear regression Using gradient Descent, Multiple Linear regression: Application,		
2.2	Classification, Logistic Regression, Log Loss. Sigmoid, K-Nearest Neighbour		
3.	Evaluation Metrics	06 Hours	10%

3.1	Regression (SSE, MSE, RMSE)		
3.2	Classification (Accuracy, precision, Recall, F1 Score, Confusion Matrix)		
4.	Unsupervised Learning	10 Hours	16%
4.1	Anomaly Detection		
5.	Neural Network	16 Hours	27%
5.1	Perceptron, Perceptron Algorithm, Single Layer NN, Back Propagation, Tuning, Application (Classification, Regression)		

C. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To solve difficult and complex problem of computer science using AI techniques (Apply).
CO2	To select any R&D field related to application of AI (Understand, Apply).
CO3	To understand soft computing and machine learning courses (Understand).
CO4	To develop software solution as per need of today's IT edge which requires high automation and less human intervention (Understand, Create, Apply)
CO5	To demonstrate working knowledge in Python in order to write and explore

	more sophisticated Python programs (Apply).
CO6	To apply knowledge representation, reasoning, and machine learning techniques to real-world problems (Apply).

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	2	-	-	2	-	-	1	2	2
CO2	3	3	2	2	2	-	-	-	-	-	-	-	-	1
CO3	2	-	-		-	-	-	-	-	-	-	-	1	1
CO4	3	3	3	2	1	1	-	-	-	-	-	-	1	1
CO5	2	2	2	2	2	1	-	-	-	-	-	2	-	1
CO6	3	3	3	3	2	2	-	-	-	-	-	1	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

E. Recommended Study Material:

❖ Text Books:

1. Machine Learning, Tom Mitchell, McGraw Hill, 1997. ISBN 0070428077
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, 2004

❖ Reference Books:

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
2. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification, Second Edition", Wiley & Sons, 2001.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of statistical learning", Springer, 2001.

- ❖ Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", MIT Press, 1998.

Web Materials:

1. <https://www.youtube.com/watch?v=foHSmB48rY&list=PLKvX2d3IUq586lc9gIhZj6ubpWV-OJfl4>

2. https://www.youtube.com/watch?v=CS4cs9xVecg&list=PLkDaE6sCZn6EcXTbcXluRg2_u4xOEky0
3. <https://www.youtube.com/watch?v=UzxYlbK2c7E>
4. <https://www.youtube.com/watch?v=fgtUFzxNztA>
5. <http://www-formal.stanford.edu/jmc/whatisai/whatisai.html>
6. https://www.webopedia.com/TERM/A/artificial_intelligence.html
7. https://en.wikipedia.org/wiki/Artificial_intelligence

❖ Software

1. Scikit Learn
2. PyTorch
3. TensorFlow
4. Colab
5. Keras

EE392: EV BATTERIES AND CHARGING SYSTEM

6th Semester and 3rd Year (Level III)

B. Tech. (Electrical Engineering)

INSTITUTE ELECTIVE – III

(For Minor Specialization in “Electric Vehicle Systems”)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of Batteries	06
2	EV Batteries	08
3	EV Charging	10
4	Power Electronic Converter for Battery Charging	15
5	Battery Management System	15
6	Fuel cells	06

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

C. Detailed Syllabus:

- 1 Basics of Batteries 06 Hours 10%
Basics – Types, Parameters – Capacity, Discharge rate, State of charge, state of Discharge, Depth of Discharge, Technical characteristics, Battery pack Design, Properties of Batteries, Cell and battery voltages, Charge (or Amphour) capacity, Energy stored, Energy density, Specific power, Amphour (or charge) efficiency, Energy efficiency, Self-discharge rates, Battery geometry, Battery temperature, heating and cooling needs, Battery life and number

of deep cycles

- 2 EV Batteries 08 Hours 13.33%
- Lead Acid Batteries :**Lead acid battery basics, Special characteristics of lead acid batteries, Battery life and maintenance, Battery charging, **Nickel-based Batteries :** Introduction, Nickel cadmium, Nickel metal hydride batteries, **Lithium Batteries :**Introduction, The lithium polymer battery, The lithium ion battery ,Recent development in EV batteries, Modeling of Batteries
- 3 EV Charging 13 Hours 21.67%
- Charging Infrastructure: Domestic Charging Infrastructure, Public Charging Infrastructure, Normal Charging Station, Occasional Charging Station, Fast Charging Station, Battery Swapping Station, Move-and-charge zone. Battery Chargers: Charge equalisation, Conductive (Basic charger circuits, Microprocessor based charger circuit. Arrangement of an off-board conductive charger, Standard power levels of conductive chargers, Inductive (Principle of inductive charging, Soft-switching power converter for inductive charging), Battery indication methods
- 4 Power Electronic Converter for Battery Charging 12 Hours 20 %
- Charging methods for battery, Termination methods, charging from grid, The Z-converter, Isolated bidirectional DC-DC converter, Design of Z-converter for battery charging, High-frequency transformer based isolated charger topology, Transformer less topology
- 5 Battery Management System 15 Hours 25%
- Introduction and BMS functionality, Battery pack topology, BMS Functionality, Voltage Sensing, Temperature Sensing, Current Sensing, BMS Functionality, High-voltage contactor control, Isolation sensing, Thermal control, Protection, Communication Interface, Range estimation, State-of charge estimation, Cell total energy and cell total

power, Battery state of charge estimation (SOC), voltage-based methods to estimate SOC, Model-based state estimation, Battery Health Estimation, Lithium-ion aging: Negative electrode, Lithium ion aging: Positive electrode, Cell Balancing, Causes of imbalance, Circuits for balancing

6 **Fuel cells**

06 Hours 10%

Thermodynamics of fuel cells; free energy change and cell potentials; effects of temperature and pressure on cell potential; energy conversion efficiency; factors affecting conversion efficiency; polarization losses; important types of fuel cells (hydrogen-oxygen, organic compounds-oxygen, carbon or carbon monoxide-air, nitrogen compounds-air); Equivalent circuits of fuel cells.

D. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realise the actual application of relay.

- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

E. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand the basics of battery and physics of battery.
CO2:	Understand and practically select the proper battery for electric vehicle.
CO3:	Analyze the different charging strategies for EVs.
CO4:	Design the converters for battery charging.
CO5:	Understand about the fuel cell and design battery management system

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-1	-	-	-	-
CO2:	3	1	1	-	1	-	-	-	-	-	-	1	-	-
CO3:	3	1	1	1	2	-	-	-	1	-	-	1	1	-
CO4:	3	3	3	1	2	-	-	-	1	-	1	1	2	-
CO5:	3	1	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:	
1.	Mehrdad Ehsani, Yimin Gao and Ali Emadi, “Modern Electric, Hybrid Electric and Fuel cell vehicles: Fundamentals, Theory and Design”, CRC Press, 2004.
2.	Plug in Electric Vehicles in Smart Grids: Charging Strategy by Sumedha Rajakaruna, Farhad Shahnia, Arindam Ghosh, Springer.

3.	Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.
Reference Books:	
1.	Deshang Sha, Guo Xu, HFrequency Isolated Bidirectional Dual Active Bridge DC–DC Converters with Wide Voltage Gain, Springer Nature Singapore Pte Ltd. 2019.
2.	Battery management systems, Volume I & II: Battery modeling. Artech House, 2015
3.	John Lowry and James Larminie , “Electric Vehicle Technology Explained”, Wiley
Web Material:	
1.	https://nptel.ac.in/courses/108/106/108106170/
2.	https://onlinecourses.nptel.ac.in/noc21_ee112/
3.	https://nptel.ac.in/courses/108/102/108102121/

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Theory	Practical	Total		Internal	External	Internal	External	
VII	EE441	Power System Operations and Control	4	2	6	5	30	70	25	25	150
	EE442	Power System Protection	4	2	6	5	30	70	25	25	150
	EE443	Electrical Machine Design	4	2	6	5	30	70	25	25	150
	EE444	Simulation Lab		4	4	2			50	50	100
	EE450	Summer Internship II		3	3	3			75	75	150
		Elective - I	3	2	5	4	30	70	25	25	150
	Assignment Practice/ Student Counselling		0	0	4	0	0	0	0	0	0
			15	15	34	24	120	280	225	225	850
VIII	EE458	Major Project	0	36	36	20	0	0	400	400	800
	Student Counselling		0	0	2	0	0	0	0	0	0
			00	00	38	20	0	0	400	400	800

Code	Elective - I	Code	Elective - I
EE446.01	Commissioning and Testing of Electrical Equipment	EE476.01	Advances in Power System
EE447.01	Modeling and Control of Renewable Energy Sources	EE481	Industry 4.0
EE472.01	Communication Engineering	EE482	Smart Grid Technologies and Applications

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE441: POWER SYSTEM OPERATIONS AND CONTROL
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Load Flow Analysis	10
2	Optimal Dispatch of Generation	08
3	Automatic Generation Control (AGC)	08
4	Transients in Power System	04
5	Short Circuit Studies using Bus Impedance Matrix	06
6	Power System Stability	20
7	Power System Security	04

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

C. Detailed Syllabus:

1	Load Flow Analysis	16.66%	10 Hrs
	Introduction, Formulation of Bus Admittance Matrix and Examples using Singular transformation method, Gauss – Siedel Load Flow Method and examples, Newton – Raphson Load Flow Method, Fast Decoupled Load Flow Method, Computer aided simulation of load flow analysis, Examples.		
2	Optimal Dispatch of Generation	13.34%	08 Hrs

	Introduction, non-linear function optimization, unconstrained and constrained parameter optimization, equality and inequality constraints, Operating cost of thermal power plant, economic dispatch neglecting losses and no generation limits and with consideration of generation limits, Economic dispatch including losses and examples, Derivation of loss formula, Unit commitment.		
3	Automatic Generation Control (AGC)	13.34%	08 Hrs
	Introduction, basic generation control loops, Load frequency control, generator model, prime mover model, load model, governor model, AGC in a single area system, AGC in multi-area system, tie-line bias control, Reactive power and voltage control, amplifier model, exciter model, generator model, sensor model, excitation system stabilizer.		
4	Transients in Power System	6.66%	04 Hrs
	Traveling waves on transmission lines, wave equations, specifications of traveling waves, surge impedance and wave velocity, Reflection and refraction of traveling waves, typical cases of line terminations, Successive reflections, Bewley lattice diagram, Lightning phenomena, protection of power system against lightning surges, insulation coordination.		
5	Short Circuit Studies using Bus Impedance Matrix	10.00%	06 Hrs
	Bus admittance and incidence matrices, Thevenin's theorem and Z_{BUS} , Modification of existing Z_{BUS} after addition or removal of branches, Direct determination of Z_{BUS} , Unbalanced fault analysis using Z_{BUS} , Examples.		
6	Power System Stability	33.34%	20 Hrs
	Derivation of swing equation, M and H constants, equivalent H constant and examples, Classification of power system stability, Steady state stability, derive the expression for natural frequency of oscillations and damped frequency of oscillations, example, Derivation of synchronizing power coefficient, examples. Transient stability and factors affecting it, Equal area criteria, Cases of sudden application of mechanical power input, application of three phase fault at the middle of the line or at the end of the line, Example based on equal area criteria, Point by point method for the solution of swing equation and examples, Transient stability of multi machine system. Voltage stability, voltage instability and voltage collapse, voltage stability phenomena illustration by radial feeder, active and reactive power transmission by		

	elementary models, difficulties with reactive power transmission.		
7	Power System Security	6.66%	04 Hrs
	Operating states of power system, Factors affecting power system security, Contingency analysis, Overview of security analysis, Linear sensitivity factors, AC power flow methods, Contingency relaxation.		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

Course Articulation Matrix

CO1	The knowledge regarding to planning and designing of the new interconnected power system through power flow analysis with practical simulation.
CO2	Carry out the short circuit studies of an inter connected power system and design the circuit breaker ratings.
CO3	Understand different types of stability phenomena being observed in the power system networks including the analysis methods and the design of control measures required for the improvement of system stability with practical simulation.
CO4	Understand the concept of travelling waves on transmission lines and design the components to protect the system against over-voltages.
CO5	Techniques to get the economic operation of generating units and modeling of the components for load frequency control with practical simulation.

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	-	-	-	-	-	-	2	-
CO2	2	1	3	1	2	-	-	-	-	-	-	-	2	-
CO3	1	3	2	1	2	-	-	-	-	-	-	-	2	-
CO4	1	1	2	-	-	-	-	-	-	-	-	-	2	-
CO5	1	-	1	-	1	-	-	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Power System Analysis and Design by B.R. Gupta, S. Chand
2. Power System Analysis by Grainger & Stevenson, Tata Mcgraw Hill.
3. P. Kundur, Power System Stability and Control, McGraw Hill, 1994.
4. Power System Analysis by Hadi Saadat, Tata Mcgraw Hill.

Reference Book:

1. J. Wood and B.F. Wollenberg, Power Generation, Operation and Control, John Wiley & Sons, New York, USA.
2. C.W. Taylor, Power System Voltage Stability, McGraw Hill.

Web Material:

1. <http://nptel.ac.in/downloads/108101040/>
2. <http://nptel.ac.in/courses/108104052/2>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE442: POWER SYSTEM PROTECTION
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Philosophy of Protective Relaying System	04
2	Instrument Transformer	05
3	Overcurrent Protection	08
4	Transformer Protection	07
5	Protection of Transmission Line	16
6	Protection of Generator	05
7	Induction Motor Protection	02
8	Busbar Protection	02
9	Testing, commissioning and Maintenance of Relays	03
10	Numerical Relaying	08

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

C. Detailed Syllabus:

- 1 Philosophy of Protective Relaying System** 08% 04 Hrs
Introduction, faults and abnormalities in power system, need for protection, functions of protective relaying.

Basic tripping circuit, required characteristic for protective system, zones of protection, primary and back up protection, unit and non-unit protection.

Phasor diagram of voltages and currents during various faults.

2 Instrument Transformer 08% 05 Hrs

Current transformer (CT), Saturation characteristic of current transformer.

Classification of CT, General rule applicable for selecting current transformer in electrical networks, difference between CT cores for measurement and protection, problems encountered in CT.

Voltage transformer (VT), General rule applicable for using voltage transformer in electrical networks, Coupling capacitor voltage transformer (CCVT) and its transient response

3 Overcurrent Protection 13% 08 Hrs

Protection by fuse, fuse characteristic, types of fuse, physics of fuse interruption, limitation of fuse.

Operating principle of induction relay, Types of overcurrent relay, choice of different inverse characteristic, operating characteristic of other electromechanical relays.

Plug setting and time setting for overcurrent relays, rules for setting IDMT relays for phase and ground relays.

Requirement of directional relaying, operating principle of directional overcurrent relay, 30° , 60° and 90° connection

4 Transformer Protection 12% 07 Hrs

Faults in transformer, differential protection, difficulties in differential protection of transformer.

Percentage biased differential protection, harmonic restraint relay.

Inherent phase shift in star-delta transformer, differential protection of 3 – phase transformer.

Restricted earth fault protection, miscellaneous protections for transformer.

Examples.

5 Protection of Transmission Line. 27% 16 Hrs

Different discrimination schemes for transmission line and their comparison.

Protection of radial feeder, two overcurrent and one earth fault scheme for radial feeder, three overcurrent and one earth fault scheme for transformer feeder. Examples.

Drawback of overcurrent protection, Distance protection of transmission line.

Impedance relay, reactance relay, ohm relay, Mho relay, performance of distance relays

for the faults involving arc resistances.

Three stepped characteristic of distance relays for single and double infeed, Examples based on setting of distance relays.

Complete distance protection of three-phase line for phase and ground faults.

Performance of distance relays in the event of power swing, out-of-step blocking and tripping scheme.

Other problems in distance relaying, examples.

Need for carrier aided protection, various option for carrier, components for carrier aided scheme, directional and phase comparison scheme.

6 Protection of Generator 08% 05 Hrs

Faults and abnormal conditions in generator, generator differential protection, stator inter-turn and earth fault protection.

Rotor earth fault protection, negative phase sequence protection.

Protection against loss of prime mover and loss of excitation, class A, B and C protection schemes for generator, Examples.

7 Induction Motor Protection 03% 02 Hrs

Faults and abnormalities in induction motor, protection against overloading, stalling, single phasing, and miscellaneous protection for induction motor.

8 Bus bar Protection 03% 02 Hrs

Differential protection of bus bar, selection of CT ratio for bus bar protection, high impedance bus bar differential protection, breaker back up protection

9 Testing, commissioning and Maintenance of Relays 05% 03 Hrs

Acceptance tests, type test, special tests, commissioning tests, laboratory set up for testing of different types of relays, dynamic testing of relays

10 Numerical Relaying 13% 08 Hrs

Advantages of numerical relaying.

Numerical relay hardware, facilities available in numerical relaying.

Digital signal processing, data acquisition system, sample and hold circuit, sampling theorem.

Anti-aliasing filter, sampling rate criteria, estimation of phasors, implementation of 2 – sample window in ideal condition and with real life conditions, three samples per data window, least square estimation.

Full cycle Fourier algorithm, half cycle Fourier algorithm.

D. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	Interpret the fundamentals of protective relaying principles and can select correct instrument transformer for protection.
CO2	Select the proper relay for over current protection. Can able to do relay coordination for transmission line protection by over current protection, by distance protection and by carrier current protection.
CO3	Create/Design protection scheme for three phase power transformer, induction motor and bus bar protection with CT and relay connection and settings.
CO4	Develop skill for different protection scheme for generator. Design the protection scheme for generator and suggest relay settings with connections diagram of CTs for the relays applied for generator protection.
CO5	Follow various types of tests of relays, commissioning and maintenance of relays. Also, recognize the construction and working of numerical relaying. Think about relays in advanced level.

Course Articulation Matrix

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	-	-	-	-	1	-	-	-	-
CO2	3	2	2	1	2	-	-	-	-	-	-	-	2	-
CO3	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO4	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO5	3	1	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realise the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

F. Recommended Study Material:

Text Book:

1. Power System Protection and Switchgear by B.A. Oza, N.C. Nair, R.P. Mehta and V.H. Makwana, Mcgraw Hill, 2010
2. Protection and Switchgear by Bhavesh Bhalja, R. P. Maheshwari and N. G. Chothani, Oxford University Press, 1st Edition, 2011
3. Fundamentals of Power System Protection by Y.G. Paithankar, S.R. Bhide, Prentice Hall, India, 2003

Reference Book:

1. Power System Protection and Switchgear by Badri Ram and D N Vishwakarma, Tata Mcgraw Hill, 2001
2. Power System Protection Static Relays by T.S. Madhavrao, Tata Mcgraw Hill, 1989

Web Material:

- http://www.cdeep.iitb.ac.in/nptel/Electrical%20Engineering/Power%20System%20Protection/TOC_M1.html
NPTEL

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE443: ELECTRICAL MACHINE DESIGN
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basic Considerations in Electrical Machine Design	04
2	Design of Transformer	19
3	Induction Machines Design	19
4	Synchronous Machines Design	18

Total hours (Theory):60Hrs
Total hours (Lab) :60Hrs
Total hours : 120Hrs

C. Detailed Syllabus:

- 1 Basic Considerations in Electrical Machines Design 16.66% 04 Hrs
Design factors, Limitations in design, Modern trends in design of electric machines, Temperature rise, Expression for temperature rise, heating & cooling time constants, examples.
- 2 Design of Transformer 31.67% 19 Hrs
Specification, Output equation of transformer, Output equation-Volt per turn, Stacking factor, Ratio of iron loss to copper loss, Relation between core area and weight of iron and copper, Optimum designs, variation of output and losses in transformer with linear dimensions ,examples, Design of core, Choice of flux density

and current density, Choice of window space factor, window dimensions, Design of yoke, Overall dimensions, examples, Design of high voltage and low voltage winding, examples, Estimation of operating characteristics: Primary & Secondary resistance, Leakage reactance of windings, Regulation, examples, Mechanical forces, No load current calculation, Change of parameters with change of frequency, Temperature rise of transformer, Design of tank, examples.

3 Induction Machines Design

31.67% 19 Hrs

Choice of specific electrical loadings, Choice of specific magnetic loadings, Output equation, Separation of D & L, Examples, Stator Design: Turns per phase, Stator conductors, Shape of stator slots, Number of stator slots, Area of stator slots, Length of mean turn, Stator teeth, Stator core, examples, Length of air gap, Relation for calculation of length of air gap, Examples, Squirrel Cage Rotor Design: Number of rotor slots, Rules for selecting rotor slots, Reduction of harmonic torques, Design of rotor bars & slots, Design of end rings, examples, Wound Rotor Design: Number of rotor slots, number of rotor turns, area of rotor conductors, Rotor windings, Rotor teeth, design of rotor core, examples, Estimation of operating characteristics- No load current calculation, short circuit current calculation, Stator and rotor resistance and leakage reactance calculation, examples, Dispersion coefficient – Effect on maximum power factor and overload capacity, Effect of change of air gap length, number of poles and frequency, Relation between D&L for best power factor, examples.

4 Synchronous Machines Design

30% 18 Hrs

Choice of specific electrical loadings, Choice of specific magnetic loadings, Design of Salient pole machines: Output equations, Main dimension, Short Circuit Ratio, Effect of SCR on machine performance, Length of air gap and shape of pole face, Examples, Armature design: Number of armature slots, Coil span, Turns per phase, Conductor section, Slots dimension, Length of mean turn, Stator Core, Elimination of harmonics, Estimation of air gap length, examples, Design of rotor, Height of pole, Design of damper winding, Height of pole shoe, Pole profile drawing, examples, Design of magnetic circuit, Determination of full load field MMF, examples, Design of field winding, examples, Determination of Direct and Quadrature axis synchronous reactance.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Interpret the fundamentals of electrical machine design, modern trends, its operational effects on the environment like heat and gases liberated and many more.
CO2	Design the overall transformer which includes the design of core, windings and tank and also estimate the resistance, reactance, regulations and the mechanical forces.
CO3	Design Induction machine which includes designing of stator, air-gap, squirrel cage rotor and wound rotor.
CO4	Design of salient pole machine, armature, rotor and field winding as well as determination of direct and quadrature axis synchronous reactance.
CO5	Acquire entrepreneurial outlook to design different machines for the small and medium scale manufacturer.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	3	2	2	-	2	-	1	1	2	-	2	--
CO2	3	2	2	2	1	-	2	-	2	-	1	1	2	-
CO3	3	2	2	2	1	-	2	-	2	-	1	1	2	-
CO4	3	2	2	2	1	-	2	-	2	-	1	1	2	-
CO5	3	3	-	-	-	2	-	1	3	3	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.

- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. A course in Electrical machine design by A. K. Sawhney & A. Chakrabarti, Dhanpat Rai & Co.

Reference Book:

1. The performance and design of alternating current machines by M. G. Say, CBS Publishers & Distributors
2. Design of rotating electrical machines by Juha Pyrhonen, Tapani Jokinen, Valeria
3. Design of electrical machines by K. G. Upadhyay, New age international publishers.
4. Design of electrical machines by V. N. Mittal & A. Mittal, Standard Publishers distributors.
5. Electric Machinery 6th Edition by A. E. Fitzgerald, Charles Kingsley, Stephen. D. Umans Tata Mcgraw Hill.
6. Elements of electrical design 2nd edition by J. G. Jamnani, Mahajan publishing house.

Web Material:

- NPTEL

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE444: SIMULATION LAB
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	4	4	2
Marks	00	50	50	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Software introduction and mathematical equations solution	10
2	Electrical Power System simulation	20
3	Electrical Machines Simulation	10
4	Power Electronics Simulation	20

Total hours (Lab) :60 Hrs
Total hours : 60Hrs

C. Detailed Syllabus:

Sr. No.	Suggested List of Practical
1	To study different types of solvers used in MATLAB software.
2	To study symbolic Toolbox in MATLAB software.
3	To perform Linear equations and matrix solution using MATLAB software.
4	Formation of bus Admittance and Impedance matrix Using MATLAB software.
5	Computation of transmission line parameter using MATLAB software.
6	To simulate single phase half wave and full wave bridge uncontrolled rectifier with R, RL & RLE load.
7	To simulate single phase half wave and full wave bridge controlled rectifier with

- R, RL & RLE load.
- 8 To simulate speed control of separately excited DC Machines.
 - 9 To simulate the three phase full wave circuit using MATLAB software.
 - 10 To perform Open Circuit, Short Circuit& Load tests on Single Phase transformer using MATLAB.
 - 11 To perform fault analysis using Power world simulator and using MATLAB software
 - 12 Calculation of leakage inductance of transformer using FEM software.
 - 13 Load flow analysis using Mi Power software.
 - 14 Perform relay Co- ordination for radial system using Mi Power software.
 - 15 Long term load forecasting using Mi Power software.
 - 16 Harmonic Analysis using Mi Power software.
 - 17 Introduction of PSIM simulation software.
 - 18 To study and simulate Boost convertor with PI controller.
 - 19 To study and simulate Multilevel inverter.
 - 20 To study and simulate Flyback convertor

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	At the end of course, the students will acquire the knowledge regarding the simulation of different electrical machines and electrical power system network using different software.
CO2	This will be very helpful to them in dealing with real system in industries and to carry out consultancy.
CO3	Use of different software makes their software skills which help them to develop their future goals.
CO4	Symbolic toolbox study help student to develop their mathematical skill using MATLAB software.
CO5	Power Electronics model of Inverter, convertor helps to understand basic of electronic circuit design in real system.

Course Articulation Matrix
Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	-	-	-	-	-	-	3	3
CO2	2	2	3	3	-	-	1	-	-	-	-	-	-	3
CO3	3	-	-	2	3	-	-	2	-	-	1	2	-	3
CO4	3	2	3	1	-	-	-	-	-	-	-	-	3	2
CO5	3	3	2	2	-	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

1. At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
2. Laboratories will be conducted with the aid of multi-media projector.
3. A student has to prepare a laboratory term work as per instruction given by lab instructor.
4. Attendance is compulsory in laboratory, which carries five marks of the overall evaluation.
5. Two viva voce will be conducted during the semester and average of two will be considered as a part of overall evaluation.

F. Recommended Study Material:

Manual of software provided by lab instructor.

Reference Book:

1. Stephen. J. Chapman, Electric Machinery Fundamentals , McGraw Hill
2. Hadi Saadat, Power System Analysis , Tata McGraw Hill

Web Material:

1. <http://www.prdcinfotech.com/business/software-engineering-group/software-products/mipower/>
2. <https://www.mathworks.com/products/matlab.html>
3. <https://www.powerworld.com/>
4. <http://www.femm.info>
5. NPTEL

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE450: SUMMER INTERNSHIP II
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week (In Summer Break)	00	30	90	3
Marks	00	150	150	

B. Instructional Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.

- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

C. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division

- Administrative tree (if available)
- Main functions of the company
- Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

D. Course Outcomes (COs):

After completion of the course students will able:

CO1	To solve engineering problems by working in a team with across other disciplines.
CO2	To apply knowledge and skills learned in company/industry/organization to real-world problems.
CO3	To use experience related to professional and ethical issues in the work environment

CO4	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context
CO5	To demonstrates knowledge of contemporary issues related with engineering in general by adopting new tools and technologies

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**EE446.01: COMMISSIONING & TESTING OF ELECTRICAL
EQUIPMENT**

**7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)**

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Extracts from Indian Electricity Rules – 1956 and Principles and Planning of commissioning, maintenance and testing	04
2	Commissioning & Testing of Transformer	15
3	Commissioning & Testing of Induction Machines	13
4	Commissioning & Testing of Alternator	13
5	Commissioning & Testing of Sub Station Equipments	15

Total hours (Theory) :45Hrs

Total hours (Lab) :30Hrs

Total hours :70Hrs

C. Detailed Syllabus:

1	Extracts from Indian Electricity Rules – 1956 and Principles and Planning of Maintenance	5%	02Hrs
	Extracts from Indian Electricity Rules Introduction, Routine Maintenance, Periodical Maintenance & Maintenance on Fault		
2	Commissioning & Testing of Transformer	29%	13Hrs

	Open circuit and Short circuit test, Impedance measurement test, Insulation resistance measurement, Polarization Index measurement, Transformer oil tests, Transformer oil purification, Power frequency voltage withstand test, Separate source voltage test, Induced overvoltage test, Impulse voltage test, Partial Discharge Test, Tan delta test, Short-circuit withstand test, Overvoltage inter-turn test, Measurement of temperature, Core insulation voltage test, drying out of transformer, Parallel operation of transformer, Parallel operation of transformer: Load sharing problem, Phase shifting/ Phase group, Correct pairing of transformer, Transformer Troubles: Switching & short circuit surges, harmonics, unbalanced loading, voltage regulation, open circuits in phase windings and lines, protective gear troubles, oil deterioration, Commissioning steps for transformer, Troubleshooting & Maintenance of transformer.		
3	Commissioning & Testing of Induction Machines	22%	10Hrs
	Commissioning of Motors: Insulation test and drying out of windings, Temperature rise test, induction motor stator and rotor interaction, balance and vibration, ventilation and cooling, Hammer test, Testing against variation of voltage/current/frequency, Testing of auxiliaries, Degree of protection(IP Grade), AC Motor Troubles: Insulation failure due to transient voltages, low starting torque, pull out torque and stalling, low power factor, excessive slip, crawling, single phasing, reversal of one phase winding, starting transients in squirrel cage induction motors, overheating, pull over, vibration having electrical origins, slip ring wear, shaft currents, ball and roller bearing trouble. Commissioning steps for Induction motor, Commissioning of Induction Generator.		
4	Commissioning & Testing of Alternator	22%	10Hrs
	Commissioning of alternator: Preparation and drying out of alternator windings before commissioning, insulation resistance measurements, high voltage tests, measurements of temperature of windings, extra high voltage alternators, alternators protective gear tests, trip circuit supplies and tripling tests, starting time and rate of picking up load, control of auxiliaries Alternator stator and rotor interaction, , effect of excitation on stator current of an alternator in parallel with a large system, Alternator troubles: Instability of excitors, complete loss of field, failure of exciter voltage to build up, motor driven excitors, alternator instability, neutral inversion with grounded voltage transformer and un-grounded power system, heating of copper conductors, shaft currents and bearing troubles, slip ring wear on alternator rotor, Unbalanced stator currents, transient torques, heating of stator core end punching and clamps		
5	Commissioning & Testing of Sub Station Equipments	22%	10Hrs
	Bus bar: Temperature rise test, Rated short time current test, Power frequency voltage		

	withstand test, Impulse / surge testing, Vibration test. Isolator Testing: Short circuit test, Charging current, Making & Breaking test, Inductive current making & breaking test. Circuit Breaker testing: Mechanical endurance test, Temperature rise test, Impulse & surge testing, Short circuit testing of circuit breakers, synthetic testing of circuit breakers, mechanical operation and adjustment, electrical auxiliaries, Troubleshooting & maintenance of circuit breakers, Protective relay tests: relay setting, C.T. & P.T. Testing, Station Batteries for D.C. Supply, Testing & Commissioning of Lightning Arrestor, and Substation Commissioning by Thermography.
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D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Memorize, recognize, create and interpret the Single Line Diagram of Sub-Stations and Indian Electricity Rules
CO2:	Understand and discriminate the designed specifications of transformer and induction motor
CO3:	Demonstrate, simulate, analyze and evaluate the performance characteristics of transformers, induction machines, alternators and switchgear equipments by conducting the routine/commissioning tests as per IEC, IEEE, CIGRE and other Indian and international standards to get the opportunity to work as a team member/leader in multi-disciplinary branches
CO4:	Prepare the test, maintenance and troubleshooting report for static and dynamic machines
CO5:	Acquire the knowledge regarding the installation and commissioning of electrical equipment.

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	2	-	2	-	-	-	-	-	-	3	-	1	1	-
CO2:	3	-	2	-	-	-	2	-	2	-	-	3	2	-
CO3:	3	2	2	3	1		2	2	2	3	-	2	3	2
CO4:	3	3	-	1	-	-	-	2	1	3	-	2	2	-

CO5:	2	1	-	-	-	2	-	-	-	-	-	2	2	-
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1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. The Commission of electrical plant by RCH Richardson, Chapman & Hall.
2. Substation Design and Equipment by P.V. Satnam and P. V. Gupta
3. Commissioning of electrical equipment, by Prof. M. A. and S. M. Chaudhari, Tech Max Publications.

Reference Book:

1. Testing, Commissioning and maintenance of electrical equipments by S. S. Rao, Khanna Publisher.

2. Switchgear and Protection by J. B. Gupta by S. K. Kataria and Sons.
3. Electric Machinery Fundamentals by Stephen. J. Chapman, McGraw Hill
4. Electric Machinery 6th Edition by A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, Tata McGraw Hill
5. The performance and design of alternating current machines by M. G. Say, CBS Publishers & Distributors
6. Electrical Machinery by Dr. P. S. Bimbhra, Khanna Publisher
7. Electrical Machines by D.P. Kothari & I. J. Nagrath, Tata McGraw Hill

Web Material:

- http://nptel.iitm.ac.in/courses/IITMADRAS/Electrical_Machines_II/Testing/index.php
- http://www.ece.ualberta.ca/~knight/ee332/synchronous/s_main.html
- http://www.ece.ualberta.ca/~knight/ee332/induction/i_main.html
- http://nptel.iitm.ac.in/courses/IIT-MADRAS/Electrical_MachinesI/Testing/index.php

**EE447.01 : MODELING AND CONTROL OF RENEWABLE ENERGY
SOURCES**

**7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)**

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction of Renewable Energy Sources	04
2	Solar Photovoltaic System	15
3	Wind Energy System	11
4	Fuel Cell	04
5	Renewable Energy Sources in Smart Grid & Electric Vehicle	11

Total hours (Theory) :45 Hrs
Total hours (Lab) :30 Hrs
Total hours :75 Hrs

C. Detailed Syllabus:

- 1 Introduction of Renewable Energy Sources** **8.9% 04 Hrs**
Global and Indian renewable energy scenario, various renewable energy sources, Energy units and conversions, Energy Storage, introduction to IEEE 1547 standard.
- 2 Solar Photovoltaic System** **33.4% 15 Hrs**
History of Solar Energy Usage, Equivalent circuit of photovoltaic cell, From cell to module and from module to array, I-V curve under standard test condition, Fill Factor, impact of temperature and insolation on I-V curve, Shading impacts on I-V curve, I-V curve for various types of loads (resistor, DC motor, battery), MPPT Concept, DC-DC converters (boost,

buck, buck-boost) for MPPT, MPPT methods: reference cell methods, Perturb & Observation method, power slope method, standalone PV system design, Grid connected PV system principle, PV-Grid connection topologies: unfold circuit, double stage, single stage, classification of grid connected inverter: centralized inverter, string inverter & micro inverter, PV-Grid connected system architecture, PV-Grid connected system design, Economics of PV System, PV system in micro-grid, Emerging PV cell technologies.

3 Wind Energy System 24.4% 11 Hrs

Introduction, Wind Characteristics, Wind turbines: fixed speed & variable speed wind turbine, components of wind energy conversion system, Types of generators: Induction generator, synchronous generator, Permanent Magnet Synchronous Generator, Doubly Fed Induction Generator, Pitch angle control, Grid Integration of wind energy system, Economics of wind energy conversion system.

4 Fuel Cell 8.9% 04Hrs

Construction & Operation of fuel cell, types of fuel cells, comparison of different component and characteristic of fuel cell, fuel cell equivalent circuit, need for dynamic modeling of fuel cell, Polymer Electrolyte Membrane Fuel Cell (PEMFC) model structure, Solid Oxide Fuel Cell (SOFC) model structure.

5 Renewable Energy Sources in Smart Grid & Electric Vehicle 24.4% 11 Hrs

Introduction of Smart Grid, History of Electric Vehicle, Architecture of PV based electric vehicle charging: Conventional method, Centralized architecture, Distributed architecture, single stage architecture, Control Strategy for EV charging, Impact of EV charging on Electrical Distribution System, V2G Concept, PV powered water pumping, Smart Meters.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Recognize the electric circuits connected to various Renewable Energy Sources
CO2:	Acquire the knowledge regarding the fundamentals of Smart Grid and Electric Vehicle
CO3:	Understand the construction, working and control of solar and wind energy system
CO4:	Aware with the performance characteristics of Renewable Energy Sources.

CO5: Design & Analyze the performance of PV system

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	-	1	-	-	-	1	1	-	1	2	-
CO2:	3	2	2	1	1	-	2	-	1	-	1	-	2	-
CO3:	3	2	2	1	1	1	2	-	1	-	1	1	2	1
CO4:	3	1	-	1	2	-	2	1	-	-	2	-	2	1
CO5:	3	1	3	3	3	1	3	-	1	-	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Renewable & Efficient Electric Power System by Gilbert M. Masters, Wiley.
2. Design of Smart Power Grid Renewable Energy Systems by Ali Keyhani, Wiley.
3. Modeling and Control of Fuel Cells by M. Hashem Nehrir, Caisheng Wang, Wiley.
4. Solar PV and Wind Energy Conversion Systems: an introduction to theory modeling with MATLAB/Simulink and role of soft computing techniques by S. Sumathi, L. Ashok Kumar, P. Surekha, Springer.
5. Plug in Electric Vehicles in Smart Grids: Charging Strategy by Sumedha Rajakaruna, Farhad Shahnia, Arindam Ghosh, Springer.
6. Smart Energy Grid Engineering by Hossam A Gabbar, ELSEVIER.

Reference Book:

1. Grid Connected PV systems Design and Installation by Global Sustainable Energy Solutions.
2. Solar Energy principles of thermal collection and storage by S.P. Sukhatme, J.K. Nayak, Tata McGraw Hill.
3. Solar Photovoltaics fundamentals, technologies and applications by Chetan Singh Solanki, PHI Learning Pvt Ltd.

Web Material:

- https://onlinecourses.nptel.ac.in/noc18_ee35/course (Design of Photovoltaic System)

EE472.01: COMMUNICATION ENGINEERING (PE III)
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Amplitude Modulation	10
2	Angle Modulation	10
3	Receivers	05
4	Probability Theory & Random Processes	10
5	Information Theory	05
6	Principle of digital data Transmission	05

Total hours (Theory) :45Hrs
Total hours (Lab) :30Hrs
Total hours :75Hrs

C. Detailed Syllabus:

1 Amplitude Modulation 22% 10Hrs

Amplitude Modulation, Amplitude Modulation Index, Modulation Index for Sinusoidal AM, Frequency Spectrum for Sinusoidal AM, Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM, Non sinusoidal Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation.

2 Angle Modulation 22% 10Hrs

Frequency Modulation, Sinusoidal FM, Frequency Spectrum for Sinusoidal FM, Average Power in Sinusoidal FM, Non-Sinusoidal Modulation: Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation, Equivalence Between PM and FM, Sinusoidal Phase Modulation, Digital Phase Modulation, Angle Modulator Circuits, FM Transmitter, Angle Modulation Detectors, Automatic Frequency Control, Amplitude Limiters, Pre-Emphasis and De-Emphasis, FM Broadcast Receivers.

3 Receivers 11% 05Hrs

Homodyne, Heterodyne and Super Heterodyne Receivers, Tuning Range, Tracking, Sensitivity and Gain, Image Rejection, Spurious Responses, Adjacent Channel Selectivity, AGC, Double Conversion, Electronically Tuned Receivers (ETRS), Integrated-Circuit Receivers.

4 Probability Theory & Random Processes 22% 10Hrs

Introduction to digital communication, Probability, Conditional Probability of Independent Events, Relation between probability and probability density, CDF, PDF, Gaussian PDF, Threshold Detection, Random Variable, Variance of a random variable, co relation between random variables, Mean and variance of sum of random variables, Statistical Average, Chebyshev In Equality, Raleigh Probability Density, The Central Limit Theorem, correlation, random processes, PSD of random processes(along with examples), Multiple random processes, band pass random processes.

5 Information Theory 11% 05Hrs

Concept & Measure Of Information, entropy, Error Free Communication Over A Noisy Channel, The Channel Capacity Of A Discrete Memory Less Channel, Channel Capacity Of A Continuous Channel, Practical Communication Systems In Light Of Shannon's Equation.

6 Principle of digital data Transmission 12% 05Hrs

Line Coding, PSD of On/Off Signal, Bipolar Signal, Duo Binary Signal, Pulse Shaping, Nyquist First and Second Criterion for Zero ISI, Regenerative Repeaters, Detection Error Probability, M-ary System, Scrambling, Digital Carrier System, Digital Multiplexing.

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Understand different techniques for amplitude modulation and demodulation
CO2:	Able to get Depth Analysis in Noise Performance of Various Receivers
CO3:	Understand and Analyze Channel characteristics
CO4:	Design a digital communication system
CO5:	Simulate communication systems and analyze their performance on MATLAB software

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2:	2	1	-	1	-	-	-	-	-	-	-	-	1	-

CO3:	2	1	-	1	-	-	-	-	-	-	-	-	1	-
CO4:	-	-	3	1	-	-	-	-	-	-	-	-	1	-
CO5:	-	-	-	1	3	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Electronic Communications by Dennis Roddy& John Coolen IV Edition PHI.
2. Modern Digital and Analog Communication System III Edition By B.P.Lathi Pub
3. B.P.Lathi ,Digital and Analog communication system , LCBS Publication

4. B. Carlson ,Communication system , TMH Publication
5. Simon Haykin, Digital and analog communication system , Willey Publication

Reference Book:

1. Electronic Communications by Dennis Roddy & John Coolen IV Edition PHI.
2. Modern Digital and Analog Communication System III Edition By B.P.Lathi Pub
3. Taub & Schilling, Principle of communication system, TMH Publication
4. Wayne Wolf, Computers as components Principles of embedded computing
5. Tammy Noergaard, Embedded Systems Architecture, Elsevier
6. Anderw N. SLOSS, Dominic SYMES, Chirs WRIGHT, ARM System Developer's Guide, Elsvier.
7. Daniel W.Lewis, Fundamentals of Embedded Software: Where C and Assembly Meet, Prentice Hall

Reading Materials, Web Materials with Full Citations:

- 1.<http://adp.mmu.edu.my/enotes/asma/dtc5038/notes/lecture%20notes%203%20chapter%203.pdf>
2. <http://web.cecs.pdx.edu/~ece2xx/ece223/slides/communicationsx4.pdf>
- 3.<http://stattrek.com/Lesson2/Binomial.aspx?Tutorial=sat>
4. <http://www.youtube.com/watch?v=IYdiKeQ9xEI>
5. <http://www.complextoreal.com/tutorial.htm>

EE476.01: ADVANCES IN POWER SYSTEM (PROGRAMME ELECTIVE-IV)
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Smart Grid	12
2	Restructuring of Power System	08
3	State Estimation in Power System	10
4	Substation Automation	05
5	Load Forecasting	10

Total hours (Theory) : 45 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 75 Hrs

C. Detailed Syllabus:

1 Introduction to Smart Grid 33.33% 12 Hrs

Introduction to Smart Grid, Architecture of Smart Grid System, Standards for Smart Grid System, Elements and Technologies of Smart Grid System, Distributed Generation Resources, Wide Area Monitoring Systems, Phasor Estimation, Digital relays for Smart Grid Protection, Islanding Detection Techniques, Smart Grid Protection, Modelling of Storage Devices, Modelling of DC Smart Grid components, Operation and control of AC Microgrid, Simulation and Case Study of AC-DC Hybrid Microgrid, Demand side management. of Smart Grid, Energy Management, Design of Smart grid and Practical Smart Grid case study, System Analysis of AC/DC

2 Restructuring of Power System 16.66% 08 Hrs

Concepts of regulation and deregulation, Characteristic of regulated power system, need to restructured the power system, Overview of a deregulated industry, Structure of deregulated power system, Different entities in deregulated power system, Responsibilities of independent system operator, Trading arrangements: Pool, Bilateral, Multilateral, Power Exchange, Energy auction and market clearing

prices, Available transfer capability, Congestion management, Ancillary services

3 State Estimation in Power System 20% 10 Hrs

Power system state estimation: Past, Present and Future, Weighted least square estimation, Hessian matrix formulation, State estimation of AC network, Detection and identification of bad data measurement, estimation of quantities not being measured, Network observability and Pseudo measurement, Application of state estimation.

4 Substation Automation 13.34% 05 Hrs

Introduction to distribution automation, Automation Systems and Functionality, Typical Integrated Distribution Utility, Key Design Parameters of SCADA/DMS system, Levels of Automation, Applications and benefits of DAS.

5 Load Forecasting 16.66% 10 Hrs

Introduction, Estimation of Average and trend terms, Estimation of periodic components, Estimation of Stochastic components: Time series approach, Auto-Regressive Model, Auto-Regressive Moving, Average Models, Kalman Filtering Approach, On-line techniques for non-stationary load prediction.

D. Course Outcomes (Learning Outcomes):

Upon successful completion of this course, a student would be able to acquire the knowledge regarding:

1. Past, present and future of Power System. Also, Restructuring of an Indian Power System. (1,2,3,11, PSO1, PSO1)
2. Requirement of power system state estimation and methodologies used for state estimation with practical simulation. (1,2,3,5, PSO1, PSO2)
3. Substation Automation and Distribution system monitoring, control and design. (1,2,3,4,5, PSO1, PSO2)
4. To learn data analysis and develop algorithm to predict future energy demand for smart grid. (1,2,3,4,5, PSO1, PSO2)
5. The fundamentals and detail knowledge regarding smart grid technologies with operation and control. (1,2,4,5,7, PSO1)

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1								1		2	1
CO2	3	2	1		2								2	1

CO3	3	2	3	1	2								2	1
CO4	3	3	2	2	3								2	2
CO5	3	2	3	1	2		1						3	

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. J. Wood and B.F. Wollenberg, Power Generation, Operation and Control, JohnWiley & Sons, New York, USA, 1996.
2. Power System Analysis by Hadi Saadat, Tata Mcgraw Hill.
3. Modern Power System Analysis by D.P. Kothari & I. J. Nagrath, Tata McgrawHill.
4. Restructured Power System by S.A. Khaparde and A.R. Abhyankar, Narosa Publication.

5. Electric Power Distribution Systems by A.S. Pabla, 6th Edition, Tata McgrawHill.

Reference Book:

1. Advances in Power Systems and Energy Management by Priyadarshi, N., P., S., Ghadai, R.K., Panda, A.R., Patel, R. (Eds.), Proceedings of ETAEERE 2020, Springer.
2. Power System Protection and Switchgear by Badri Ram and D N Vishwakarma, Tata Mcgraw Hill, 2001
3. Power System Protection Static Relays by T.S. Madhavrao, Tata Mcgraw Hill, 1989

Web Material:

1. <http://www.inase.org/library/2014/interlaken/POW.pdf>
2. https://onlinecourses.nptel.ac.in/noc18_ee42/
3. A. P. Sakis Meliopoulos, George J. Cokkinides, Xing Yong Chao “A NEW PROBABILISTIC POWER FLOW ANALYSIS METHOD”, IEEE Transactions on Power Systems, Vol. 5, No. 1, February 1990.

EE481: INDUSTRY 4.0
7th Semester and 4rd Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Industry 4.0	08
2	A Conceptual Framework for Industry 4.0:	08
3	Technology Roadmap for Industry 4.0 :	07
4	Talent Development for Industry 4.0	07
5	Technologies and Applications	15

Total hours (Theory) : 45 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 75 Hrs

C. Detailed Syllabus:

1	Introduction to Industry 4.0	17.78%	8 Hrs
	Historical context, what is this industry 4.0, different changes in the revolution, transform the industries towards Industry 4.0		
2	A Conceptual Framework for Industry 4.0:	17.78%	8 Hrs
	Introduction, Main Concepts and Components of Industry 4.0, State of Art, Supportive Technologies, Proposed Framework for Industry 4.0.		
3	Technology Roadmap for Industry 4.0 :	15.56%	7 Hrs
	Introduction, Proposed Framework for Technology Roadmap, Strategy Phase, Strategy Phase, New Product and Process Development Phase.		
4	Talent Development for Industry 4.0	15.56%	7 Hrs
	Introduction, Skill Requirements in the Digital World, Talent Development Practices for Industry 4.0, Conclusion		

5	Technologies and Applications	33.32%	15 Hrs
	Internet of Things and New Value Proposition: Examples for IoTs Value Creation in Different Industries: Smart Agriculture, Smart City, Smart Life, Wearable Technologies, Smart Health etc. IoTs Value Creation Barriers: Standards, Security and Privacy Concerns Advances in Robotics in the Era of Industry4.0: Introduction, Recent Technological Components of Robots- Advanced Sensor Technologies, Internet of Robotic Things, Cloud Robotics, and Cognitive Architecture for Cyber-Physical Robotics, Industrial Robotic Applications- Manufacturing, Maintenance and Assembly.		

D. Course Outcomes (Learning Outcomes):

Upon successful completion of this course, a student will be able to

CO1:	Describe Industry 4.0 and its revolution.
CO2:	Main concepts and supportive technologies of Industry 4.0.
CO3:	Product and Process development including industry 4.0.
CO4:	Describes robotics technologies and IOT values in industry 4.0.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	2	2	-	-	-	-	-	-	-	-
CO3	-	3	3	-	-	-	-	-	-	-	2	-	3	-
CO4	-	2	-	-	2	2	-	-	-	-	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Alp Ustundag and Emre Cevikcan, "Industry 4.0: Managing the Digital Transformation".
2. Bartodziej, Christoph Jan, "The Concept Industry 4.0".
3. Klaus Schwab, "The Fourth Industrial Revolution".
4. Christian Schröder, "The Challenges of Industry 4.0 for Small and Medium-sized Enterprises".

Web Material:

<https://nptel.ac.in/courses/106/105/106105195/>

(Videos and Notes by Prof. Sudip Misra, IIT Kharagpur)

EE482: SMART GRID TECHNOLOGIES AND APPLICATIONS
(PROGRAMME ELECTIVE-IV)
7th Semester and 4th Year (Level IV)
B.Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

The objectives of the course are:

1. To understand the basic concepts, components and architecture of smart grid.
2. To understand the various measurement technologies in smart grid.
3. To educate the importance of renewable energy in smart grid.
4. To know about battery technology and energy storage.
5. To brief about role of Electric Vehicles in smart grid.

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction	09
2	Sensors and Measurement	09
3	Distributed Generation	09
4	Energy Storage	09
5	Electric Vehicles	09

Total hours (Theory) : 45 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 75 Hrs

E. Detailed Syllabus:

- 1 Introduction 20% 09 Hrs
Today's Grid Versus Smart Grid, Rationale for Smart Grid, Computational Intelligence, Power System Enhancement, Communication and Standards, Environment and Economics, Shareholders Roles and Function, Architecture, Functions of Components
- 2 Sensors and Measurement 20% 09 Hrs
Sensors for Smart Grid, Monitoring and Measurement Technologies, PMU, Smart meters, Smart Appliances, Multi Agent Systems (MAS) Technology, Micro grid and Smart grid comparison, Wide Area Measurement
- 3 Distributed Generation 20% 09 Hrs
Solar Energy, PV Systems, Wind turbine Systems, Biomass, Small and Micro Hydro Power, Fuel Cell, Geothermal heat pumps.
- 4 Energy Storage 20% 09 Hrs
Batteries, Flow Batteries, Fuel Cell and hydrogen electrolytes, Flywheel, Super conduction magnetic energy storage systems, super capacitors, Simulation and case studies
- 5 Electric Vehicles 20% 09 Hrs
Plugin Electric Vehicles and hybrid, Vehicle classes, Vehicle Architecture, Grid to Vehicle (G2V) Charging, Grid Impacts, Vehicle to Grid (V2G)

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
10	20	25	20	15	10

G. Course Outcomes (Learning Outcomes):

Upon successful completion of this course, a student would be able to acquire the knowledge regarding:

1. Summarize the basic concepts, components and architecture of smart grid.
2. Compare the various measurement technologies in smart grid.

3. Demonstrate modelling of renewable energy sources which are used in smart grid.
4. Illustrate about battery technology and energy storage.
5. Interpret about role of Electric Vehicles in smart grid.

Mapping of course outcomes with program outcomes

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	1	-	2	-	-	-	1	1	3	2
CO2	3	3	1	1	3	-	1	-	-	-	2	-	3	1
CO3	3	3	1	2	-	-	-	-	-	-	2	-	3	2
CO4	3	2	2	-	1	-	-	-	-	-	-	-	2	-
CO5	3	2	2	2	-	1	2	-	1	-	1	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

I. Recommended Study Material:

Text Book:

1. James Momoh, “Smart Grid: Fundamentals of design and analysis”, John Wiley & sons Inc, IEEE press 2012.
2. Janaka Ekanayake, Nick Jenkins, Kithsiri Liyanage, Jianzhong Wu, Akihiko
3. Yokoyama, “Smart Grid: Technology and Applications”, John Wiley & Sons Inc, 2012.
4. Lars. T. Berger, K. Iniewski, “Smart Grid: Applications, Communications & Security” Wiley India Pvt. Ltd, Reprint 2015.

Reference Book:

1. Fereidoon P. Sioshansi, “Smart Grid: Integrating Renewable, Distributed & Efficient Energy”, Academic Press, 2012.
2. Clark W. Gellings, “The smart grid: Enabling energy efficiency and demand response”, Fairmont Press Inc, 2009.
3. Qi Huang, Shi Jing “Innovative Testing and Measurement Solutions for Smart Grid”, John Wiley & Sons Inc, 2015.

Web Material:

1. Introduction to Smart Grid: - <https://nptel.ac.in/courses/108107113>.
2. Fundamentals of Electric Vehicles: Technology & Economics: - <https://nptel.ac.in/courses/108106170>.

J. List of Experiments:

1. Development of hybrid smart grid system using MATLAB/SIMULINK.
2. Simulation of 1-phase, 240 Vrms, 3500 W Transformer less Grid-Connected PV Array.
3. 24-hour Simulation of a Vehicle-to-Grid (V2G) System.
4. Simulation of a 9 MW wind farm using Doubly-Fed Induction Generator (DFIG) driven by a wind turbine.

5. Simulation of Solid-Oxide Fuel Cell Connected to Three-Phase Electrical Power System.

K. Course Assignments:

	% weightage (Approx.)
Midterm exams	20
Final exams	70
Assignments	10

L. Course Assessments & Course Outcomes matrix (LOs or CLOs)

Assessment Methods	COs or LOs
1 st Midterm exam	CO No: 1, 2
2 nd Midterm exam	CO No: 3,4
Final exam	CO No: 1 to 5

M. Teaching Methods & Learning Activities: (tick the applicable activities)

Telling/Explaining	✓	Collaborating	
Questioning	✓	Experiments	✓
Reading	✓	Oral Presentations/Reports	
Demonstrating	✓	Web Searching	✓
Problem Solving	✓		

N. Assessment Methods (Formal & Informal) (tick the applicable activities)

Test/Exam	✓
Quiz	✓

O. Student Work load: (Total 135 Hours)

Course Readings	20 hrs	Exams/Quizzes	10 hrs
Problem Solving	30 hrs	Lectures	45 hrs
Lab Work	30 hrs		

CE491: DEEP LEARNING APPLICATIONS AND AI

B.TECH 7TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – IV

(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

A. Credits and Hours:

Teaching Scheme	Theory	Tutorial/Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	10
2	Deep neural network	12
3	Improving deep neural network	10
4	Convolution Neural Network	10
5	Sequence models	10
6	Structuring Deep learning project	08

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

C. Detailed Syllabus:

1	Introduction	10 Hours	16%
1.1	Introduction: deep learning		
1.2	Tools for deep learning		
1.3	Shallow and deep neural network		
1.4	Tensor flow		
2	Deep Neural Network	12 Hours	20%
2.1	Forward propagation		
2.2	Backpropagation		
2.3	Application: Load Forecasting		

3	Improving Deep Neural Network	10 Hours	17%
3.1	Learning rate, Activation Function		
3.2	Number of Layers and Neurons		
3.3	Overfitting and Underfitting Test		
4	Convolution Neural Network	10 Hours	17%
4.1	Convolution operation		
4.2	Padding and strides		
4.3	Activation functions		
4.4	Pooling and dropout layers		
5	Sequence models	10 Hours	17%
5.1	Time series and sequence data		
5.2	Modelling data for sequence learning		
5.3	Recurrent neural network		
5.4	Long short-term memory network		
6	Structuring Deep learning project	08 Hours	13%

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and practical session which carries 10 Marks and 5 marks weightage respectively.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a practical session where students have an opportunity to build an appreciation for the concepts being taught in lectures.

E. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To understand fundamental of Deep Learning and Artificial Intelligence (Understand).
CO2	To solve difficult and complex problem of computer science using AI techniques (Understand, Apply).
CO3	To identify fields where AI can be applied (Understand, Apply).

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	1	1	2	-	-	-	-	-	-	1	1	1
CO2	2	2	2	2	1	1	-	-	-	-	-	1	1	1
CO3	2	1	1	2	1	1	-	-	-	-	-	1	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Reference Books:

1. Stuart Russell, Artificial Intelligence – A Modern Approach (3 Edition), Pearson, U.K., 2016.
2. Tariq Rashid, Make your own neural network, CreateSpace Independent Publishing Platform, U.S., 2016.
3. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, The MIT Press, Cambridge, England, 2017.
4. Seth Weidman, Deep Learning from scratch, O'Reilly media, U.S.A., 2019.

Web Materials:

1. <https://www.udemy.com/course/artificial-intelligence-for-civil-engineers-part-1/>
2. <https://www.udemy.com/course/artificial-intelligence-for-civil-engineers-part-2/>

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VIII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE458: MAJOR PROJECT
8th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	36	36	20
Marks	00	800	800	

B. Course Outcomes (COs):

On successful completion of the course, students would be able to:

CO1	To apply knowledge and skills learned from project to real-world problems.
CO2	To solve engineering problems by the teamwork and work with other discipline.
CO3	Use experience related to professional and ethical issues in the work environment.
CO4	To explain the impact of engineering solutions developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.
CO5	To implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

C. Instructional Method and Pedagogy:

- Major project has 20 credit subject and 36 hours per week.
- Department/Institute will help students to find appropriate major projects.
- The student must be find the appropriate guide for their project and make the project
- Proposal and submit the project proposal to departmental co-coordinator.
- Project must be from the electrical bag round other project must be rejected.
- Student must meet at least once in week to his/her guide.
- The student shall submit two documents to the Coordinator for the evaluation of the major project:
 - Project report
 - Project presentation
- Upon the completion of project, a hard copy of “project Report” must be submitted to the Coordinator by the end of the semester.
- The report must outline with practical knowledge which student gain during project making. The guide must verify report.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory, the student shall be requested to repeat the project.

D. Format of Project Report:

Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12 pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the

purpose and scope of internship, the actual work done during project, and conclusions arrived at.

- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- INTRODUCTION: In this section, give the purpose of the project, reasons for choosing the project, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the project. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the project. Present your observations, contributions and intellectual benefits.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.